Supplementary Material

Multi-dimensional spectral geometry of biological knowledge in single-cell transformer representations

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Each table below summarises one analysis; the complete values are in the file named in its caption (folder `supplementary_tables/`). Extractions: lung (Tabula Sapiens lung, primary), lung (ext.) (lung atlas of Travaglini et al., replication), kidney B, immune and kidney A (see the main text, Table 1). L_ℓ is layer ℓ (0–11); SV_k is the k-th direction of the singular value decomposition of the centred observed-gene matrix at that layer. Unless stated otherwise intervals are 95%; network-statistic intervals use vertex subsampling with delete-d scaling; edge-statistic intervals use the two-way (factor × target) pigeonhole bootstrap; node-statistic intervals resample genes.

Table S1: **Spectral concentration with depth, in every extraction.** Singular values $S_1 \geq S_2 \geq \ldots$ of the centred observed-gene matrix ($|V| \times 512$) at each layer. ER, effective rank $\exp(-\sum_k p_k \log p_k)$ with $p_k = S_k / \sum_j S_j$ (Roy and Vetterli, 2007); $\mathrm{ER}_{\mathrm{var}}$, exponential of the Shannon entropy of the variance shares $q_k = S_k^2 / \sum_j S_j^2$; PR, participation ratio $(\sum_k S_k^2)^2 / \sum_k S_k^4$; SV_1, share of variance q_1 on the first direction (%); $\mathrm{SV}_{1\text{–}10}$, cumulative share of the first ten (%). **A**, the five main extractions at five layers ($|V|$ = 5,648, 5,616, 4,780, 2,039 and 1,972 genes for lung, lung (ext.), kidney B, immune and kidney A). Fold, L_0 value over L_{11} value for ER, $\mathrm{ER}_{\mathrm{var}}$ and PR, L_{11} over L_0 for the variance shares. Intervals for SV_1: 95% gene-bootstrap percentile intervals ($B = 200$; genes resampled with replacement). No bootstrap interval is given for ER: resampling genes with replacement duplicates rows and biases it downwards, so its sampling variability is taken from the without-replacement subsamples of panel D. **B**, all sixteen extractions. Cell draw, an independent draw of 256 immune cells; donor, 256 cells of one donor; B/CD4 T/CD8 T cells, 256 cells of one type; ckpt, one draw of 192 kidney cells (the 8,143-gene axis of kidney B) passed through the whole-human checkpoint used for every other extraction, the kidney-fine-tuned and the brain scGPT checkpoint. Rises, number of the eleven consecutive layer steps at which ER / $\mathrm{ER}_{\mathrm{var}}$ / PR increased (0, strictly decreasing). **C**, identifiability and controls. Near-degenerate pairs: number of layers (of 12) at which the 95% basic bootstrap interval of the gap ratio S_k/S_{k+1} ($B = 200$ gene resamples) reaches 1; last column, count and percentage over all 192 extraction-layers of the sixteen extractions (all 2,112 pairs in the last of these rows). The nine such pairs among SV_1–SV_4 ($k \leq 3$) of the five main extractions are flagged again at B = 1,000. Feature shuffle: ER after each of the 512 features is permuted independently across genes (mean of 20 draws; SD ≤ 0.12), divided by the observed ER, and the shuffled ER itself at L_{11}; last column, range over the sixteen extractions. Null direction: singular values below $10^{-3} S_1$ per layer, and the absolute cosine between the last right singular vector and the uniform vector $\mathbf{1}/\sqrt{512}$ at L_0 / L_{11}, with the relative size of the last singular value. **D**, detection controls. Matched: 1,300 genes subsampled without replacement within common deciles of $\log n_{\mathrm{obs}}$ pooled over the five extractions, mean ± SD over 200 draws; $n_{\mathrm{obs}} \geq 3$: the spectrum at L_{11} after excluding genes contributed by fewer than three sampled cells (thresholds 5 and 10 in the CSV). Last column, range over the five extractions. Full values: `Table_S1_compression.csv`.

A Spectral summaries at five layers, five main extractions

Extraction	Measure	L_0	L_3	L_6	L_9	L_{11}	Fold
lung	ER	457.4	435.1	402.4	354.2	324.1	1.41×
	$\mathrm{ER}_{\mathrm{var}}$	257.2	199.9	128.4	69.6	47.3	5.43×
	PR	86.4	65.3	36.0	18.4	12.3	7.04×
	SV_1 (%)	7.3	7.3	11.2	16.9	24.0	3.27×
	95% CI	7.2–7.5	7.2–7.6	10.7–11.8	16.0–18.2	22.5–25.4	
	$\mathrm{SV}_{1\text{–}10}$ (%)	24.1	30.1	40.8	54.0	61.3	2.54×
lung (ext.)	ER	459.4	439.0	409.8	363.3	340.8	1.35×
	$\mathrm{ER}_{\mathrm{var}}$	263.1	206.3	140.5	77.1	65.2	4.03×
	PR	87.8	63.7	37.2	18.2	17.8	4.94×
	SV_1 (%)	7.3	7.7	11.4	18.7	17.9	2.45×
	95% CI	7.1–7.5	7.5–8.0	11.1–11.8	18.3–19.2	17.5–18.4	
	$\mathrm{SV}_{1\text{–}10}$ (%)	23.7	29.4	38.4	50.6	54.6	2.31×
kidney B	ER	456.2	434.0	401.0	352.8	324.1	1.41×
	$\mathrm{ER}_{\mathrm{var}}$	257.3	196.7	127.5	70.5	50.6	5.09×
	PR	88.5	63.3	36.1	19.1	14.8	5.96×
	SV_1 (%)	7.0	7.1	10.9	16.5	17.6	2.50×
	95% CI	6.9–7.2	7.0–7.3	10.6–11.2	16.1–17.0	17.1–18.1	
	$\mathrm{SV}_{1\text{–}10}$ (%)	24.0	30.8	41.2	53.8	60.8	2.53×

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Table S1 (continued)

immune	ER	442.4	420.5	388.5	342.8	309.6	1.43×
	ER_{var}	236.9	186.5	125.5	74.7	48.7	4.86×
	PR	90.1	68.1	40.2	22.8	15.2	5.93×
	SV_1 (%)	5.6	6.5	10.3	14.3	18.6	3.32×
	95% CI	5.4–6.0	6.2–7.0	9.8–11.0	13.7–15.0	16.7–20.9	
	$SV_{1–10}$ (%)	25.4	31.2	40.7	52.2	61.3	2.42×
kidney A	ER	437.4	415.2	381.0	329.2	297.1	1.47×
	ER_{var}	209.5	158.0	98.1	52.0	34.8	6.01×
	PR	64.8	46.3	25.5	14.3	10.7	6.08×
	SV_1 (%)	8.1	9.8	14.7	20.6	23.1	2.85×
	95% CI	7.6–8.6	9.4–10.3	14.2–15.5	19.7–21.6	22.3–24.1	
	$SV_{1–10}$ (%)	28.7	35.8	46.9	60.6	68.4	2.38×

B From the first to the last layer, all sixteen extractions

			Fold L_0/L_{11}			SV_1 (%)	Rises
Extraction	Design	$\|V\|$	ER	ER_{var}	PR	$L_0 \to L_{11}$	$ER/ER_{var}/PR$
lung	primary	5,648	1.41	5.43	7.04	7.3 → 24.0	0/0/0
lung (ext.)	replication	5,616	1.35	4.03	4.94	7.3 → 17.9	0/0/1
kidney B	second tissue	4,780	1.41	5.09	5.96	7.0 → 17.6	0/0/0
immune	continuity	2,039	1.43	4.86	5.93	5.6 → 18.6	0/0/1
kidney A	continuity	1,972	1.47	6.01	6.08	8.1 → 23.1	0/0/1
immune	cell draw, seed 43	2,080	1.43	5.17	6.68	6.0 → 21.9	0/0/0
	cell draw, seed 44	1,972	1.41	4.60	5.59	5.7 → 17.5	0/0/0
	donor TSP14	1,964	1.47	6.35	8.14	7.4 → 30.0	0/0/0
	donor TSP21	1,765	1.37	3.96	5.08	5.6 → 16.3	0/0/1
	donor TSP25	1,791	1.38	4.04	5.03	6.0 → 16.6	0/0/1
	B cells	1,398	1.47	8.37	13.35	6.4 → 38.2	0/0/0
	CD4 T cells	1,465	1.40	4.64	6.27	5.9 → 18.7	0/0/0
	CD8 T cells	1,595	1.44	6.37	9.77	6.1 → 30.1	0/0/0
kidney (192 cells)	whole-human ckpt	4,923	1.40	5.08	5.94	7.1 → 16.8	0/0/0
	kidney ckpt	4,923	2.49	23.88	13.04	8.8 → 34.0	0/0/0
	brain ckpt	4,923	1.64	9.73	10.56	7.0 → 28.9	0/0/2

C Near-degenerate adjacent singular values, feature shuffle and the null direction

	lung	lung (ext.)	kidney B	immune	kidney A	All 16
Near-degenerate pairs, S_k/S_{k+1}						
$k = 1$ (SV_1/SV_2)	0	0	0	4	0	8/192 (4.2%)
$k = 2$ (SV_2/SV_3)	0	0	1	0	1	6/192 (3.1%)
$k = 3$ (SV_3/SV_4)	1	1	0	1	0	24/192 (12.5%)
$k = 4$ (SV_4/SV_5)	0	0	1	1	3	25/192 (13.0%)
$k = 5$ (SV_5/SV_6)	0	0	2	5	1	42/192 (21.9%)
$k = 6$ (SV_6/SV_7)	3	0	0	2	2	34/192 (17.7%)
$k = 7$ (SV_7/SV_8)	0	2	2	2	4	31/192 (16.1%)
$k = 8$ (SV_8/SV_9)	2	2	2	9	4	67/192 (34.9%)
$k = 9$ (SV_9/SV_{10})	5	3	6	3	3	60/192 (31.2%)
$k = 10$ (SV_{10}/SV_{11})	4	3	1	3	6	74/192 (38.5%)
$k = 11$ (SV_{11}/SV_{12})	5	1	5	2	3	58/192 (30.2%)
all pairs, $k = 1$–11	20/132	12/132	20/132	32/132	27/132	429/2,112 (20.3%)
Feature shuffle						
shuffled/observed ER, L_0	1.09×	1.09×	1.10×	1.11×	1.12×	1.06–1.12
shuffled/observed ER, L_{11}	1.48×	1.40×	1.48×	1.52×	1.57×	1.40–2.51
shuffled ER, L_{11}	479.6	478.7	479.8	470.8	467.5	454.1–479.9
Null direction						
singular values $< 10^{-3} S_1$	1	1	1	1	1	1 at all 192
$\|\cos\|$ with uniform, L_0/L_{11}		0.9993 / 0.9991 in each of the five				
S_{512}/S_1 (L_0, L_{11})		$3.3–7.0 \times 10^{-8}$				

D Detection controls, five main extractions

	lung	lung (ext.)	kidney B	immune	kidney A	Range
ER, all V, L_{11} (panel A)	324.1	340.8	324.1	309.6	297.1	297.1–340.8
ER, matched 1,300 genes, L_0	432.0 ± 0.4	435.7 ± 0.3	433.1 ± 0.4	431.4 ± 0.3	427.4 ± 0.3	427.4–435.7
ER, matched 1,300 genes, L_{11}	296.3 ± 1.1	318.3 ± 0.5	302.8 ± 0.8	299.5 ± 0.8	290.0 ± 0.6	290.0–318.3
ER, matched, fold L_0/L_{11}	1.46×	1.37×	1.43×	1.44×	1.47×	1.37–1.47
genes with $n_{obs} \geq 3$	4,221	4,529	3,528	1,341	1,346	1,341–4,529
$n_{obs} \geq 3$: ER, L_{11}	335.1	340.0	325.2	318.1	301.2	301.2–340.0
$n_{obs} \geq 3$: PR, L_{11}	20.5	17.6	13.4	20.6	9.3	9.3–20.6
$n_{obs} \geq 3$: SV_1 (%), L_{11}	13.7	17.7	18.8	15.5	27.0	13.7–27.0

Table S2: **The early compartment axis.** Each of the first twelve directions (SV_1–SV_{12}) of the observed-gene decomposition at each layer, tested for separating genes directly annotated to extracellular space (GO:0005615) or cytosol (GO:0005829) from all other observed genes. AUROC, sign-free AUROC $\max(A, 1-A)$; z, AUROC standardised by the mean and SD of its permutation null; p_{adj}, Westfall–Young max-T adjusted p over the 288 tests of one extraction (12 directions × 12 layers × 2 terms; 2,000 joint permutations of the gene labels), 0.0005 = 1/2,001 being its floor. **A**, the cell (layer, direction) with the largest AUROC for each term; n, observed genes annotated to the term; under each extraction, $|V|$ / genes with any GO annotation. CI, 95% gene-bootstrap percentile interval (B = 2,000, direction held fixed), extracellular cells only. 95th percentile of the null maximum of z: 4.68, 4.66, 4.59, 4.67, 4.55 (row order). Strat., AUROC within deciles of $\log n_{\mathrm{obs}}$, oriented like the AUROC (own max-T family, labels permuted within deciles; p_{adj} at the floor except kidney A cytosol, 0.0355); $|\rho|$, absolute Spearman correlation of the coordinate with $\log n_{\mathrm{obs}}$; Annot., AUROC of the same cell among annotated genes only (p_{adj} at the floor); OR, Fisher odds ratio (Haldane) for the term in the 27% pole where it is enriched (two-sided p) and in the opposite pole. **B**, extracellular AUROC of the compartment direction (SV_1; SV_2 in immune) at every layer; for kidney A the row shows SV_1, which is not the lung compartment direction (r and lung-matched cell in D); p_{adj} as in A; each layer is decomposed separately; permutation-null mean 0.510–0.520. **C**, extracellular space on one common gene set per extraction (observed genes in the scGPT vocabulary and the ESM2 file; n, with extracellular genes in parentheses). Probe, out-of-fold AUROC of L2 logistic regression on the first 256 principal components (5 outer folds, regularisation by inner 5-fold cross-validation), with 95% gene-bootstrap percentile interval (B = 2,000 resamples of the out-of-fold scores); Δ, paired difference from scGPT L_3 on the same resamples; detection-stratified probe AUROCs differ from these by ≤ 0.013 (cytosol probes in the CSV). Own-SVD, sign-free AUROC of SV_1 of the representation's own decomposition of these genes, and the best of its first twelve directions (index; optimistic by construction). Input-token embedding, scGPT's gene-embedding table, before any cell is processed; ESM2-15B, the 5,120-dimensional per-protein embedding of the 15-billion-parameter protein language model ESM2. **D**, on the extracellular cell of A, oriented so that extracellular genes score high: AUROC of cytosol, mitochondrion (GO:0005739) and endoplasmic-reticulum lumen (GO:0005788) against all other genes (below 0.5, the opposite end; p_{adj} for cytosol as in A, for the other two over the 288 tests of these two terms, same permutations); r, Pearson correlation with the lung coordinate on the n shared genes; lung-matched cell, the extraction's cell most correlated with the lung coordinate ($|r|$), and its extracellular AUROC oriented like the lung coordinate. Full values: `Table_S2_compartment.csv`.

A Best cell for each term (288 tests per extraction)

Extraction	Term (n)	Cell	AUROC	95% CI	z	p_{adj}	Strat.	$\lvert\rho\rvert$	Annot.	OR, enriched pole (p)	OR, opposite
lung	extracellular (591)	L_3 SV_1	0.773	0.754–0.792	34.5	0.0005	0.776	0.12	0.779	5.52 (7.9×10^{-81})	0.13
5,648 / 5,343	cytosol (1,989)	L_4 SV_2	0.633	–	26.5	0.0005	0.589	0.53	0.615	1.96 (7.8×10^{-28})	0.42
lung (ext.)	extracellular (554)	L_3 SV_1	0.756	0.736–0.777	30.8	0.0005	0.767	0.20	0.763	5.02 (1.0×10^{-68})	0.16
5,616 / 5,239	cytosol (1,978)	L_4 SV_2	0.640	–	26.2	0.0005	0.594	0.40	0.619	2.14 (8.3×10^{-35})	0.37
kidney B	extracellular (354)	L_2 SV_1	0.707	0.681–0.733	20.4	0.0005	0.700	0.25	0.719	3.77 (1.1×10^{-31})	0.24
4,780 / 4,544	cytosol (1,876)	L_{11} SV_2	0.603	–	19.2	0.0005	0.554	0.58	0.580	1.80 (5.4×10^{-19})	0.51
immune	extracellular (258)	L_0 SV_2	0.760	0.732–0.788	21.6	0.0005	0.761	0.00	0.752	4.61 (1.5×10^{-28})	0.09
2,039 / 1,389	cytosol (427)	L_0 SV_1	0.714	–	21.2	0.0005	0.615	0.64	0.610	3.52 (9.4×10^{-28})	0.17
kidney A	extracellular (146)	L_0 SV_1	0.759	0.724–0.793	15.9	0.0005	0.804	0.56	0.658	3.79 (4.2×10^{-14})	0.04
1,972 / 1,352	cytosol (466)	L_3 SV_1	0.740	–	24.8	0.0005	0.563	0.80	0.586	3.25 (2.4×10^{-25})	0.04

B Extracellular AUROC of the compartment direction at every layer

Extraction		L_0	L_1	L_2	L_3	L_4	L_5	L_6	L_7	L_8	L_9	L_{10}	L_{11}
lung	AUROC	0.730	0.751	0.762	0.773	0.751	0.646	0.633	0.612	0.594	0.591	0.629	0.611
SV_1	p_{adj}	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005
lung (ext.)	AUROC	0.715	0.725	0.747	0.756	0.729	0.667	0.672	0.625	0.599	0.556	0.529	0.548
SV_1	p_{adj}	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0015	0.9170	0.0370
kidney B	AUROC	0.682	0.693	0.707	0.703	0.666	0.626	0.623	0.587	0.565	0.532	0.508	0.523
SV_1	p_{adj}	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0035	0.9705	1.0000	1.0000
immune	AUROC	0.760	0.740	0.643	0.563	0.620	0.647	0.632	0.588	0.583	0.546	0.542	0.596
SV_2	p_{adj}	0.0005	0.0005	0.0005	0.1824	0.0005	0.0005	0.0005	0.0015	0.0040	0.8211	0.9465	0.0005
kidney A	AUROC	0.759	0.744	0.714	0.707	0.722	0.722	0.721	0.701	0.690	0.671	0.659	0.646
SV_1	p_{adj}	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005

C Linear probe and own-decomposition AUROC for extracellular space, common gene sets

Representation	lung n = 5,413 (591)	lung (ext.) n = 5,306 (554)	kidney B n = 4,602 (354)	immune n = 1,554 (258)
Probe, out-of-fold AUROC (95% CI)				
scGPT L_0	0.837 (0.819–0.855)	0.834 (0.813–0.853)	0.799 (0.771–0.824)	0.832 (0.803–0.859)
scGPT L_3	0.836 (0.818–0.854)	0.831 (0.810–0.851)	0.806 (0.779–0.832)	0.824 (0.792–0.852)
scGPT L_{11}	0.833 (0.815–0.850)	0.829 (0.808–0.848)	0.799 (0.772–0.825)	0.827 (0.797–0.854)
scGPT input-token embedding	0.838 (0.820–0.856)	0.834 (0.814–0.854)	0.797 (0.770–0.824)	0.826 (0.796–0.853)
ESM2-15B	0.900 (0.884–0.914)	0.910 (0.895–0.924)	0.868 (0.845–0.890)	0.900 (0.877–0.921)
Δ *probe AUROC from scGPT* L_3 *(95% CI)*				
scGPT L_0	+0.001 (−0.005, +0.006)	+0.002 (−0.003, +0.008)	−0.007 (−0.015, +0.001)	+0.008 (−0.004, +0.021)
scGPT L_{11}	−0.003 (−0.011, +0.005)	−0.003 (−0.011, +0.005)	−0.007 (−0.018, +0.004)	+0.003 (−0.010, +0.016)
scGPT input-token embedding	+0.002 (−0.004, +0.008)	+0.003 (−0.004, +0.010)	−0.008 (−0.018, +0.002)	+0.002 (−0.009, +0.012)
ESM2-15B	+0.064 (+0.048, +0.081)	+0.079 (+0.062, +0.095)	+0.062 (+0.040, +0.085)	+0.076 (+0.050, +0.105)
Own-SVD: SV_1 *AUROC; best of* SV_1–SV_{12} *(index)*				
scGPT L_0	0.746; 0.746 (1)	0.730; 0.730 (1)	0.694; 0.694 (1)	0.635; 0.715 (3)
scGPT L_3	0.779; 0.779 (1)	0.766; 0.766 (1)	0.724; 0.724 (1)	0.642; 0.706 (4)
scGPT L_{11}	0.604; 0.705 (3)	0.528; 0.722 (2)	0.525; 0.660 (3)	0.542; 0.729 (6)
scGPT input-token embedding	0.545; 0.758 (2)	0.530; 0.743 (2)	0.505; 0.705 (2)	0.516; 0.723 (3)
ESM2-15B	0.542; 0.758 (8)	0.539; 0.762 (8)	0.542; 0.685 (8)	0.557; 0.751 (8)

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Table S2 (continued)

D Other compartments on the extracellular cell, and replication of the direction

		Oriented AUROC (p_{adj})				Lung-matched cell	
Extraction	Cell	Cytosol	Mitochondrion	ER lumen	r with lung (n)	Cell ($\lvert r\rvert$)	Extracellular (p_{adj})
lung	L_3 SV_1	0.392 (0.0005)	0.400 (0.0005)	0.667 (0.0005)	–	–	–
lung (ext.)	L_3 SV_1	0.395 (0.0005)	0.401 (0.0005)	0.660 (0.0005)	0.985 (4,535)	L_3 SV_1 (0.985)	0.756 (0.0005)
kidney B	L_2 SV_1	0.426 (0.0005)	0.438 (0.0010)	0.613 (0.0420)	0.970 (4,023)	L_2 SV_1 (0.970)	0.707 (0.0005)
immune	L_0 SV_2	0.465 (0.9330)	0.420 (0.7406)	0.665 (0.0315)	0.920 (856)	L_1 SV_2 (0.929)	0.740 (0.0005)
kidney A	L_0 SV_1	0.690 (0.0005)	0.578 (0.0525)	0.798 (0.0005)	0.118 (840)	L_0 SV_2 (0.886)	0.568 (0.5027)

Table S3: **H1: physical-interaction partners share spectral coordinates (significance, nulls, hubs, co-expression, co-complex pairs, pole size and STRING confidence).** r, scalar assortativity of a spectral coordinate over the STRING v12 physical edges (combined score ≥ 700) whose two ends are observed genes (V; n_V genes, m edges), i.e. the Pearson correlation of the coordinate across the two ends of every edge. Cell, one direction at one layer; the family is SV_1–SV_{10} at L_0–L_{11} (120 cells). MS, Maslov–Sneppen degree-preserving rewiring of the same edges (double-edge swaps; burn-in $10m$ accepted swaps, then m between draws); N, joint null draws applied to all 120 cells; Sig., number of cells (of 120) with two-sided Westfall–Young max-T adjusted $p < 0.05$ (the family minimum is at the permutation floor $1/(N+1)$ in every family shown, and every significant cell has $r > 0$); $z = (r - \text{null mean})/\text{null SD}$; strongest cell, the cell with the largest $|z|$ under MS. 95% CI: vertex subsampling ($n_V/2$ genes without replacement, $B = 500$) with delete-d scaling. **(A)** Robustness extractions: two further 256-cell immune draws, one draw per donor for three donors, three immune cell types, and one further 192-cell draw of kidney B's cells embedded with each of three scGPT checkpoints (whole-human, kidney, brain); no interval was computed for them. **(B)** Label shuffle, permutation of coordinates across V; degree strata, permutation within deciles of degree in the V network or in the genome-wide STRING network; detection strata, permutation within deciles of $\log n_{obs}$, the number of sampled cells that contributed to a gene. **(C)** The 5% or 10% most connected genes of V are removed with their edges; genes, number removed; edges kept, fraction of the m edges remaining; max. change, largest decrease of r when one gene and its edges are removed; jackknife SE, delete-1 jackknife standard error of r. **(D)** Upper part: every STRING edge is compared with five gene pairs of V that have no physical STRING link at any score, drawn from the same decile of |co-expression| (Pearson correlation of log-normalised expression over the extraction's cells); AUROC stratified by decile, for three scores: cosine of the two genes' centred 512-dimensional vectors at the strongest cell's layer, $-|\Delta x|$ (negative absolute difference of the two genes' coordinates on the strongest cell), and |co-expression| itself (the co-expression left after matching); 95% CI as above; mean |coexpr|, over edges / matched non-edges; 50 bins, the same with 50 quantile bins (immune: 5,687 of 6,255 non-edges available); 3,000 cells, matching on co-expression computed on 3,000 cells of the same population. Lower part: co-complex pairs within V (ComplexPortal; hu.MAP 2.0; each also without the pairs that are STRING physical ≥ 700 edges) as the edge set of the same 120-cell family; r and z at the STRING strongest cell. **(E)** Co-pole rate, fraction of edges whose two ends fall in the same pole (top or bottom K genes of the coordinate, $K = \lfloor (K/n)\, n_V \rfloor$); ratio, observed rate / null mean (N as in A). Upper part: MS null; the interval is for the ratio of the observed rate to its configuration-model expectation, which differs from the MS null mean by 0.1–0.9% in these cells. Lower part: $K/n = 0.27$ under both nulls. **(F)** ρ, Spearman correlation of a link's STRING score with the cosine of its two genes' centred, L2-normalised 512-dimensional vectors; ρ_{adj}, partial Spearman correlation controlling for the ranks of the summed log degree (score ≥ 150 network within V), of the summed $\log(1 + n_{obs})$ and of |co-expression| (3,000 cells); 95% CI as above. Band AUROC, links of each combined-score band against one non-linked pair per link (no physical link at any score) matched on the degree deciles of both ends, drawn once (no interval). Rows 'all 12': ranges over layers; for ρ_{adj} the smallest lower confidence bound over layers in brackets; monotone, layers at which the four band AUROCs increase strictly. MS chains: mean acceptance 0.90–0.96, lag-1 autocorrelation of the strongest cell's null statistic 0.09–0.14. Vertex-subsampling intervals with the pre-registered scaling factor $\sqrt{m/n}$, which omits the finite-population term and so gives half-widths smaller by about a factor $\sqrt{2}$, are in the CSV (columns ending `_prereg`). Full values: `Table_S3_interaction.csv`.

A H1 family against Maslov–Sneppen rewiring (MS), per extraction

Extraction	n_V	m	N	Sig.	Strongest cell	r	95% CI	Null mean	z	Median r
lung	5,648	14,474	10,000	120	$L_3\,SV_1$	0.435	0.388–0.487	−0.001	53.1	0.181
lung (ext.)	5,616	14,132	10,000	120	$L_8\,SV_2$	0.406	0.343–0.462	−0.001	49.2	0.170
kidney B	4,780	12,434	2,000	120	$L_9\,SV_3$	0.404	0.351–0.454	−0.001	45.9	0.209
immune	2,039	1,251	2,000	118	$L_4\,SV_6$	0.566	0.465–0.688	−0.003	20.9	0.318
kidney A	1,972	1,390	2,000	120	$L_5\,SV_2$	0.702	0.602–0.809	−0.011	28.2	0.353
Robustness extractions (MS, $N = 1{,}000$)										
immune, draw (seed 43)	2,080	1,293	1,000	117	$L_6\,SV_5$	0.562	–	−0.004	20.7	0.266
immune, draw (seed 44)	1,972	1,263	1,000	117	$L_4\,SV_6$	0.577	–	−0.005	20.3	0.306
immune, donor TSP14	1,964	1,030	1,000	111	$L_6\,SV_7$	0.559	–	−0.004	19.0	0.294
immune, donor TSP21	1,765	1,050	1,000	109	$L_2\,SV_2$	0.525	–	−0.005	16.9	0.247
immune, donor TSP25	1,791	1,064	1,000	107	$L_2\,SV_2$	0.521	–	−0.005	16.8	0.228
immune, B cells	1,398	615	1,000	102	$L_2\,SV_2$	0.547	–	−0.007	14.0	0.265
immune, CD4 T cells	1,465	749	1,000	109	$L_1\,SV_2$	0.502	–	−0.003	14.2	0.251
immune, CD8 T cells	1,595	896	1,000	109	$L_3\,SV_3$	0.588	–	−0.004	18.4	0.257
kidney cells, whole-human ckpt.	4,923	12,689	1,000	120	$L_9\,SV_3$	0.383	–	0.000	44.2	0.216
kidney cells, kidney ckpt.	4,923	12,689	1,000	120	$L_2\,SV_3$	0.418	–	−0.001	47.4	0.187
kidney cells, brain ckpt.	4,923	12,689	1,000	120	$L_1\,SV_4$	0.302	–	−0.002	35.3	0.147

continued on next page

Table S3 (continued)

B Null mean of r at the strongest cell under each null ($N = 10{,}000$)

	lung, $L_3\,SV_1$ ($r = 0.435$)				lung (ext.), $L_8\,SV_2$ ($r = 0.406$)			
Null	Null mean	Null SD	z	Sig.	Null mean	Null SD	z	Sig.
Label shuffle over V	−0.001	0.008	53.1	120	−0.001	0.008	49.0	120
Degree strata (V network)	0.005	0.009	50.4	120	0.007	0.009	45.3	120
Degree strata (genome-wide STRING)	0.021	0.010	41.0	120	0.025	0.010	36.5	120
Detection strata ($\log n_{\text{obs}}$)	0.000	0.008	52.0	117	0.005	0.009	45.5	120
MS rewiring (primary)	−0.001	0.008	53.1	120	−0.001	0.008	49.2	120

C Hub removal (MS null on the remaining graph, $N = 2{,}000$) and leave-one-gene-out influence, strongest cell

			Top 5% by degree removed				Top 10% by degree removed				Leave one gene out	
Extraction	Cell	r	Genes	Edges kept	r	z	Genes	Edges kept	r	z	Max. change	Jackknife SE
lung	$L_3\,SV_1$	0.435	283	0.42	0.515	40.4	565	0.25	0.511	30.7	0.0045	0.025
lung (ext.)	$L_8\,SV_2$	0.406	281	0.42	0.518	39.5	562	0.25	0.487	28.5	0.0042	0.028
kidney B	$L_9\,SV_3$	0.404	239	0.44	0.499	36.4	478	0.26	0.488	27.9	0.0036	0.026
immune	$L_4\,SV_6$	0.566	102	0.29	0.537	10.0	204	0.12	0.528	6.7	0.0141	0.052
kidney A	$L_5\,SV_2$	0.702	99	0.32	0.569	11.9	198	0.12	0.512	6.9	0.0075	0.050

D STRING edges against co-expression-matched non-edges (stratified AUROC), and co-complex pairs as edge sets

			Deciles of |co-expression|, extraction cells			50 bins		3,000 cells	
Extraction	Cell	Mean |coexpr|	Cosine	$-\lvert\Delta x\rvert$	|coexpr|	Cosine	|coexpr|	Cosine	|coexpr|
lung	$L_3\,SV_1$	0.125/0.108	0.725 (0.706–0.743)	0.631 (0.612–0.647)	0.560 (0.539–0.579)	0.740	0.555	0.725	0.561
lung (ext.)	$L_8\,SV_2$	0.085/0.077	0.679 (0.662–0.697)	0.617 (0.600–0.634)	0.534 (0.519–0.549)	0.690	0.527	0.690	0.551
kidney B	$L_9\,SV_3$	0.121/0.111	0.681 (0.660–0.699)	0.625 (0.610–0.641)	0.538 (0.521–0.554)	0.695	0.530	0.683	0.542
immune	$L_4\,SV_6$	0.216/0.158	0.810 (0.770–0.855)	0.577 (0.525–0.625)	0.663 (0.617–0.715)	0.800	0.553	0.803	0.658
kidney A	$L_5\,SV_2$	0.282/0.269	0.873 (0.838–0.912)	0.733 (0.687–0.784)	0.567 (0.529–0.604)	0.860	0.528	0.872	0.554

	lung, $L_3\,SV_1$				lung (ext.), $L_8\,SV_2$			
Co-complex edge set (MS, $N = 2{,}000$)	Pairs	r	z	Sig.	Pairs	r	z	Sig.
ComplexPortal	1,934	0.588	26.1	119	1,863	0.524	23.3	120
ComplexPortal, not STRING ≥ 700	246	0.423	7.0	79	220	0.278	4.6	38
hu.MAP 2.0	6,782	0.265	21.5	113	6,639	0.246	20.0	119
hu.MAP 2.0, not STRING ≥ 700	5,159	0.196	14.1	95	5,039	0.172	12.0	95

E Co-pole rate as a function of pole size, and label shuffle against rewiring at $K/n = 0.27$

		Observed / MS-null co-pole rate (95% CI) at pole size K/n				
Extraction	Cell	0.10	0.20	0.27	0.33	0.50
lung	$L_3\,SV_1$	2.27 (1.84–2.58)	1.88 (1.69–2.05)	1.70 (1.57–1.82)	1.60 (1.50–1.69)	1.32 (1.25–1.37)
lung (ext.)	$L_8\,SV_2$	2.10 (1.71–2.37)	1.85 (1.66–2.01)	1.70 (1.58–1.82)	1.56 (1.45–1.66)	1.30 (1.23–1.35)
kidney B	$L_9\,SV_3$	2.06 (1.72–2.35)	1.86 (1.68–2.03)	1.66 (1.53–1.77)	1.55 (1.44–1.65)	1.26 (1.21–1.31)

			Label shuffle				MS rewiring			
Extraction	Cell	Observed	Null mean	Null SD	Ratio	z	Null mean	Null SD	Ratio	z
lung	$L_3\,SV_1$	0.256	0.146	0.0074	1.76	14.9	0.150	0.0024	1.70	44.2
lung (ext.)	$L_8\,SV_2$	0.258	0.146	0.0076	1.77	14.9	0.152	0.0024	1.70	44.1
kidney B	$L_9\,SV_3$	0.248	0.146	0.0077	1.71	13.4	0.150	0.0025	1.66	39.3
kidney B	$L_{11}\,SV_1$	0.116	0.146	0.0078	0.79	−3.9	0.082	0.0021	1.40	15.8
immune	$L_4\,SV_6$	0.344	0.145	0.0176	2.37	11.3	0.182	0.0088	1.88	18.4
kidney A	$L_5\,SV_2$	0.574	0.146	0.0184	3.93	23.2	0.284	0.0099	2.02	29.3
kidney A	$L_2\,SV_2$	0.614	0.145	0.0180	4.23	26.1	0.294	0.0101	2.09	31.8
kidney A	$L_4\,SV_2$	0.563	0.146	0.0181	3.87	23.1	0.275	0.0097	2.05	29.6
kidney A	$L_0\,SV_1$	0.212	0.145	0.0190	1.46	3.5	0.148	0.0063	1.43	10.1
kidney A	$L_{11}\,SV_1$	0.403	0.145	0.0187	2.77	13.8	0.255	0.0070	1.58	21.2

F STRING confidence gradient (physical links, combined score ≥ 150, within V)

					AUROC by STRING combined-score band				
Extraction	Links	Layer	ρ (95% CI)	ρ_{adj} (95% CI)	150–399	400–699	700–899	900–999	Monotone
lung	132,077	L_0	0.098 (0.081–0.116)	0.091 (0.075–0.108)	0.569	0.597	0.642	0.692	
		L_{11}	0.050 (0.032–0.069)	0.051 (0.036–0.067)	0.545	0.558	0.584	0.630	
		all 12	0.050–0.102	0.051–0.096 [0.036]	0.545–0.569	0.558–0.600	0.584–0.643	0.630–0.698	12/12
lung (ext.)	127,663	L_0	0.092 (0.073–0.110)	0.096 (0.078–0.112)	0.568	0.590	0.643	0.683	
		L_{11}	0.051 (0.032–0.069)	0.060 (0.044–0.076)	0.550	0.561	0.596	0.629	
		all 12	0.051–0.095	0.060–0.099 [0.044]	0.550–0.568	0.561–0.592	0.596–0.644	0.629–0.686	12/12
kidney B	112,823	L_0	0.097 (0.078–0.117)	0.097 (0.079–0.115)	0.556	0.584	0.639	0.694	
		L_{11}	0.043 (0.023–0.063)	0.046 (0.028–0.064)	0.544	0.551	0.587	0.633	
		all 12	0.043–0.099	0.046–0.098 [0.028]	0.542–0.557	0.551–0.586	0.587–0.640	0.633–0.696	12/12
immune	7,803	L_0	0.147 (0.101–0.195)	0.111 (0.066–0.157)	0.624	0.637	0.736	0.767	
		L_{11}	0.092 (0.038–0.145)	0.055 (0.006–0.104)	0.590	0.585	0.636	0.691	
		all 12	0.092–0.153	0.055–0.115 [0.006]	0.589–0.624	0.584–0.641	0.636–0.742	0.691–0.768	8/12
kidney A	7,938	L_0	0.188 (0.120–0.264)	0.190 (0.128–0.244)	0.661	0.659	0.749	0.820	
		L_{11}	0.085 (0.007–0.169)	0.115 (0.050–0.178)	0.647	0.613	0.699	0.761	
		all 12	0.085–0.188	0.115–0.190 [0.050]	0.636–0.663	0.606–0.659	0.691–0.750	0.758–0.822	0/12

Table S4: **Step-by-step bridge on kidney A from the screen's interaction co-pole statistic to the H1 statistic.** Each step changes one factor of the previous step (cumulatively, except the off-path check S1′, shown in italics, which branches from S1). Every step evaluates 36 cells, SV_2–SV_4 at L_0–L_{11} of that step's decomposition (at S0 the zero-row SVD, whose SV_2–SV_4 correspond to observed SV_1–SV_3; from S1 the observed-gene SVD), with one null draw applied jointly to all 36. Co-pole rate, fraction of pairs whose two genes fall in the same pole (the top or bottom $K = \lfloor 0.27\,n \rfloor$ genes of the coordinate, n the number of genes in the universe); label shuffle, permutation of coordinates (or pole labels) across the universe; MS, Maslov–Sneppen degree-preserving rewiring; r, scalar assortativity of the coordinate over the pairs. Zero-row SVD, SVD of the full 4,803-gene matrix including the all-zero rows of unobserved genes. Genes/pairs, size of the gene universe and number of gene pairs; N, null draws. Cells, number of the 36 with raw $p < 0.05$ / raw $p < 0.05/183$ (the screen's Bonferroni threshold over its 183 hypotheses; n.r., not resolvable because $1/(N+1) > 0.05/183$) / max-T adjusted $p < 0.05$ over the 36 cells (descriptive). The minimum adjusted p is at the permutation floor $1/(N+1)$ at every step except S1′ (0.017); at S5, 36 of the 36 cells are at the floor. Tracked direction: $L_5\,SV_3$ of the zero-row SVD at S0, which is $L_5\,SV_2$ of the observed-gene SVD (the zero-row SVD's leading direction separates unobserved from observed genes and shifts the low indices by one; observed-gene $L_5\,SV_3$ at S1 has a co-pole rate of 0.135, $z = -0.25$). Value, statistic of that direction; null, null mean; ratio, value / null mean (co-pole steps); z = (value − null mean)/null SD; best cell, the cell of the 36 with the largest z (largest $|z|$ for two-sided steps). S0 does not use the screen's own decomposition, of a 209-gene subset that includes 22 zero rows and maps only weakly onto the observed-gene basis (Table S14); the screen's values on that subset are in Table S13 (claim C21). At S3 the co-pole null mean of the tracked direction under MS rewiring is about twice its label-shuffle value, so the label-shuffle ratio overstates the enrichment about twofold (Table S3, panel E); S4 and S5 use assortativity, whose null mean is close to zero under both nulls. Full values: `Table_S4_bridge.csv`.

						Tracked direction					Best cell of 36		
Step	Change	Genes/pairs	Statistic; null	N	Cells	Cell	Value	Null	Ratio	z	Cell	Value	z
S0	The screen's ingredients (the 187 observed genes of its curated list; functional STRING pairs, score ≥ 0.7, among them), with coordinates from the zero-row SVD	187/809	co-pole rate; label shuffle, one-sided	10,000	23/13/15	$L_5\,SV_3$	0.333	0.141	2.36	7.5	$L_5\,SV_3$	0.333	7.5
S1	SVD of the observed-gene matrix (V, 1,972 genes); coordinates still restricted to the 187 genes	187/809	co-pole rate; label shuffle, one-sided	10,000	13/10/11	$L_5\,SV_2$	0.333	0.141	2.36	7.4	$L_5\,SV_2$	0.333	7.4
S1′	*Off-path check: S1 with physical STRING ≥ 700 pairs among the 187 genes*	*187/163*	*co-pole rate; label shuffle, one-sided*	*10,000*	*9/0/3*	$L_5\,SV_2$	*0.166*	*0.141*	*1.18*	*0.7*	$L_{11}\,SV_2$	*0.276*	*3.9*
S2	Universe V: poles and shuffle over all 1,972 observed genes; same 809 functional pairs	1,972/809	co-pole rate; label shuffle, one-sided	10,000	19/13/15	$L_5\,SV_2$	0.600	0.146	4.12	13.9	$L_6\,SV_2$	0.656	15.7
S3	Physical STRING ≥ 700 edges within V instead of functional pairs	1,972/1,390	co-pole rate; label shuffle, one-sided	10,000	29/23/26	$L_5\,SV_2$	0.574	0.145	3.95	23.3	$L_2\,SV_2$	0.614	25.3
S4	Scalar assortativity r instead of the co-pole rate	1,972/1,390	r; label shuffle, two-sided	10,000	36/36/36	$L_5\,SV_2$	0.702	−0.004	–	27.7	$L_5\,SV_2$	0.702	27.7
S5	Maslov–Sneppen rewiring instead of the label shuffle (the primary H1 run of kidney A)	1,972/1,390	r; MS rewiring, two-sided	2,000	36/n.r./36	$L_5\,SV_2$	0.702	−0.011	–	28.2	$L_5\,SV_2$	0.702	28.2

Table S5: **H1 conditioned on compartment co-membership and on the detection threshold.** Lung, lung (ext.) and kidney B; r, scalar assortativity of a spectral coordinate over the STRING physical edges (combined score ≥ 700) among observed genes, as in Table S3; MS, Maslov–Sneppen degree-preserving rewiring; Sig., number of the 120 cells (SV_1–$SV_{10} \times L_0$–L_{11}) with two-sided max-T adjusted $p < 0.05$ (the family minimum is at the permutation floor $1/(N+1)$ in every family; every significant cell has $r > 0$). 95% CI: vertex subsampling ($n_V/2$ genes, $B = 500$) with delete-d scaling, coordinates and residuals held fixed. **(A)** Each gene is assigned one class from its direct GO cellular-component annotations, in the priority order listed (extracellular space GO:0005615, plasma membrane GO:0005886, endoplasmic reticulum, ER lumen or Golgi, mitochondrion, nucleus, cytosol, any other direct GO annotation, none). Edge share, fraction of all STRING edges with both genes in that class; expected, its stub-matching (configuration-model) expectation; MS null, the same-class share in $N = 2{,}000$ rewired graphs. **(B)** Cells, the three cells with the largest $|z|$ under MS in Table S3 (*lung $L_7\,SV_6$, a cell with little compartment content, chosen post hoc). MS null, null mean of r in an MS replica with the same N; permutation within classes, one permutation of coordinates within each class applied to all 120 cells (preserves class, not degree); class-pair-preserving MS, double-edge swaps accepted only if every edge keeps its pair of classes (preserves every degree and the multiset of edge class pairs; acceptance 0.360–0.363, lag-1 autocorrelation of the strongest cell's null statistic 0.18–0.23); retained, $(r - \text{null mean})/(r - \text{MS null mean})$, the share of the excess over MS that remains (no interval). Rows '120 cells': largest null mean over cells; Sig.; median (minimum) retained share. **(C)** Ordinary least squares of each coordinate on the regressors listed; entries are residual r / raw r at the strongest cell (with 95% CI for the two tested designs) and the median over the 120 cells; Sig. for the residual coordinates. Class factor, seven class indicators and an intercept; 30 GO-CC/BP/MF terms, indicators of the 30 most frequent direct cellular-component, biological-process or molecular-function annotations among the genes of V; 8 class-defining terms, the eight class definitions of panel A as overlapping indicators; random gene sets, 30 sets of the same sizes as the CC terms; for these, ranges over the five seeds (two decimals). The 30-CC design explains 20–26% of the strongest coordinate's variance. Share of STRING edges whose two genes share at least one term: CC 0.89–0.90 (random pairs 0.41–0.45), BP 0.39–0.41 (0.06), MF 0.92 (0.54–0.59; its most frequent term, 'protein binding', is largely interaction-derived). **(D)** STRING edges split into those whose two genes share a class and the rest, each evaluated with its own MS chain (which does not keep edges within or between classes); class-centred, coordinates centred within each class before computing r. **(E)** V restricted to genes with n_{obs} (number of sampled cells contributing to the gene) at or above the threshold, SVD recomputed on the restricted genes and edges among them; rows 'all (V)', the unrestricted analysis (Table S3; MS $N = 10{,}000$ for lung and lung (ext.), interval from the draws of this analysis); strongest direction of the full-V analysis, the restricted direction at the same layer with the largest |correlation| of coordinates on the kept genes. Full values: `Table_S5_compartment_conditioning.csv`.

A Compartment classes and the share of STRING edges whose two genes fall in the same class

	lung			lung (ext.)			kidney B		
Class	Genes	Edge share	Expected	Genes	Edge share	Expected	Genes	Edge share	Expected
Extracellular space	591	0.043	0.016	554	0.041	0.014	354	0.023	0.009
Plasma membrane	1,392	0.164	0.088	1,370	0.161	0.086	1,137	0.146	0.072
ER or Golgi	368	0.011	0.004	359	0.011	0.004	326	0.012	0.004
Mitochondrion	284	0.010	0.002	283	0.010	0.002	312	0.018	0.004
Nucleus	1,362	0.200	0.112	1,378	0.203	0.117	1,253	0.222	0.136
Cytosol	509	0.019	0.007	516	0.020	0.007	493	0.020	0.007
Other annotated	837	0.008	0.003	779	0.007	0.003	669	0.007	0.003
Unannotated	305	0.000	0.000	377	0.000	0.000	236	0.000	0.000
All classes (same-class edges)	5,648	0.455	0.231	5,616	0.451	0.233	4,780	0.448	0.235
MS null: mean (SD); z		0.231 (0.0031)	$z = 72.7$		0.232 (0.0032)	$z = 69.4$		0.234 (0.0033)	$z = 63.8$

B Compartment-aware nulls ($N = 2{,}000$) at the three strongest cells of each extraction

				Permutation within classes				Class-pair-preserving MS			
Extraction	Cell	r	MS null	Null mean	Null SD	z	Retained	Null mean	Null SD	z	Retained
lung	$L_3\,SV_1$	0.435	−0.001	0.075	0.0139	25.9	0.825	0.083	0.0077	46.0	0.807
	$L_6\,SV_2$	0.418	−0.001	0.066	0.0136	25.8	0.839	0.074	0.0078	44.0	0.819
	$L_8\,SV_3$	0.418	−0.001	0.067	0.0135	26.1	0.839	0.094	0.0077	42.1	0.774
	$L_7\,SV_6$*	0.311	−0.001	0.005	0.0090	33.7	0.979	0.013	0.0082	36.3	0.954
	120 cells	–	–	max 0.075		119/120	0.970 (0.755)	max 0.094		120/120	0.957 (0.753)
lung (ext.)	$L_8\,SV_2$	0.406	−0.001	0.064	0.0134	25.6	0.840	0.078	0.0079	41.5	0.806
	$L_3\,SV_1$	0.407	−0.001	0.065	0.0136	25.1	0.837	0.069	0.0080	42.4	0.829
	$L_7\,SV_2$	0.393	−0.001	0.063	0.0134	24.5	0.837	0.069	0.0080	40.6	0.823
	120 cells	–	–	max 0.065		120/120	0.958 (0.837)	max 0.081		120/120	0.956 (0.785)
kidney B	$L_9\,SV_3$	0.404	−0.001	0.056	0.0136	25.6	0.860	0.075	0.0084	39.3	0.811
	$L_8\,SV_3$	0.385	−0.001	0.052	0.0138	24.1	0.863	0.059	0.0086	38.0	0.844
	$L_7\,SV_2$	0.376	−0.001	0.048	0.0136	24.2	0.870	0.051	0.0087	37.5	0.861
	120 cells	–	–	max 0.056		120/120	0.972 (0.860)	max 0.075		120/120	0.963 (0.801)

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Table S5 (continued)

C Residualisation: residual r / raw r after regressing each coordinate on gene annotations (MS null, $N = 2{,}000$)

	lung, $L_3\,\mathrm{SV}_1$			lung (ext.), $L_8\,\mathrm{SV}_2$			kidney B, $L_9\,\mathrm{SV}_3$		
Design (regressors)	Strongest cell	Median	Sig.	Strongest cell	Median	Sig.	Strongest cell	Median	Sig.
Class factor (8 levels) $+ \log n_{\mathrm{obs}}$	0.591 (0.535–0.649)	0.873	119	0.602 (0.527–0.677)	0.848	120	0.629 (0.569–0.688)	0.848	120
30 GO-CC terms $+ \log n_{\mathrm{obs}}$ (pre-specified)	0.404 (0.327–0.477)	0.702	116	0.412 (0.337–0.498)	0.684	120	0.471 (0.395–0.552)	0.723	120
Post hoc controls (no test)									
$\log n_{\mathrm{obs}}$ alone	1.006	0.999	–	0.980	0.994	–	0.967	0.980	–
30 GO-CC terms, without $\log n_{\mathrm{obs}}$	0.390	0.714	–	0.405	0.706	–	0.471	0.750	–
8 class-defining terms, multi-label $+ \log n_{\mathrm{obs}}$	0.587	0.849	–	0.612	0.834	–	0.658	0.820	–
30 GO-BP terms $+ \log n_{\mathrm{obs}}$	0.583	0.704	–	0.539	0.713	–	0.609	0.688	–
30 GO-MF terms $+ \log n_{\mathrm{obs}}$	0.711	0.785	–	0.707	0.784	–	0.708	0.787	–
30 random gene sets $+ \log n_{\mathrm{obs}}$ (5 seeds)	0.99–1.01	0.99–1.00	–	0.97–0.98	0.98–0.99	–	0.96–0.97	0.97–0.98	–

D Edges within one class and between classes, each against its own MS null ($N = 1{,}000$)

				Strongest cell, r (95% CI)		Median r over 120 cells		Sig.	
Extraction	Cell	Edges	m	Raw	Class-centred	Raw	Class-centred	Raw	Class-centred
lung	$L_3\,\mathrm{SV}_1$	Same class	6,582	0.497 (0.432–0.567)	0.309 (0.241–0.372)	0.197	0.175	117	115
		Different classes	7,892	0.377 (0.315–0.438)	0.204 (0.159–0.258)	0.155	0.145	120	117
lung (ext.)	$L_8\,\mathrm{SV}_2$	Same class	6,380	0.477 (0.404–0.548)	0.295 (0.227–0.363)	0.186	0.166	120	120
		Different classes	7,752	0.340 (0.277–0.397)	0.198 (0.145–0.250)	0.145	0.134	119	119
kidney B	$L_9\,\mathrm{SV}_3$	Same class	5,565	0.481 (0.416–0.549)	0.312 (0.248–0.381)	0.225	0.199	120	120
		Different classes	6,869	0.331 (0.262–0.397)	0.204 (0.147–0.256)	0.181	0.166	116	116

E Gene set restricted to genes observed in at least 3, 5 or 10 sampled cells (SVD recomputed; MS null, $N = 2{,}000$)

					Strongest cell		Strongest direction of the full-V analysis			
Extraction	n_{obs}	n_V	m	Sig.	Cell	r (95% CI)	Median r	Cell	|corr|	r (95% CI)	z
lung	all (V)	5,648	14,474	120	$L_3\,\mathrm{SV}_1$	0.435 (0.384–0.486)	0.181	$L_3\,\mathrm{SV}_1$	1	0.435 (0.384–0.486)	53.1
	≥ 3	4,221	10,685	120	$L_4\,\mathrm{SV}_1$	0.418 (0.365–0.470)	0.189	$L_3\,\mathrm{SV}_1$	0.999	0.406 (0.353–0.463)	41.8
	≥ 5	3,558	8,244	120	$L_4\,\mathrm{SV}_1$	0.397 (0.339–0.457)	0.191	$L_3\,\mathrm{SV}_1$	0.996	0.393 (0.336–0.452)	36.5
	≥ 10	2,580	5,228	119	$L_5\,\mathrm{SV}_3$	0.401 (0.330–0.477)	0.198	$L_3\,\mathrm{SV}_1$	0.987	0.366 (0.295–0.436)	27.8
lung (ext.)	all (V)	5,616	14,132	120	$L_8\,\mathrm{SV}_2$	0.406 (0.351–0.459)	0.170	$L_8\,\mathrm{SV}_2$	1	0.406 (0.351–0.459)	49.2
	≥ 3	4,529	11,604	120	$L_3\,\mathrm{SV}_1$	0.401 (0.344–0.458)	0.163	$L_8\,\mathrm{SV}_2$	0.986	0.389 (0.336–0.443)	43.0
	≥ 5	3,976	9,981	118	$L_3\,\mathrm{SV}_1$	0.396 (0.338–0.455)	0.160	$L_8\,\mathrm{SV}_2$	0.975	0.381 (0.323–0.441)	38.4
	≥ 10	3,060	7,341	117	$L_{10}\,\mathrm{SV}_2$	0.398 (0.336–0.463)	0.174	$L_8\,\mathrm{SV}_2$	0.962	0.379 (0.305–0.442)	32.7
kidney B	all (V)	4,780	12,434	120	$L_9\,\mathrm{SV}_3$	0.404 (0.351–0.457)	0.209	$L_9\,\mathrm{SV}_3$	1	0.404 (0.351–0.457)	45.9
	≥ 3	3,528	8,812	120	$L_5\,\mathrm{SV}_4$	0.369 (0.294–0.445)	0.186	$L_9\,\mathrm{SV}_3$	0.990	0.350 (0.289–0.406)	33.8
	≥ 5	2,914	6,676	120	$L_5\,\mathrm{SV}_4$	0.397 (0.319–0.488)	0.198	$L_9\,\mathrm{SV}_3$	0.976	0.334 (0.263–0.402)	27.2
	≥ 10	2,023	4,134	120	$L_5\,\mathrm{SV}_3$	0.452 (0.373–0.532)	0.216	$L_9\,\mathrm{SV}_3$	0.787	0.304 (0.229–0.377)	20.0

Table S6: **Transcription-factor identity (H2): primary statistic, baselines, adjustment, spectral bands and the reported 0.744.** Score: leave-one-out (LOO) mean cosine of a gene's V-centred vector to the census TFs of Lambert et al. (excluding itself); statistic: AUROC of TFs against all other observed genes V, stratified by deciles of $\log n_{\text{obs}}$ (n_{obs}, cells whose input contained the gene) unless marked unstratified. $|V|$, observed genes; L_ℓ, layer; z, standardised distance from the within-stratum permutation null; max-T, Westfall–Young correction over the 12 layers (primary, N = 10,000 joint permutations, floor $1/10{,}001 = 1.0 \times 10^{-4}$) or over the 936 band cells (Panels D, E; N = 2,000, floor 5.0×10^{-4}). Intervals are 95% gene-bootstrap percentile intervals (B = 1,000; LOO scores held fixed), written [lower, upper]; ranges over layers or extractions use an en-dash (or 'to' when a bound is negative). Because the TF centroid is not re-estimated in the bootstrap, a half-sampling SE that re-estimates it is 0.83–0.84 times the bootstrap SE in lung and lung (ext.) but 1.15–1.32 times it in the two immune extractions checked (immune, CD4 T cells), so intervals in the immune extractions are about 15–30% too narrow. **A**: all 16 extractions (the 11 robustness extractions below the rule; the three checkpoints are applied to the same 192 kidney cells); the decision uses lung and lung (ext.). **B**: the same statistic for single-gene scores (higher value scores higher) and for static embeddings scored by the same LOO cosine; unstratified AUROC in parentheses for detection; ESM2-15B is the 5,120-dimensional per-protein embedding (UCE file), matched for 94–96% of V in the lung extractions; paired differences (static minus residual-stream layer) use identical genes, labels and bootstrap draws, with the range over the 12 layers and the lowest lower bound; 'all 16' gives the range over extractions (and, for the paired rows, how many of the 192 extraction–layer intervals include 0). **C**: standardised coefficient of the score, its smallest lower 95% Wald bound and the largest likelihood-ratio p over the 12 layers; restricted-V rows re-centre and re-stratify on the retained genes (N = 2,000; genes / TFs below the range). **D**: each direction scored by its raw coordinate with a sign-free AUROC, at the layer where it is highest; p is the 936-cell max-T p of that cell; [a]primary analysis (Panel A): z and 12-layer max-T p from N = 10,000 permutations. **E**: cos variant (LOO cosine within the band) unless stated; the centroid (LDA) variant is fitted in-sample and refitted in each null draw, so its maximum is optimistic. Screen bands are those of the automated screen, indexed on the observed-gene basis; in parentheses, the number of layers with $p < 0.05$ on the band's own 12-layer max-T and the smallest p in the 936-cell max-T; [b]kidney A: the band read in the reference decomposition of the full kidney A matrix, zero rows included, and mapped to the observed-gene basis, in which its SV_k is observed SV_{k-1} (Table S14). The screen assembled SV_2–SV_4 on a 209-gene decomposition whose band also maps to observed SV_1–SV_3, and its later kidney A iterations on mean embeddings used SV_5–SV_7 of the observed genes, which maps to itself: AUROC 0.429–0.696 over the 12 layers (10 layers with $p < 0.05$; smallest p of 936, 0.003). Cross-fit: the best cell chosen on a random half of the genes and scored on the other half, 200 halvings; the brackets are the 2.5–97.5% quantiles over halvings of one data set, not sampling intervals. Immune and kidney A are the extractions in which the screen's results were found. **F**: 361 genes named by the screen, 295 observed; per-layer values in the CSV. TRRUST task: TRRUST TFs against genes that are only ever TRRUST targets, same machinery as Panel A. Per-layer values of every analysis and all 936 band cells of the five main extractions are in the CSV. Full values: `Table_S6_tf_identity.csv`.

A. Pre-registered H2 statistic (full 512-dimensional representation), per extraction

Extraction	$\lvert V\rvert$ / TFs	AUROC, 12 layers	Best layer: AUROC [95% CI]	z, 12 layers	Smallest max-T p	Layers max-T $p < 0.05$
lung	5,648 / 583	0.625–0.665	L_5: 0.665 [0.641, 0.688]	8.8–11.5	1.0×10^{-4}	12/12
lung (ext.)	5,616 / 623	0.627–0.662	L_1: 0.662 [0.637, 0.686]	8.9–11.6	1.0×10^{-4}	12/12
kidney B	4,780 / 511	0.615–0.654	L_1: 0.654 [0.628, 0.681]	7.3–10.6	1.0×10^{-4}	12/12
immune	2,039 / 113	0.566–0.720	L_2: 0.720 [0.657, 0.772]	2.3–6.5	1.0×10^{-4}	12/12
kidney A	1,972 / 90	0.678–0.783	L_3: 0.783 [0.732, 0.829]	5.4–7.6	1.0×10^{-4}	12/12
immune (seed 43)	2,080 / 105	0.549–0.657	L_2: 0.657 [0.596, 0.719]	1.8–4.8	1.0×10^{-4}	10/12
immune (seed 44)	1,972 / 105	0.604–0.720	L_2: 0.720 [0.667, 0.771]	3.4–6.6	1.0×10^{-4}	12/12
immune (donor TSP14)	1,964 / 103	0.584–0.704	L_2: 0.704 [0.640, 0.759]	2.8–6.0	1.0×10^{-4}	12/12
immune (donor TSP21)	1,765 / 95	0.575–0.693	L_2: 0.693 [0.631, 0.748]	2.4–5.5	1.0×10^{-4}	12/12
immune (donor TSP25)	1,791 / 98	0.547–0.649	L_0: 0.649 [0.584, 0.715]	1.7–4.5	1.0×10^{-4}	9/12
immune (B cells)	1,398 / 85	0.615–0.694	L_2: 0.694 [0.613, 0.772]	3.3–5.0	1.0×10^{-4}	12/12
immune (CD4 T cells)	1,465 / 78	0.576–0.663	L_2: 0.663 [0.586, 0.739]	2.1–4.0	2.0×10^{-4}	12/12
immune (CD8 T cells)	1,595 / 96	0.599–0.707	L_0: 0.707 [0.640, 0.768]	3.0–5.5	1.0×10^{-4}	12/12
kidney (whole-human ckpt)	4,923 / 545	0.612–0.652	L_1: 0.652 [0.626, 0.676]	7.3–10.5	1.0×10^{-4}	12/12
kidney (kidney ckpt)	4,923 / 545	0.565–0.595	L_3: 0.595 [0.567, 0.622]	4.7–6.9	1.0×10^{-4}	12/12
kidney (brain ckpt)	4,923 / 545	0.562–0.607	L_0: 0.607 [0.579, 0.633]	4.2–6.7	1.0×10^{-4}	12/12

B. Baselines: detection-stratified AUROC of single-gene scores and static embeddings on the same genes and strata

	lung	lung (ext.)	kidney B	immune	kidney A	all 16
Detection alone, $\log n_{\text{obs}}$ (unstratified AUROC)	0.503 (0.458) [0.484, 0.522]	0.531 (0.454) [0.509, 0.552]	0.510 (0.461) [0.490, 0.529]	0.522 (0.650) [0.485, 0.560]	0.540 (0.555) [0.490, 0.590]	0.466–0.540
Mean expression, extraction cells	0.588 [0.565, 0.611]	0.595 [0.571, 0.620]	0.532 [0.504, 0.560]	0.486 [0.424, 0.544]	0.588 [0.518, 0.658]	0.435–0.595

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Table S6 (continued)

Mean expression, population	0.579 [0.556, 0.602]	0.576 [0.552, 0.601]	0.524 [0.496, 0.550]	0.512 [0.459, 0.569]	0.606 [0.546, 0.665]	0.506–0.606
scGPT input gene embedding, V-centred	0.760 [0.740, 0.781]	0.744 [0.722, 0.765]	0.742 [0.718, 0.765]	0.750 [0.684, 0.806]	0.770 [0.724, 0.812]	0.678–0.770
scGPT input gene embedding, uncentred	0.668	0.638	0.649	0.738	0.747	0.638–0.762
ESM2-15B per-protein embedding, V-centred	0.881 [0.865, 0.897]	0.883 [0.868, 0.898]	0.862 [0.844, 0.880]	0.938 [0.909, 0.963]	0.917 [0.877, 0.948]	0.862–0.943
ESM2-15B per-protein embedding, uncentred	0.761	0.761	0.725	0.865	0.844	0.725–0.866
Genes with an ESM2-15B vector	5,413	5,306	4,602	1,554	1,399	1,075–5,413
Input embedding minus layer (paired; 12 layers)	0.095–0.135 lower $\geq$ 0.074	0.082–0.118 lower $\geq$ 0.063	0.089–0.127 lower $\geq$ 0.068	0.030–0.184 lower ≥ -0.005	-0.013 to $+0.092$ lower ≥ -0.044	-0.013 to $+0.184$ 19/192 CIs include 0
ESM2-15B minus layer (paired; 12 layers)	0.216–0.263 lower $\geq$ 0.188	0.209–0.245 lower $\geq$ 0.183	0.208–0.248 lower $\geq$ 0.175	0.238–0.400 lower $\geq$ 0.188	0.195–0.296 lower $\geq$ 0.139	0.195–0.402 0/192 CIs include 0

C. Adjustment: logistic regression TF $\sim$ score $+ \log n_{\text{obs}} + \log$ mean expression (standardised), per layer; and V restricted by detection

	lung	lung (ext.)	kidney B	immune	kidney A	all 16
β per SD of score, extraction-cell expression	0.367–0.560	0.454–0.565	0.462–0.598	0.464–1.158	1.036–2.029	0.177–2.029
Smallest lower 95% bound of β	0.265	0.365	0.363	0.172	0.762	0.086
Largest likelihood-ratio p, extraction-cell expression	6.1×10^{-13}	1.8×10^{-23}	8.0×10^{-20}	0.002	1.5×10^{-16}	0.008
β per SD of score, population expression	0.365–0.549	0.477–0.579	0.456–0.589	0.484–1.168	1.000–2.083	0.176–2.083
Largest likelihood-ratio p, population expression	2.5×10^{-13}	1.2×10^{-26}	3.6×10^{-20}	0.001	1.4×10^{-13}	0.008
H2 AUROC, genes with $n_{\text{obs}} \geq 3$ (12 layers)	0.646–0.658 4,221 / 407	0.627–0.662 4,529 / 502	0.611–0.649 3,528 / 367	0.625–0.723 1,341 / 90	0.660–0.764 1,346 / 79	0.517–0.764
H2 AUROC, genes with $n_{\text{obs}} \geq 5$ (12 layers)	0.649–0.670 3,558 / 329	0.635–0.663 3,976 / 420	0.607–0.654 2,914 / 296	0.632–0.716 1,149 / 85	0.613–0.729 1,208 / 74	0.508–0.730
H2 AUROC, genes with $n_{\text{obs}} \geq 10$ (12 layers)	0.668–0.694 2,580 / 223	0.666–0.682 3,060 / 297	0.598–0.661 2,023 / 200	0.660–0.730 965 / 77	0.645–0.736 1,041 / 64	0.510–0.736

D. Single directions SV_1–SV_{12} (raw coordinate, sign-free AUROC), best of the 12 layers

Direction	lung: AUROC (layer)	lung: z	lung: p (936 cells)	lung (ext.): AUROC (layer)	lung (ext.): z	lung (ext.): p (936 cells)
SV_1	0.572 (L_3)	8.0	5.0×10^{-4}	0.561 (L_4)	6.4	5.0×10^{-4}
SV_2	0.567 (L_5)	7.5	5.0×10^{-4}	0.558 (L_1)	6.1	5.0×10^{-4}
SV_3	0.559 (L_8)	6.4	1.0×10^{-3}	0.570 (L_1)	7.8	5.0×10^{-4}
SV_4	0.564 (L_0)	7.1	5.0×10^{-4}	0.572 (L_0)	8.1	5.0×10^{-4}
SV_5	0.586 (L_{10})	9.9	5.0×10^{-4}	0.582 (L_4)	9.6	5.0×10^{-4}
SV_6	0.598 (L_4)	11.2	5.0×10^{-4}	0.654 (L_3)	18.7	5.0×10^{-4}
SV_7	0.559 (L_8)	6.4	1.0×10^{-3}	0.637 (L_8)	15.9	5.0×10^{-4}
SV_8	0.631 (L_0)	15.2	5.0×10^{-4}	0.624 (L_2)	14.7	5.0×10^{-4}
SV_9	0.616 (L_4)	13.7	5.0×10^{-4}	0.593 (L_4)	10.7	5.0×10^{-4}
SV_{10}	0.601 (L_8)	11.4	5.0×10^{-4}	0.600 (L_{10})	11.8	5.0×10^{-4}
SV_{11}	0.569 (L_{10})	7.0	5.0×10^{-4}	0.631 (L_{11})	15.4	5.0×10^{-4}
SV_{12}	0.560 (L_1)	6.3	1.0×10^{-3}	0.617 (L_7)	13.5	5.0×10^{-4}
Full 512, best layer[a]	0.665 (L_5)	11.3	1.0×10^{-4}	0.662 (L_1)	11.6	1.0×10^{-4}

E. All 78 contiguous bands of SV_1–SV_{12} at 12 layers (936 cells), screen bands, and cross-fitted band choice

	lung	lung (ext.)	kidney B	immune	kidney A	all 16
Cells with max-T $p < 0.05$ (of 936)	633	690	591	71	359	4–690
Best cell by AUROC	SV_3–SV_{12}, L_2 0.676	SV_3–SV_{12}, L_2 0.711	SV_7–SV_{11}, L_2 0.662	SV_3–SV_{12}, L_0 0.730	SV_1–SV_9, L_2 0.738	0.592–0.738
Cell with the largest z: AUROC (z)	SV_8, L_0 0.631 (15.2)	SV_6, L_3 0.654 (18.7)	SV_7, L_2 0.646 (16.9)	SV_{11}, L_9 0.674 (8.1)	SV_5, L_3 0.699 (8.4)	–
Null 97.5% quantile of the maximum AUROC	0.548	0.547	0.551	0.613	0.623	0.547–0.641
Centroid (LDA) variant, best cell (in-sample)	SV_1–SV_{12}, L_2 0.700	SV_1–SV_{12}, L_2 0.734	SV_1–SV_{12}, L_2 0.693	SV_1–SV_{12}, L_0 0.763	SV_1–SV_{12}, L_0 0.805	0.619–0.805
Screen band SV_2–SV_4: AUROC, 12 layers (layers $p < 0.05$; smallest p of 936)	0.522–0.590 (11; 0.003)	0.530–0.588 (12; 5.0×10^{-4})	0.519–0.587 (10; 0.002)	0.503–0.593 (3; 0.40)	SV_1–SV_3[b]: 0.595–0.638 (12; 0.040)	–

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Table S6 (continued)

Screen band SV_5–SV_7: AUROC, 12 layers (layers $p < 0.05$; smallest p of 936)	0.497–0.597 (11; 5.0×10^{-4})	0.550–0.654 (12; 5.0×10^{-4})	0.546–0.641 (12; 5.0×10^{-4})	0.256–0.612 (4; 0.35)	SV_4–SV_6 [b]: 0.566–0.669 (11; 0.030)	–
Screen band SV_2–SV_7: AUROC, 12 layers (layers $p < 0.05$; smallest p of 936)	0.574–0.611 (12; 5.0×10^{-4})	0.580–0.653 (12; 5.0×10^{-4})	0.555–0.645 (12; 5.0×10^{-4})	0.515–0.622 (4; 0.20)	SV_1–SV_6 [b]: 0.628–0.715 (12; 5.0×10^{-4})	–
Cross-fitted band, held-out AUROC (median [2.5–97.5% of 200 halvings])	0.669 [0.644, 0.692]	0.702 [0.678, 0.725]	0.648 [0.618, 0.676]	0.681 [0.578, 0.745]	0.695 [0.572, 0.760]	0.571–0.702
Full 512, held-out AUROC (median [2.5–97.5% of 200 halvings])	0.660 [0.643, 0.685]	0.659 [0.636, 0.680]	0.651 [0.630, 0.674]	0.687 [0.600, 0.773]	0.761 [0.699, 0.814]	0.589–0.761
Band minus full 512, held-out (median [2.5–97.5% of 200 halvings])	+0.007 [−0.014, 0.029]	+0.042 [0.019, 0.065]	−0.004 [−0.032, 0.024]	−0.002 [−0.143, 0.058]	−0.058 [−0.232, −0.012]	−0.058 to +0.042
Optimism (in-sample minus held-out), median	0.007	0.005	0.019	0.040	0.055	0.005–0.101
Most frequent choice (of 200)	SV_4–SV_{12} at L_2 (57)	SV_3–SV_{12} at L_2 (133)	SV_7–SV_{11} at L_2 (64)	SV_3–SV_{12} at L_0 (77)	SV_2–SV_5 at L_3 (35)	–

F. The reported 0.744 (immune; screen labels: 73 TFs against 222 target-only genes) and the TRRUST task

Step	Layer-mean AUROC	Procedure and remarks
Value reported by the screen	0.744	As recorded in the screen's ledger.
Screen procedure, reproduced	0.747	Logistic regression ($C = 1$) on standardised SV_2–SV_7 coordinates, 5-fold stratified CV (seed 42), fold-mean AUROC averaged over 12 layers; over CV seeds 0–49 median 0.744, 2.5–97.5% 0.727–0.759; permutation $p = 1/1{,}001$ ($N = 1{,}000$, the floor); best layer L_2 0.792.
Same out-of-fold predictions, pooled	0.745	Unstratified AUROC of the pooled out-of-fold predictions.
Same out-of-fold predictions, stratified	0.683	Detection-stratified AUROC (deciles of $\log n_{\text{obs}}$).
LOO cosine in SV_2–SV_7, unstratified	0.644	Unsupervised score; layer-mean permutation p 1.0×10^{-4} ($N = 10{,}000$).
LOO cosine in SV_2–SV_7, stratified	0.553	Layer-mean permutation p 0.042; per layer highest at L_2 (0.666), lowest at L_9 (0.398).
LOO cosine, full 512, unstratified	0.702	Layer-mean permutation p 1.0×10^{-4}.
LOO cosine, full 512, stratified	0.629	Layer-mean permutation p 5.0×10^{-4}.
Attribution of the 0.747	–	Order-dependent: removing supervision first costs 0.10 and then stratifying 0.09 (0.747 → 0.644 → 0.553); stratifying first costs 0.06 and then removing supervision 0.13 (0.747 / 0.745 → 0.683 → 0.553). Supervision accounts for 0.10–0.13, detection for 0.06–0.09.

TRRUST task: TRRUST TFs against target-only genes	lung	lung (ext.)	kidney B	immune	kidney A	all 16
Genes / TFs	1,956 / 555	1,971 / 579	1,577 / 480	283 / 68	180 / 55	173–1,971 / 50–579
Full 512, AUROC over 12 layers (layers max-T $p < 0.05$)	0.646–0.713 (12/12)	0.643–0.710 (12/12)	0.618–0.692 (12/12)	0.519–0.671 (8/12)	0.602–0.718 (12/12)	0.487–0.718
Mean expression alone	0.628	0.609	0.567	0.458	0.608	0.445–0.628
Detection alone	0.502	0.520	0.510	0.551	0.615	0.444–0.615
Cross-fitted band minus full 512 (median [2.5–97.5%])	−0.047 [−0.095, −0.032]	−0.029 [−0.073, −0.011]	−0.061 [−0.126, −0.034]	+0.039 [−0.139, 0.185]	−0.036 [−0.265, 0.118]	−0.071 to +0.056

Table S7: **Regulatory edges (H3): the benchmark in every extraction and every edge-level sensitivity analysis.** Positives: TRRUST pairs with both genes observed; candidates: up to ten per positive, pairs of the same factor whose target is a documented target of another factor and that are absent, in either direction, from TRRUST and from DoRothEA at any confidence level, and matched where possible to the positive on the target's decile of TRRUST in-degree and of n_{obs} (on both for 75–90% of candidates, otherwise on in-degree or not at all; CSV column `frac_cand_fully_matched`); factors with ≥ 3 of each. Statistic: within-factor stratified AUROC of the signed cosine of the V-centred 512-dimensional vectors at layer L_ℓ. CI, 95% interval; UB, one-sided 95% upper bound; both from the two-way (factor × target) pigeonhole bootstrap, $B = 2{,}000$ unless stated (1,000 for some strata, CSV column `B`); Max. UB, the largest UB over the twelve layers. $|r|$, absolute Pearson co-expression on the same pairs, on the extraction's own cells (extr.; 256, or 192 for kidney B and the checkpoints) or on 3,000 cells of the same population (3k); Δ, cosine minus $|r|$ AUROC on the same bootstrap draws. **A**: max-T p, within-factor label permutation applied jointly to the twelve layers ($N = 10{,}000$; floor 1.0×10^{-4}); MDE, the AUROC (and Δ) detected with 80% power at two-sided $\alpha = 0.05$ in a simulation that resamples factors as clusters with the observed between-factor heterogeneity and tests with the two-way bootstrap. Verdict: D, detected (max-T $p < 0.05$); U, every layer's UB < 0.60 (AUROC ≥ 0.60 excluded); C, every layer's Δ UB < 0.05 (not above co-expression), against the extraction cells (C_{3k}: against the 3k tier only). *0.0502 at L_2; two independent bootstrap seeds give 0.0528 and 0.0537, so C fails by less than the Monte-Carlo error. Pre-declared replication (same sign, and a lung (ext.) CI excluding 0.5): the sign agrees at 12/12 layers and the CI excludes 0.5 at 0/12, so it is not met. **B**: Shortcut rows score a pair by its target's TRRUST in-degree (no layer). ChIP-seq rows: positives, DoRothEA ChIP-seq pairs (level D) absent from TRRUST and from DoRothEA levels A–C; candidates constructed as for TRRUST, excluding in either direction every pair documented in any regulatory resource at any level, the ChIP-seq positives included; about half of these candidates (49–55%) could not be matched on in-degree, so the fully matched rows keep only candidates matched on both deciles (and the positives of factors that have at least one; CSV column `chip_candidates`). [a]Positives in strata that hold both classes. [b]$|r|$ AUROC, and Δ, computed with exactly tied $|r|$ values ordered by floating-point noise rather than counted as one half (effect on the $|r|$ AUROC, and hence on Δ, ≤ 0.00012 where checked). Contamination: AUROC and UB corrected for a fraction π of undocumented edges among the candidates, $0.5 + (A - 0.5)/(1 - \pi)$; seventh column, layers whose corrected UB is below 0.60. [c]Mean of the per-factor AUROCs with a factor-bootstrap CI; last column, fraction of factors above 0.5 (range over layers). [d]Covariates: the pair's log $|r|$ and the target's log mean expression, log n_{obs}, STRING physical degree and GO annotation count (factor-level covariates cancel within the factor stratum); two-way bootstrap with refitting, $B = 500$. Residualised: AUROC of the cosine after OLS residualisation on the covariates. Conditional logit: odds ratio (OR) per pooled SD of the cosine, adjusted (unadjusted in the eighth column); layer mean, mean of the twelve coefficients. [e]Effective number of factors under each statistic's weights ($n_{\text{pos}} n_{\text{cand}}$ per factor; Fisher information) and the share of the five largest. **C**: post hoc subsets, no multiplicity adjustment; Δ_{3k}, difference from $|r|$ on 3,000 cells. PAX5→CD79A (illustration): detected, fraction of the extraction's cells expressing the gene; rank 1 is the highest cosine among PAX5's documented and candidate pairs (n); candidates below, share of PAX5's candidates scored lower. **D**: each split holds out a random fifth of the 245 kidney A TRRUST pairs (49 positives, three candidate non-edges each); nothing is trained, so the spread over splits is the sampling distribution of the test statistic, not a CI. Uncentred, cosine of raw vectors, as the screen scored pairs; zero-row SVD, the reference decomposition of the full kidney A matrix including the unobserved all-zero gene rows (Table S14). Reference split: one fixed 80/20 split of these pairs; its value (first row with its i.i.d. CI) and its percentile among the 500 splits. MDE as in A, for the pooled AUROC (positives against all held-out non-edges) and for the within-factor AUROC; size, rejection rate at a population AUROC of 0.5 (nominal 0.05); perm. SD, Wald test with the permutation SD; oracle SD, the simulated sampling SD. **E**: regulator importance, normalised per target, of each factor for each target: GENIE3, the extra-trees variant (300 trees, scikit-learn); GRNBoost2, GRNBoost2-style gradient boosting (scikit-learn, early stopping on a tenth of the cells); T1, five seeds on the extraction cells; T2, two seeds on 3,000 cells; AUROC of the seed-averaged score, and mean (SD) of the per-seed AUROCs; Δ, the method's AUROC minus that of $|r|$ on the same cells or of the cosine at the stated layer, on the same bootstrap draws. Cosine L_0 (first column): scGPT on the same pairs. **F**: L2-penalised logistic probes on features of the factor's and the target's vectors x_i, x_j (bilinear: outer product of 16 principal components fitted on training genes; $[x_j, \cdot]$: target block added); AUROC, mean over 25 test folds; CI, corrected resampled t (Nadeau–Bengio) with the test-to-training ratio counted in pairs (about 0.0625) or in genes (0.25, the cautious reading). − target-only: minus, fold by fold, the better of the target-only and the factor-only probe (the latter scores 0.5 within factor). Rewired labels: degree-preserving swaps of targets between factors' positives, 100 draws, each retrained on one repetition's folds; $p = (b + 1)/(N + 1)$, floor 0.0099; max-T over layers. Difference in differences: bilinear minus concatenation AUROC on the true labels against the same difference on each rewired draw. Input gene-token embedding: scGPT whole-human checkpoint; ESM2-15B: 5,120-dimensional per-protein embedding (UCE); both without layers. Non-linear: histogram gradient boosting and a two-layer perceptron, one repetition; rows, probe × layer × feature rows whose interval (pair ratio) lies above or below the model's own target-only control. **G**: fractions among untied pairs; sign test, exact two-sided; regulator-cluster CI, bootstrap over source factors; probe CIs, binomial (pairs independent) or gene-cluster bootstrap ($B = 2{,}000$); strict subset, reverse absent from every regulatory set; out-/in-degree, the same fraction for TRRUST degrees (no model). Full values: `Table_S7_edges.csv`.

A. Primary benchmark: within-factor AUROC of the full-dimensional signed cosine in all 16 extractions

Extraction	Factors; positives; candidates	AUROC, range L_0–L_{11}	Best layer: AUROC [95% CI]	Max. UB	Min. max-T p	MDE: AUROC; Δ	$\lvert r\rvert$ AUROC: extr.; 3k	Max. Δ UB: extr.; 3k	Verdict
lung	312; 5,266; 47,160	0.516–0.542	L_1 0.542 [0.496, 0.589]	0.583	1.0×10^{-4}	0.564; 0.071	0.508; 0.515	0.074; 0.068	D, U
lung (ext.)	315; 5,093; 45,854	0.518–0.538	L_1 0.538 [0.493, 0.588]	0.581	1.0×10^{-4}	0.562; 0.072	0.524; 0.551	0.058; 0.033	D, U; C_{3k}
kidney B	254; 3,668; 32,712	0.524–0.554	L_0 0.554 [0.499, 0.603]	0.597	1.0×10^{-4}	0.576; 0.085	0.515; 0.522	0.090; 0.084	D, U
kidney (whole-human ckpt)	274; 4,111; 36,798	0.519–0.542	L_0 0.542 [0.491, 0.593]	0.587	1.0×10^{-4}	0.575; 0.091	0.510; 0.522	0.078; 0.067	D, U
kidney (kidney ckpt)	274; 4,111; 36,798	0.513–0.521	L_2 0.521 [0.475, 0.568]	0.561	0.0020	0.567; 0.084	0.510; 0.522	0.050*; 0.040	D, U; C_{3k}
kidney (brain ckpt)	274; 4,111; 36,798	0.516–0.523	L_4 0.523 [0.464, 0.573]	0.566	4.0×10^{-4}	0.564; 0.090	0.510; 0.522	0.069; 0.055	D, U
immune	36; 547; 3,144	0.577–0.638	L_0 0.638 [0.574, 0.715]	0.700	1.0×10^{-4}	0.601; 0.162	0.559; 0.562	0.151; 0.141	D

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Table S7 (continued)

immune (seed 43)	34; 515; 2,877	0.584–0.651	L_0 0.651 [0.575, 0.726]	0.712	1.0×10^{-4}	0.583; 0.113	0.565; 0.582	0.156; 0.140	D
immune (seed 44)	36; 538; 3,074	0.570–0.644	L_0 0.644 [0.572, 0.712]	0.701	1.0×10^{-4}	0.590; 0.122	0.545; 0.574	0.166; 0.140	D
immune (donor TSP14)	34; 489; 2,711	0.588–0.650	L_1 0.650 [0.575, 0.722]	0.709	1.0×10^{-4}	0.597; 0.131	0.580; 0.572	0.139; 0.149	D
immune (donor TSP21)	34; 498; 2,796	0.571–0.646	L_1 0.646 [0.580, 0.726]	0.709	1.0×10^{-4}	0.595; 0.135	0.547; 0.561	0.197; 0.157	D
immune (donor TSP25)	35; 467; 2,498	0.562–0.627	L_1 0.627 [0.558, 0.692]	0.684	1.0×10^{-4}	0.589; 0.111	0.572; 0.574	0.143; 0.136	D
immune (B cells)	27; 330; 1,531	0.549–0.633	L_0 0.633 [0.555, 0.711]	0.697	1.0×10^{-4}	0.608; 0.163	0.531; 0.560	0.187; 0.159	D
immune (CD4 T cells)	29; 384; 1,791	0.561–0.618	L_1 0.618 [0.534, 0.697]	0.685	1.0×10^{-4}	0.603; 0.191	0.508; 0.528	0.193; 0.167	D
immune (CD8 T cells)	31; 440; 2,330	0.589–0.647	L_1 0.647 [0.577, 0.731]	0.717	1.0×10^{-4}	0.593; 0.169	0.553; 0.551	0.213; 0.209	D
kidney A	29; 213; 1,446	0.601–0.652	L_1 0.652 [0.561, 0.747]	0.731	1.0×10^{-4}	0.632; 0.157	0.559; 0.577	0.195; 0.169	D

B. Sensitivity analyses in the two lung extractions (cosine at L_0, and over the twelve layers)

Analysis	Extraction	Factors; positives	L_0 [95% CI]	Range L_0–L_{11}	Max. UB	Layers with CI > 0.5	$\lvert r\rvert$ AUROC, same pairs	Δ vs $\lvert r\rvert$, range	Max. Δ UB
All pairs (primary)	lung	312; 5,266	0.541 [0.497, 0.588]	0.516–0.542	0.583	0/12	0.508	+0.008 to +0.034	0.074
	lung (ext.)	315; 5,093	0.538 [0.496, 0.586]	0.518–0.538	0.581	0/12	0.524	−0.006 to +0.014	0.058
Shortcut: target in-degree, incl. own edge	lung	312; 5,266	0.733 [0.658, 0.799]	(no layer)	0.787	–	–	–	–
	lung (ext.)	315; 5,093	0.750 [0.675, 0.818]	(no layer)	0.803	–	–	–	–
Shortcut: target's other-factor in-degree	lung	312; 5,266	0.626 [0.551, 0.702]	(no layer)	0.687	–	–	–	–
	lung (ext.)	315; 5,093	0.633 [0.554, 0.711]	(no layer)	0.696	–	–	–	–
Strata: factor × other-factor in-degree	lung	–; 3,564[a]	0.530 [0.471, 0.607]	0.510–0.531	0.597	0/12	0.504[b]	+0.005 to +0.027[b]	0.093[b]
	lung (ext.)	–; 3,195[a]	0.542 [0.485, 0.619]	0.522–0.542	0.606	0/12	0.524[b]	−0.002 to +0.018[b]	0.084[b]
ChIP-seq positives (not in TRRUST)	lung	225; 57,017	0.528 [0.506, 0.551]	0.528–0.538	0.559	12/12	0.542	−0.014 to −0.004	0.018
	lung (ext.)	218; 54,332	0.533 [0.510, 0.557]	0.533–0.541	0.561	12/12	0.515	+0.018 to +0.026	0.047
ChIP-seq positives, fully matched candidates	lung	224; 57,014	0.529 [0.511, 0.545]	0.521–0.531	0.544	12/12	0.516	+0.005 to +0.014	0.030
	lung (ext.)	218; 54,332	0.528 [0.510, 0.545]	0.526–0.531	0.546	12/12	0.507	+0.019 to +0.024	0.040
≥ 2 publications	lung	131; 579	0.591 [0.507, 0.677]	0.535–0.593	0.669	2/12	0.511[b]	+0.024 to +0.081[b]	0.163[b]
	lung (ext.)	129; 553	0.588 [0.506, 0.672]	0.539–0.588	0.672	2/12	0.551[b]	−0.012 to +0.037[b]	0.135[b]
1 publication	lung	312; 4,687	0.531 [0.483, 0.576]	0.512–0.532	0.571	0/12	0.508[b]	+0.005 to +0.024[b]	0.064[b]
	lung (ext.)	315; 4,540	0.528 [0.485, 0.575]	0.513–0.528	0.570	0/12	0.519[b]	−0.005 to +0.010[b]	0.051[b]
Factor is input in $\geq 10\%$ of cells	lung	94; 1,753	0.566 [0.515, 0.642]	0.557–0.584	0.633	12/12	0.528	+0.029 to +0.056	0.117
	lung (ext.)	111; 1,921	0.560 [0.514, 0.613]	0.541–0.567	0.610	12/12	0.528	+0.012 to +0.038	0.090
Factor is input in $< 10\%$ of cells	lung	218; 3,513	0.527 [0.479, 0.584]	0.493–0.527	0.578	0/12	0.497	−0.004 to +0.030	0.075
	lung (ext.)	204; 3,172	0.524 [0.474, 0.590]	0.503–0.524	0.582	0/12	0.521	−0.019 to +0.003	0.056
Factor expressed in $\geq 10\%$ of cells	lung	265; 4,669	0.539 [0.488, 0.585]	0.515–0.540	0.582	0/12	0.506	+0.008 to +0.034	0.078
	lung (ext.)	227; 3,955	0.530 [0.483, 0.586]	0.512–0.530	0.582	0/12	0.517	−0.005 to +0.014	0.067
Factor expressed in $< 10\%$ of cells	lung	47; 597	0.577 [0.507, 0.651]	0.536–0.577	0.641	3/12	0.538	−0.002 to +0.038	0.111
	lung (ext.)	88; 1,138	0.581 [0.527, 0.648]	0.550–0.583	0.637	8/12	0.567	−0.016 to +0.016	0.075
Contamination $\pi = 0.05$	lung	–	0.543	0.517–0.544	0.587	UB < 0.60: 12/12	–	–	–
	lung (ext.)	–	0.540	0.518–0.540	0.585	UB < 0.60: 12/12	–	–	–
Contamination $\pi = 0.10$	lung	–	0.546	0.518–0.547	0.592	UB < 0.60: 12/12	–	–	–
	lung (ext.)	–	0.542	0.520–0.542	0.590	UB < 0.60: 12/12	–	–	–

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Table S7 (continued)

Contamination $\pi = 0.20$	lung	–	0.551	0.520–0.552	0.604	UB < 0.60: 9/12	–	–	–
	lung (ext.)	–	0.547	0.522–0.548	0.601	UB < 0.60: 9/12	–	–	–
Macro AUROC[c]	lung	312; –	0.580 [0.565, 0.596]	0.554–0.580	0.593	12/12	0.518	factors > 0.5: 68–75%	
	lung (ext.)	315; –	0.573 [0.558, 0.588]	0.553–0.573	0.586	12/12	0.520	factors > 0.5: 69–72%	
Residualised[d]	lung	312; 5,266	0.536 [0.495, 0.581]	0.512–0.537	0.577	0/12	–	resid. − raw: −0.005 to −0.004	
	lung (ext.)	315; 5,093	0.533 [0.489, 0.582]	0.515–0.533	0.579	0/12	–	resid. − raw: −0.005 to −0.002	
Conditional logit OR[d]	lung	312; 5,266	1.238 [1.148, 1.348]	1.123–1.239	–	CI > 1: 12/12	unadj. 1.132–1.247	layer mean 1.189 [1.114, 1.281]; unadj. 1.198	
	lung (ext.)	315; 5,093	1.218 [1.119, 1.336]	1.132–1.219	–	CI > 1: 12/12	unadj. 1.140–1.230	layer mean 1.184 [1.098, 1.294]; unadj. 1.193	
Effective no. of factors[e]	lung	312; –	pooled AUROC 14.9 (top 5: 51%)		cond. logit 75.8 (top 5: 19%)				
	lung (ext.)	315; –	pooled AUROC 14.6 (top 5: 51%)		cond. logit 73.5 (top 5: 19%)				

C. Cell-type extractions restricted to factors detected in $\geq 10\%$ of their cells, and the PAX5→CD79A example

Subset	Factors; positives; candidates	L_0 [95% CI]	Best layer [95% CI]	Range L_0–L_{11}	Max. UB	$\lvert r \rvert$ extr. [95% CI]	Δ at L_0 [95% CI]	Layers with Δ CI > 0	Δ_{3k} at L_0 [95% CI]
immune (B cells)	21; 292; 1,200	0.629 [0.537, 0.707]	L_0 0.629 [0.537, 0.707]	0.543–0.629	0.694	0.519 [0.423, 0.621]	+0.110 [−0.010, 0.221]	0/12	+0.080 [−0.040, 0.187]
immune (CD4 T cells)	22; 319; 1,290	0.605 [0.520, 0.685]	L_1 0.606 [0.520, 0.685]	0.555–0.606	0.670	0.487 [0.414, 0.569]	+0.118 [0.014, 0.222]	3/12 (L_0–L_2)	+0.092 [−0.023, 0.185]
immune (CD8 T cells)	24; 353; 1,610	0.622 [0.550, 0.689]	L_1 0.627 [0.556, 0.693]	0.576–0.627	0.683	0.559 [0.462, 0.663]	+0.064 [−0.053, 0.180]	0/12	+0.073 [−0.030, 0.171]

PAX5→CD79A: rank of the documented pair among all of PAX5's pairs (documented and candidate), over L_0–L_{11}

Extraction	Detected: PAX5; CD79A	Rank (n pairs)	Candidates below	$\lvert r \rvert$ rank: extr.; 3k	Extraction	Detected: PAX5; CD79A	Rank (n pairs)	Candidates below	$\lvert r \rvert$ rank: extr.; 3k
immune (B cells)	78.5%; 91.4%	5–7 (94)	95–98%	5; 53	immune (CD4 T cells)	0.8%; 3.1%	3–5 (86)	96–99%	57; 73
immune (CD8 T cells)	0.4%; 0.4%	1–4 (69)	97–100%	61; 6	immune	13.3%; 21.1%	5 (108)	98%	3; 3
immune (seed 43)	14.8%; 24.6%	4 (110)	99%	3; 3	immune (seed 44)	15.6%; 21.1%	5 (110)	98%	2; 3
immune (donor TSP14)	9.0%; 15.2%	4 (87)	99%	3; 3	immune (donor TSP21)	25.4%; 30.1%	4–5 (103)	98–99%	3; 3
immune (donor TSP25)	26.6%; 34.0%	5–8 (108)	95–98%	1; 3	lung	1.6%; 0.4%	2–28 (176)	86–100%	54; 1
lung (ext.)	1.2%; 2.7%	4–6 (198)	99–100%	1; 1	kidney B	6.2%; 6.8%	4–7 (176)	98–100%	2; 2
kidney A	3.5%; 2.7%	5–19 (74)	77–98%	3; 3	kidney (whole-human ckpt)	7.8%; 6.8%	4–9 (187)	96–99%	1; 2
kidney (kidney ckpt)	7.8%; 6.8%	2–14 (187)	93–99%	1; 2	kidney (brain ckpt)	7.8%; 6.8%	5–97 (187)	49–99%	1; 2

All nine immune extractions: rank 1–8 at every layer.

D. An 80/20 held-out split of the kidney A pairs (49 positives, 147 candidates, 27 factors, 117 targets): 500 random splits

Scorer (L_0)	Quantity	Mean	SD	2.5–97.5%	Min–max	Fraction > 0.5 (> 0 for Δ)	Reference split (percentile)
Full 512-d, uncentred (screen convention)	AUROC	0.580	0.043	0.501–0.664	0.458–0.699	0.974	0.591 [0.499, 0.684] (60.8th)
	Δ vs $\lvert r \rvert$	+0.081	0.047	−0.014 to +0.169	−0.075 to +0.230	0.954	+0.035 (15.4th)
Full 512-d, V-centred	AUROC	0.633	0.040	0.553–0.710	0.514–0.740	1.000	–
	Δ vs $\lvert r \rvert$	+0.134	0.049	+0.036 to +0.230	−0.023 to +0.296	0.996	–
SV_2–SV_7, zero-row SVD (reference basis)	AUROC	0.598	0.040	0.525–0.673	0.450–0.714	0.994	0.589
	Δ vs $\lvert r \rvert$	+0.098	0.054	−0.006 to +0.202	−0.058 to +0.270	0.962	–
SV_2–SV_7, observed-gene SVD	AUROC	0.607	0.040	0.532–0.682	0.461–0.729	0.998	–
	Δ vs $\lvert r \rvert$	+0.107	0.053	+0.000 to +0.206	−0.046 to +0.291	0.974	–

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Table S7 (continued)

SV_5–SV_7, zero-row SVD (reference basis)	AUROC	0.545	0.043	0.463–0.627	0.424–0.663	0.854	0.541
	Δ vs $\|r\|$	+0.046	0.060	−0.070 to +0.158	−0.135 to +0.232	0.800	–
SV_5–SV_7, observed-gene SVD	AUROC	0.567	0.044	0.485–0.648	0.438–0.689	0.926	–
	Δ vs $\|r\|$	+0.068	0.057	−0.042 to +0.189	−0.103 to +0.234	0.872	–
$\|r\|$, 3,000 cells	AUROC	0.500	0.044	0.408–0.581	0.344–0.599	0.496	0.556 (89.6th)
Random scores	AUROC	0.500	0.049	0.407–0.600	0.349–0.642	0.486	0.513
	Δ vs $\|r\|$	+0.000	0.067	−0.130 to +0.132	−0.191 to +0.178	0.492	–
Full 512-d, uncentred (screen convention), within-factor AUROC	AUROC	0.579	0.053	0.484–0.682	0.367–0.741	0.944	–

Minimum detectable effect of that design (80% power, two-sided $\alpha = 0.05$; target ICC 0.47, factor ICC 0.17; 400 simulations per grid point)

Statistic	Test	MDE AUROC	Size at null	MDE Δ	Size at null (Δ)
Pooled AUROC (positives against all held-out non-edges)	two-way bootstrap	0.716	0.022	0.254	0.007
	factor-only bootstrap	0.651	0.163	0.173	0.152
	i.i.d. DeLong	0.644	0.195	0.165	0.158
	Wald, perm. SD	0.643	0.188	–	–
	oracle SD	0.700	–	0.216	–
Within-factor AUROC	two-way bootstrap	0.795	0.003	0.359	0.003
	factor-only bootstrap	0.716	0.025	0.261	0.028
	Wald, perm. SD	0.671	0.092	–	–
	oracle SD	0.695	–	0.253	–
Either statistic	Hanley–McNeil	0.634	–	0.161	–

E. GENIE3 (extra-trees) and GRNBoost2-style boosting on the same benchmark (T1, the extraction's own cells; T2, 3,000 cells)

Extraction (cosine L_0)	Method, tier (cells)	AUROC, seed ensemble [95% CI]	UB	Seed mean (SD; seeds)	$\|r\|$, same cells	Δ vs $\|r\|$ [95% CI]	Δ vs cosine L_0 [95% CI]	Δ vs cosine, range L_0–L_{11}	T2 − T1 [95% CI]
lung (0.541)	GENIE3 T1 (256)	0.516 [0.488, 0.550]	0.545	0.516 (0.0031; 5)	0.508	+0.008 [−0.017, 0.036]	−0.025 [−0.065, 0.013]	−0.026 to +0.000	–
	GENIE3 T2 (3,000)	0.513 [0.479, 0.550]	0.544	0.513 (0.0027; 2)	0.515	−0.002 [−0.032, 0.032]	−0.028 [−0.068, 0.011]	−0.029 to −0.003	−0.003 [−0.029, 0.022]
	GRNBoost2 T1 (256)	0.518 [0.494, 0.546]	0.541	0.518 (0.0059; 5)	0.508	+0.010 [−0.022, 0.038]	−0.023 [−0.071, 0.026]	−0.024 to +0.002	–
	GRNBoost2 T2 (3,000)	0.512 [0.483, 0.544]	0.539	0.515 (0.0020; 2)	0.515	−0.003 [−0.031, 0.025]	−0.029 [−0.073, 0.014]	−0.030 to −0.004	−0.006 [−0.038, 0.026]
lung (ext.) (0.538)	GENIE3 T1 (256)	0.535 [0.508, 0.565]	0.559	0.533 (0.0028; 5)	0.524	+0.011 [−0.017, 0.036]	−0.002 [−0.051, 0.036]	−0.003 to +0.018	–
	GENIE3 T2 (3,000)	0.538 [0.501, 0.585]	0.576	0.538 (0.0003; 2)	0.551	−0.013 [−0.044, 0.022]	+0.001 [−0.047, 0.043]	+0.000 to +0.021	+0.003 [−0.028, 0.041]
	GRNBoost2 T1 (256)	0.521 [0.493, 0.551]	0.547	0.520 (0.0070; 5)	0.524	−0.003 [−0.030, 0.025]	−0.016 [−0.063, 0.020]	−0.017 to +0.004	–
	GRNBoost2 T2 (3,000)	0.535 [0.504, 0.570]	0.564	0.533 (0.0014; 2)	0.551	−0.017 [−0.048, 0.019]	−0.003 [−0.053, 0.036]	−0.004 to +0.017	+0.013 [−0.022, 0.049]
kidney B (0.554)	GENIE3 T1 (192)	0.529 [0.496, 0.564]	0.559	0.527 (0.0024; 5)	0.515	+0.014 [−0.021, 0.049]	−0.026 [−0.081, 0.023]	−0.026 to +0.005	–
	GENIE3 T2 (3,000)	0.528 [0.497, 0.563]	0.558	0.527 (0.0003; 2)	0.522	+0.006 [−0.034, 0.052]	−0.027 [−0.077, 0.023]	−0.027 to +0.004	−0.001 [−0.036, 0.038]
	GRNBoost2 T1 (192)	0.519 [0.492, 0.548]	0.542	0.513 (0.0027; 5)	0.515	+0.004 [−0.029, 0.035]	−0.035 [−0.089, 0.017]	−0.035 to −0.004	–

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Table S7 (continued)

	GRNBoost2 T2 (3,000)	0.528 [0.501, 0.557]	0.553	0.527 (0.0003; 2)	0.522	+0.006 [−0.028, 0.044]	−0.027 [−0.076, 0.022]	−0.027 to +0.004	+0.008 [−0.025, 0.045]
immune (0.638)	GENIE3 T1 (256)	0.578 [0.508, 0.664]	0.649	0.577 (0.0048; 5)	0.559	+0.019 [−0.045, 0.082]	−0.060 [−0.137, 0.020]	−0.060 to +0.000	–
	GENIE3 T2 (3,000)	0.610 [0.532, 0.693]	0.677	0.609 (0.0015; 2)	0.562	+0.048 [−0.031, 0.121]	−0.028 [−0.115, 0.055]	−0.028 to +0.032	+0.032 [−0.027, 0.083]
	GRNBoost2 T1 (256)	0.574 [0.502, 0.667]	0.651	0.572 (0.0067; 5)	0.559	+0.015 [−0.048, 0.091]	−0.064 [−0.140, 0.021]	−0.064 to −0.003	–
	GRNBoost2 T2 (3,000)	0.605 [0.529, 0.685]	0.672	0.604 (0.0087; 2)	0.562	+0.044 [−0.025, 0.115]	−0.032 [−0.116, 0.051]	−0.032 to +0.028	+0.031 [−0.039, 0.095]
kidney A (0.647)	GENIE3 T1 (256)	0.557 [0.460, 0.677]	0.649	0.554 (0.0042; 5)	0.559	−0.002 [−0.096, 0.097]	−0.091 [−0.205, 0.030]	−0.095 to −0.044	–
	GENIE3 T2 (3,000)	0.596 [0.504, 0.701]	0.683	0.597 (0.0030; 2)	0.577	+0.019 [−0.096, 0.144]	−0.051 [−0.153, 0.056]	−0.056 to −0.005	+0.040 [−0.053, 0.132]
	GRNBoost2 T1 (256)	0.554 [0.449, 0.670]	0.647	0.544 (0.0126; 5)	0.559	−0.005 [−0.104, 0.097]	−0.094 [−0.214, 0.038]	−0.098 to −0.047	–
	GRNBoost2 T2 (3,000)	0.595 [0.502, 0.700]	0.678	0.589 (0.0108; 2)	0.577	+0.018 [−0.089, 0.120]	−0.052 [−0.151, 0.053]	−0.057 to −0.006	+0.041 [−0.054, 0.128]

Two-sided intervals excluding 0: Δ vs $|r|$, 0 of 20; Δ vs cosine, 0 of 240 (extraction × method × tier × layer); one one-sided bound lies below 0 (kidney A GENIE3 T1 vs L_1, UB −0.0008; unadjusted). One-sided UB of Δ vs $|r|$ in lung, lung (ext.) and kidney B: at most 0.044.

F. Supervised probes on pair-disjoint folds (neither gene of a test pair seen in training; 5 × 5 folds)

Quantity	lung	lung (ext.)	kidney B
Benchmark (each test fold holds about 4% of the pairs)	52,426 pairs; 5,266 positives	50,947 pairs; 5,093 positives	36,380 pairs; 3,668 positives
Best scGPT probe: feature, layer	bilinear, L_2	bilinear, L_8	[x_j, bilinear], L_0
AUROC [95% CI, pair ratio]; upper end of the 95% CI, gene ratio	0.597 [0.573, 0.621]; 0.638	0.591 [0.557, 0.624]; 0.647	0.589 [0.555, 0.624]; 0.647
Target-only probe: AUROC, range over layers (factor-only: 0.500 by construction)	0.540–0.554	0.558–0.573	0.553–0.573
Concatenation (additive) − target-only: range	−0.001 to +0.004	−0.001 to +0.003	−0.004 to +0.004
Hadamard − target-only: range; layers with CI < 0 (pair; gene ratio)	−0.009 to +0.002; 0; 0	−0.036 to −0.016; 3; 0	−0.041 to −0.017; 0; 0
$\lvert x_i - x_j \rvert$ − target-only: range; layers with CI < 0 (pair; gene ratio)	−0.044 to −0.019; 7; 0	−0.068 to −0.034; 12; 8	−0.074 to −0.033; 9; 5
Bilinear − target-only: range over layers	−0.004 to +0.051	+0.002 to +0.029	−0.031 to +0.006
largest gain [95% CI, pair ratio] [gene ratio]	L_2: +0.051 [0.026, 0.076] [0.009, 0.093]	L_8: +0.029 [0.008, 0.051] [−0.007, 0.065]	L_2: +0.006 [−0.019, 0.031] [−0.037, 0.049]
layers with CI > 0: pair; gene ratio	7; 1	2; 0	0; 0
[x_j, bilinear] − target-only: range; layers with CI > 0 (pair; gene ratio)	+0.001 to +0.025; 8; 3	+0.009 to +0.022; 9; 3	−0.004 to +0.016; 0; 0
Bilinear vs the same probe on rewired labels ($N = 100$): layers $p < 0.05$; max-$T < 0.05$	12/12; 11/12	8/12; 4/12	4/12; 1/12
Concatenation vs rewired labels: layers $p < 0.05$; max-$T < 0.05$; AUROC true vs rewired (mean)	9/12; 1/12; true 0.560–0.571 vs 0.534–0.542	0/12; 0/12; true 0.547–0.563 vs 0.562–0.574	0/12; 0/12; true 0.547–0.573 vs 0.553–0.563
Difference in differences, bilinear − concatenation: true labels, range (rewired-label mean); layers max-$T < 0.05$	−0.011 to +0.046 (rewired −0.022 to −0.008); 7/12 (L_1–L_7)	+0.000 to +0.028 (rewired −0.035 to −0.024); 7/12 (L_4–L_{10})	−0.053 to +0.008 (rewired −0.036 to −0.026); 0/12
Relational signal (probe − max[rewired, target-only]): layers with CI > 0, pair; gene ratio	1; 0	0; 0	0; 0
Bilinear − signed cosine on the same folds: range; layers with CI > 0 (pair; gene ratio)	+0.020 to +0.059; 10; 3	+0.035 to +0.062; 12; 0	−0.009 to +0.016; 0; 0
scGPT input gene-token embedding (no cell context), bilinear: AUROC; − target-only [pair ratio] [gene ratio]	0.581; +0.026 [0.003, 0.050] [−0.013, 0.066]	0.584; +0.016 [−0.014, 0.046] [−0.034, 0.066]	0.577; +0.005 [−0.022, 0.031] [−0.040, 0.049]
p vs rewired labels; difference-in-differences p	0.0099; 0.0693	0.32; 0.0990	0.0099; 0.21
ESM2-15B per-protein embedding (UCE), bilinear: AUROC; − target-only [pair ratio] [gene ratio]	0.567; +0.005 [−0.021, 0.031] [−0.039, 0.049]	0.563; +0.012 [−0.012, 0.035] [−0.027, 0.051]	0.548; −0.013 [−0.040, 0.014] [−0.058, 0.032]
p vs rewired labels; difference-in-differences p	0.0099; 0.0495	0.0198; 0.0198	0.12; 0.0792
scGPT bilinear − input-embedding bilinear (paired): layers with CI > 0; < 0	0; 4	0; 0	0; 5
scGPT gain over its target-only control − the same gain of the input embedding; of ESM2-15B: layers with CI > 0	1; 1	0; 0	0; 0
Gradient boosting and MLP (fixed and tuned; L_0, L_5, L_{11}; concatenation, Hadamard): AUROC range; rows above; below own target-only	0.496–0.565; 0; 2 of 24	0.491–0.555; 0; 4 of 24	0.493–0.548; 0; 0 of 24

continued on next page

Table S7 (continued)

held-out factors (targets seen): rows above own target-only	1 of 24	3 of 24	1 of 24

G. Direction: documented pairs $i \to j$ of two factors whose reverse $j \to i$ is in no regulatory resource

GENIE3 and GRNBoost2 importances, seed ensemble: fraction of untied pairs with importance($i \to j$) > importance($j \to i$)

Extraction	Method	Pairs; untied T1; T2	T1 fraction	T1 sign-test p	T1 regulator-cluster CI	T2 fraction	T2 sign-test p	T2 regulator-cluster CI	T1 per-seed range
lung	GENIE3	855; 834; 845	0.511	0.56	[0.464, 0.559]	0.509	0.63	[0.461, 0.557]	0.480–0.513
	GRNBoost2	855; 762; 723	0.528	0.14	[0.486, 0.566]	0.523	0.23	[0.473, 0.571]	0.488–0.510
lung (ext.)	GENIE3	886; 867; 882	0.525	0.15	[0.479, 0.567]	0.505	0.81	[0.453, 0.559]	0.491–0.521
	GRNBoost2	886; 710; 705	0.501	0.97	[0.462, 0.538]	0.487	0.50	[0.438, 0.534]	0.469–0.533
kidney B	GENIE3	669; 661; 668	0.523	0.24	[0.470, 0.574]	0.551	0.0095	[0.492, 0.609]	0.530–0.551
	GRNBoost2	669; 592; 586	0.502	0.97	[0.462, 0.541]	0.505	0.84	[0.458, 0.550]	0.457–0.548
immune	GENIE3	62; 60; 60	0.517	0.90	[0.387, 0.633]	0.467	0.70	[0.347, 0.607]	0.448–0.542
	GRNBoost2	62; 57; 58	0.544	0.60	[0.407, 0.673]	0.483	0.90	[0.356, 0.632]	0.463–0.589
kidney A	GENIE3	21; 21; 21	0.429	0.66	[0.222, 0.654]	0.524	1.00	[0.263, 0.789]	0.381–0.524
	GRNBoost2	21; 21; 21	0.571	0.66	[0.350, 0.800]	0.619	0.38	[0.364, 0.857]	0.421–0.600

Sign tests with $p < 0.05$: 1 of 20 (none below the Bonferroni threshold $0.05/20 = 0.0025$). $|r|$ and the cosine are symmetric and cannot score direction.

Logistic probe on the ordered concatenation, pair-disjoint folds: fraction of pairs scored forward > reverse

Extraction	Pairs; genes	Fraction, range L_0–L_{11}	Layers, binomial CI excl. 0.5	Layers, gene-cluster CI excl. 0.5	Largest fraction [gene-cluster CI]	Strict subset: pairs; range	Out-degree $i > j$	In-degree $j > i$
lung	435; 241	0.494–0.543	0/12	0/12	L_6 0.543 [0.460, 0.626]	343; 0.499–0.563	0.559	0.547
lung (ext.)	453; 251	0.536–0.594	11/12	2/12 (L_{10}–L_{11})	L_{11} 0.594 [0.506, 0.677]	357; 0.529–0.599	0.532	0.586
kidney B	325; 197	0.563–0.628	12/12	1/12 (L_7)	L_7 0.628 [0.512, 0.740]	260; 0.565–0.631	0.550	0.549

Gene-cluster intervals excluding 0.5: 3 of 36 extraction × layer combinations.

Table S8: **Rank reduction (H4): truncation to the leading spectral directions against the full representation.** m, layer-mean task metric (mean over the 12 layers) of the representation restricted to the first k directions $U_k S_k$ of the observed-gene SVD at each layer ($m(512)$, the full representation). Tasks: interaction, AUROC of the cosine of STRING physical pairs (combined score ≥ 700) against one degree-preserving rewired set of non-pairs; TF identity, the H2 statistic (detection-stratified LOO-cosine AUROC, Lambert TFs against all other observed genes); regulatory edges, the H3 statistic (within-TF stratified AUROC of the TF–target cosine, TRRUST pairs against matched candidates). **A**: k^* maximises $m(k)$ over $k \in \{1, 2, 3, 5, 7, 10, 16, 32, 64, 128, 256, 512\}$ in the selection draw (ties to the smallest k); selection gap $= m(k^*) - m(512)$ there. **B**: Δ in the evaluation extractions with paired 95% intervals ($B = 2{,}000$; the same resamples for k^* and 512): interaction, vertex subsampling ($m = n/2$) with delete-d scaling; TF identity, gene bootstrap; edges, two-way TF × target bootstrap. For TF identity the selection draw chose $k^* = 512$, so $\Delta = 0$ by construction. Last column: number of the 12 per-layer intervals entirely below / above 0. **C**–**E**: descriptive, not pre-registered; kidney A and immune (seed 43, the selection draw) are shown for completeness; the last column has no interval. **D**: removing the leading directions instead of keeping them; intervals as in B (interaction delete-d). **E**: screen bands SV_2–SV_4, SV_5–SV_7, SV_2–SV_7 on the observed-gene basis against the first 3 or 6 directions (the screen used SV_5–SV_7 of the observed genes in its later kidney A iterations; Table S14); [b]SV_5–SV_7 shifted down by one index, as when it is read in the reference decomposition of the full kidney A matrix, zero rows included (there SV_2–SV_4 and SV_2–SV_7 become the first 3 and 6 directions, a difference of 0 by construction), shown in every extraction. **F**: one vector per gene: input gene-token embeddings (scGPT, Geneformer, tGPT); gene outputs of one forward pass on a neutral input (AIDO.Cell-100M, scFoundation); ESM2-15B, the 5,120-dimensional per-protein embedding (UCE file), and residue-mean-pooled ESM2-650M; 32 sequence descriptors; and expression across 154 Human Protein Atlas cell types (128 principal components). Each is mean-centred on the 5,226 lung genes present in all eleven vocabularies and scored on identical genes, pairs and labels; d, dimension; best k over the same grid, gain = best minus full (no interval). [c]Reference row: the scGPT lung residual stream on the same genes, layer mean. No truncation exceeds the full embedding by more than 0.022 for any static representation and task. Per-layer values, all k, the descriptive any-k permutation test and, for interaction, the intervals with the pre-registered scaling factor $\sqrt{m/n}$, which omits the finite-population term and so gives half-widths smaller by a factor $\sqrt{2}$ (columns `ci_lo`/`ci_hi`, beside the reported `ci_lo_ms`/`ci_hi_ms`), are in the CSV. Full values: `Table_S8_rank.csv`.

A. Truncation k^* chosen in the selection draw (immune, seed 43), before any evaluation extraction

Task	k^*	$m(k^*)$	$m(512)$	Selection gap	$m(3) - m(512)$	$m(10) - m(512)$
Interaction	64	0.7047	0.7021	+0.0026	−0.099	−0.018
TF identity	512	0.6088	0.6088	+0.0000	−0.093	−0.055
Regulatory edges	128	0.6199	0.6198	+0.0001	−0.087	−0.037

B. Pre-registered estimand $\Delta = m(k^*) - m(512)$, layer mean, in the four evaluation extractions

Task, extraction	$m(k^*)$	$m(512)$	Δ	95% CI	Layers with CI < 0 / > 0

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Table S8 (continued)

Interaction, $k^* = 64$					
lung	0.6539	0.6549	−0.0011	[−0.0024, 0.0003]	6 / 0
lung (ext.)	0.6573	0.6579	−0.0006	[−0.0019, 0.0007]	0 / 0
kidney B	0.6474	0.6486	−0.0013	[−0.0026, −0.0000]	7 / 0
immune	0.7020	0.6998	+0.0022	[−0.0018, 0.0064]	0 / 0
TF identity, $k^* = 512$					
lung	0.6542	0.6542	0 (identical)	–	–
lung (ext.)	0.6436	0.6436	0 (identical)	–	–
kidney B	0.6343	0.6343	0 (identical)	–	–
immune	0.6553	0.6553	0 (identical)	–	–
Regulatory edges, $k^* = 128$					
lung	0.5304	0.5300	+0.0004	[−0.0022, 0.0026]	0 / 0
lung (ext.)	0.5303	0.5300	+0.0003	[−0.0021, 0.0026]	0 / 0
kidney B	0.5411	0.5413	−0.0002	[−0.0027, 0.0023]	0 / 0
immune	0.6092	0.6096	−0.0004	[−0.0070, 0.0061]	0 / 0

C. The leading k directions only: layer-mean value at $k = 3$ and $k = 10$ (difference from $m(512)$ in parentheses)

Task, k	lung	lung (ext.)	kidney B	immune	kidney A	immune (seed 43)
Interaction, $k = 512$ (full)	0.655	0.658	0.649	0.700	0.767	0.702
Interaction, $k = 3$	0.614 (−0.041)	0.626 (−0.032)	0.610 (−0.038)	0.609 (−0.091)	0.718 (−0.049)	0.604 (−0.099)
Interaction, $k = 10$	0.641 (−0.014)	0.647 (−0.011)	0.634 (−0.014)	0.682 (−0.018)	0.760 (−0.008)	0.684 (−0.018)
TF identity, $k = 512$ (full)	0.654	0.644	0.634	0.655	0.742	0.609
TF identity, $k = 3$	0.526 (−0.128)	0.556 (−0.087)	0.547 (−0.088)	0.513 (−0.142)	0.620 (−0.122)	0.516 (−0.093)
TF identity, $k = 10$	0.612 (−0.042)	0.625 (−0.018)	0.601 (−0.034)	0.571 (−0.084)	0.700 (−0.041)	0.554 (−0.055)
Regulatory edges, $k = 512$ (full)	0.530	0.530	0.541	0.610	0.632	0.620
Regulatory edges, $k = 3$	0.496 (−0.034)	0.492 (−0.038)	0.504 (−0.037)	0.527 (−0.083)	0.568 (−0.063)	0.533 (−0.087)
Regulatory edges, $k = 10$	0.516 (−0.014)	0.519 (−0.011)	0.527 (−0.014)	0.566 (−0.044)	0.603 (−0.028)	0.583 (−0.037)

D. Removing the j leading directions (keeping SV_{j+1}–SV_{512}): layer-mean difference from $m(512)$ [95% CI]

Task, directions removed	lung	lung (ext.)	kidney B	immune	kidney A	immune (seed 43)
Interaction, $j = 1$	−0.007 [−0.012, −0.002]	+0.002 [−0.004, 0.009]	+0.001 [−0.005, 0.007]	+0.027 [0.011, 0.042]	+0.031 [0.015, 0.046]	+0.022
Interaction, $j = 2$	−0.010 [−0.020, 0.000]	−0.023 [−0.036, −0.008]	−0.006 [−0.018, 0.005]	+0.018 [−0.004, 0.041]	+0.012 [−0.015, 0.037]	+0.012
Interaction, $j = 3$	−0.024 [−0.037, −0.010]	−0.030 [−0.044, −0.014]	−0.019 [−0.033, −0.005]	+0.015 [−0.008, 0.038]	+0.004 [−0.026, 0.032]	+0.013
Interaction, $j = 5$	−0.027 [−0.042, −0.013]	−0.036 [−0.051, −0.021]	−0.017 [−0.033, −0.002]	−0.000 [−0.027, 0.027]	−0.024 [−0.061, 0.012]	+0.004
Interaction, $j = 10$	−0.041 [−0.056, −0.026]	−0.049 [−0.065, −0.033]	−0.033 [−0.048, −0.016]	−0.045 [−0.081, −0.011]	−0.068 [−0.110, −0.028]	−0.041
TF identity, $j = 1$	+0.036 [0.027, 0.043]	+0.028 [0.019, 0.037]	+0.025 [0.015, 0.035]	+0.043 [0.006, 0.070]	+0.006 [−0.047, 0.027]	+0.034
TF identity, $j = 2$	+0.060 [0.047, 0.070]	+0.098 [0.079, 0.108]	+0.070 [0.050, 0.079]	+0.082 [0.024, 0.112]	−0.016 [−0.067, 0.028]	+0.092
TF identity, $j = 3$	+0.091 [0.074, 0.102]	+0.099 [0.080, 0.112]	+0.091 [0.065, 0.104]	+0.086 [0.021, 0.110]	−0.002 [−0.055, 0.044]	+0.101
TF identity, $j = 5$	+0.099 [0.080, 0.111]	+0.107 [0.087, 0.122]	+0.108 [0.084, 0.122]	+0.092 [0.035, 0.122]	−0.007 [−0.090, 0.044]	+0.117
TF identity, $j = 10$	+0.089 [0.063, 0.103]	+0.092 [0.060, 0.105]	+0.108 [0.077, 0.123]	+0.099 [0.037, 0.146]	−0.001 [−0.100, 0.059]	+0.130
Regulatory edges, $j = 1$	+0.009 [−0.007, 0.021]	+0.011 [−0.005, 0.026]	+0.009 [−0.005, 0.024]	+0.005 [−0.019, 0.031]	+0.003 [−0.046, 0.046]	−0.000
Regulatory edges, $j = 2$	+0.019 [−0.007, 0.045]	+0.031 [−0.006, 0.067]	+0.019 [−0.007, 0.043]	+0.020 [−0.025, 0.061]	−0.005 [−0.095, 0.065]	+0.010
Regulatory edges, $j = 3$	+0.033 [−0.009, 0.071]	+0.037 [−0.007, 0.075]	+0.023 [−0.014, 0.060]	+0.025 [−0.017, 0.064]	−0.009 [−0.109, 0.072]	+0.014
Regulatory edges, $j = 5$	+0.041 [−0.006, 0.081]	+0.039 [−0.007, 0.078]	+0.023 [−0.018, 0.067]	+0.026 [−0.021, 0.070]	−0.012 [−0.109, 0.076]	+0.018
Regulatory edges, $j = 10$	+0.017 [−0.034, 0.060]	+0.014 [−0.033, 0.056]	+0.003 [−0.040, 0.047]	+0.022 [−0.058, 0.086]	−0.025 [−0.116, 0.067]	+0.005

E. The screen's bands against the leading directions of the same width: layer-mean difference [95% CI]

Task, band vs partner	lung	lung (ext.)	kidney B	immune	kidney A	immune (seed 43)
Interaction, SV_2–SV_4 vs top 3	−0.018 [−0.024, −0.012]	−0.014 [−0.022, −0.006]	−0.009 [−0.017, −0.001]	+0.036 [0.009, 0.063]	+0.016 [−0.005, 0.038]	+0.032
Interaction, SV_5–SV_7 vs top 3	−0.036 [−0.053, −0.019]	−0.049 [−0.067, −0.030]	−0.039 [−0.055, −0.022]	+0.037 [−0.002, 0.075]	−0.021 [−0.061, 0.016]	+0.042
Interaction, SV_2–SV_7 vs top 6	−0.013 [−0.018, −0.008]	−0.007 [−0.014, −0.000]	−0.005 [−0.011, 0.001]	+0.031 [0.010, 0.051]	+0.023 [0.008, 0.038]	+0.033

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Table S8 (continued)

Interaction, SV_4–SV_6[b] vs top 3	−0.036 [−0.053, −0.018]	−0.051 [−0.069, −0.033]	−0.039 [−0.056, −0.021]	+0.026 [−0.009, 0.063]	−0.019 [−0.057, 0.019]	+0.027
TF identity, SV_2–SV_4 vs top 3	+0.023 [0.008, 0.036]	+0.003 [−0.012, 0.020]	+0.004 [−0.010, 0.019]	+0.039 [−0.015, 0.103]	−0.060 [−0.124, −0.024]	+0.004
TF identity, SV_5–SV_7 vs top 3	+0.045 [0.012, 0.083]	+0.063 [0.035, 0.103]	+0.044 [0.012, 0.083]	−0.037 [−0.094, 0.066]	+0.005 [−0.057, 0.078]	+0.026
TF identity, SV_2–SV_7 vs top 6	+0.015 [0.006, 0.022]	+0.026 [0.014, 0.043]	+0.020 [0.005, 0.036]	+0.019 [−0.026, 0.065]	−0.014 [−0.060, 0.013]	+0.037
TF identity, SV_4–SV_6[b] vs top 3	+0.051 [0.017, 0.090]	+0.008 [−0.024, 0.046]	−0.001 [−0.036, 0.041]	−0.012 [−0.073, 0.079]	+0.008 [−0.064, 0.082]	−0.023
Regulatory edges, SV_2–SV_4 vs top 3	+0.000 [−0.020, 0.015]	+0.008 [−0.020, 0.040]	+0.006 [−0.016, 0.027]	−0.015 [−0.064, 0.045]	−0.030 [−0.099, 0.034]	−0.016
Regulatory edges, SV_5–SV_7 vs top 3	+0.045 [−0.009, 0.095]	+0.050 [−0.003, 0.104]	+0.036 [−0.008, 0.078]	+0.016 [−0.062, 0.080]	−0.036 [−0.186, 0.095]	+0.006
Regulatory edges, SV_2–SV_7 vs top 6	+0.007 [−0.011, 0.021]	+0.011 [−0.008, 0.029]	+0.013 [−0.004, 0.029]	−0.008 [−0.049, 0.030]	−0.015 [−0.068, 0.033]	−0.012
Regulatory edges, SV_4–SV_6[b] vs top 3	+0.036 [−0.015, 0.084]	+0.042 [−0.012, 0.091]	+0.029 [−0.017, 0.082]	+0.009 [−0.062, 0.074]	−0.050 [−0.180, 0.060]	+0.000

F. Eleven static gene representations on the common lung gene set (5,226 genes): full embedding and best truncation

Representation (d)	Interaction: full	Interaction: best k (gain)	TF identity: full	TF identity: best k (gain)	Edges: full	Edges: best k (gain)
scGPT input embedding (512)	0.661	64 (+0.001)	0.751	512 (+0.000)	0.566	128 (+0.004)
Geneformer V2-316M (1,152)	0.650	5 (+0.004)	0.821	1152 (+0.000)	0.561	16 (+0.004)
Geneformer V1-10M (256)	0.583	256 (+0.000)	0.725	256 (+0.000)	0.518	256 (+0.000)
Geneformer V2-104M (768)	0.651	7 (+0.017)	0.825	512 (+0.001)	0.562	64 (+0.010)
AIDO.Cell-100M (640)	0.605	256 (+0.000)	0.627	64 (+0.022)	0.546	640 (+0.000)
scFoundation (512)	0.541	32 (+0.000)	0.454	5 (+0.003)	0.502	32 (+0.002)
tGPT (1,024)	0.574	1024 (+0.000)	0.612	512 (+0.001)	0.519	512 (+0.001)
ESM2-15B (5,120)	0.639	512 (+0.000)	0.880	32 (+0.002)	0.498	1 (+0.009)
ESM2-650M (1,280)	0.615	512 (+0.000)	0.846	64 (+0.001)	0.497	1 (+0.007)
Sequence descriptors (32)	0.667	32 (+0.000)	0.867	32 (+0.000)	0.499	5 (+0.001)
HPA cell types (128)	0.595	16 (+0.002)	0.573	128 (+0.000)	0.515	16 (+0.001)
scGPT lung, layer mean[c] (512)	0.649	512 (+0.000)	0.657	128 (+0.000)	0.530	128 (+0.001)

Table S9: **Differences between extractions: variance components, transfer of a TF read-out, lineage TFs, detection, and the two kidney extractions.** Statistics (all AUROCs, V-centred embeddings, layer mean over the 12 layers unless stated): S1, cosine of STRING physical pairs against one degree-preserving rewired non-pair set; S2, the H2 statistic (detection-stratified LOO-cosine AUROC, Lambert TFs against all other observed genes); S2b, TRRUST TFs against target-only genes, in the full representation or in the band SV_2–SV_7; S3, the H3 statistic (within-TF stratified AUROC of the TF–target cosine). **A**: moment estimate of the between-extraction SD τ (AUROC units; 0 when the estimate of τ^2 is negative) among the k extractions of each group: cell draws (immune, seed 43, seed 44; kidney B and the whole-human checkpoint run), donors TSP14/21/25, B/CD4 T/CD8 T cells, kidney A vs B, the three checkpoints on the same 192 cells, and the four tissue-level extractions (immune, kidney B, lung, lung (ext.)). Sampling covariances come from one Poisson gene bootstrap shared by all extractions (B = 1,000; S3 two-way TF × target; S1 node bootstrap, whose SE is within 5% of the delete-d vertex-subsampling SE). A tissue effect is claimed only if D, the tissue component minus the pooled within-tissue floor (cell draws, donors, gene axis), has a 95% lower bound > 0 under a parametric random-effects bootstrap (R = 10,000); the lower bound exceeds 0 neither for the layer mean nor at any single layer (0 of 12 layers for every statistic). Rows marked $\times 10^3$ are in units of 10^{-3} AUROC^2. Q-profile bounds treat the extractions as random draws; with 2–4 extractions per group they are wide. **B**: L2-regularised logistic regression (balanced classes; C chosen by 5-fold CV from $\{0.01, 0.1, 1, 10\}$) on unit-normalised V-centred embeddings, fitted per layer on immune genes; novel genes are not observed in immune, shared genes are; the in-context reference is a 5-fold out-of-fold classifier within the test extraction scored on the same genes; intervals are 95% gene-bootstrap intervals (B = 2,000); AUROCs without stratification, with the detection-stratified AUROC (within deciles of the test extraction's $\log n_{\text{obs}}$, as for S2) in parentheses; below the range over layers, the smallest lower bound over layers; the difference, its intervals and, in parentheses, the layers whose interval of the difference lies below 0 are without stratification; per-layer values in the CSV. **C**: sets of the 30 Lambert TFs most specific to each tissue in raw expression (18 for lung, which must be expressed in both lung studies), selected from raw expression alone, without any embedding statistic; for example immune BACH2, ETS1, BCL11B, IKZF1, PAX5, EBF1, IRF4; kidney PAX2, PAX8, HNF1B, HNF4A, EMX2, SALL1; lung NKX2-1, FOXA2, FOXF1, GATA6, EPAS1 (full lists in the CSV). Score: LOO cosine to the observed set members; at least 5 observed members required; centred Poisson gene-bootstrap intervals (B = 500) clipped to $[0, 1]$; observed, members in V; random, mean AUROC of 50 random Lambert-TF subsets of the same size; [h]the set's own tissue; [d]interval unreliable (fewer than 10 observed members). Across sets and extractions the AUROC falls with the set's median n_{obs} (Spearman $\rho = -0.39$, n = 29; rows not independent). **D**: AUROC of $\log n_{\text{obs}}$ alone (no stratification), gene-bootstrap 95% intervals (B = 2,000); values above 0.5 mean TFs are observed in more cells. Cells whose input exceeded scGPT's 512-gene cap are truncated. **E**: M, genes observed in both extractions, with labels recomputed on M; paired gene bootstrap on M (B = 2,000), Δ = kidney A − kidney B. Curated-gene comparison: TRRUST TFs / targets of the curated gene list derived from the screen's kidney A edge list, with the SVD fitted on the full matrix including genes no sampled cell contained (zero rows); the Hanley–McNeil SE is that of a single AUROC at the stated value and sample sizes. Full values: `Table_S9_cross_extraction.csv`.

A. Variance components of the layer-mean statistic across the 16 extractions: between-extraction SD τ within each group, and the tissue decision

Component	k	S1 interaction	S2 TF identity	S2b TRRUST, full	S2b TRRUST, band	S3 edges
Cell draw, immune	3	0.000	0.027	0.016	0.033	0.000
Cell draw, kidney	2	0.000	0.000	0.000	0.000	0.000
Donor (immune)	3	0.000	0.019	0.000	0.078	0.000
Cell type (immune)	3	0.000	0.000	0.000	0.070	0.000
Gene axis (kidney A vs B)	2	0.075	0.073	0.000	0.000	0.052
Checkpoint (same cells)	3	0.023	0.027	0.043	0.030	0.000
Tissue	4	0.020	0.000	0.000	0.000	0.033
Tissue, without lung (ext.)	3	0.025	0.000	0.000	0.000	0.037
Study (lung vs lung (ext.))	2	0.000	0.005	0.000	0.012	0.000
Tissue τ, upper Q-profile 95% bound		0.086	0.034	0.029	0.050	0.139
τ^2_{tissue} ($\times 10^3$)		0.42	−0.15	−0.26	−0.28	1.10
Pooled within-tissue floor τ^2_{within} ($\times 10^3$)		0.89	1.25	−0.15	2.17	0.38
$D = \tau^2_{\text{tissue}} - \tau^2_{\text{within}}$ ($\times 10^3$) [95% interval]		−0.47 [−2.33, 3.71]	−1.40 [−3.15, 2.68]	−0.12 [−1.51, 0.84]	−2.45 [−6.10, 5.00]	+0.72 [−2.69, 3.84]
p (H_0: tissue = floor); tissue effect claimed		0.73; no	0.83; no	0.45; no	0.79; no	0.18; no

B. A TF classifier trained on immune genes, applied unchanged to the other extractions (layer mean)

Test extraction	TFs / non-TFs (novel genes)	Novel genes: AUROC (detection-stratified)	Novel genes: range over layers	In-context classifier, same genes (detection-stratified)	Transfer minus in-context (layers CI < 0)	Shared genes: AUROC
immune (training; 5-fold out-of-fold)	113 / 1,926	0.835 (0.796)	–	–	–	–
kidney B	438 / 3,632	0.652 (0.672)	0.625–0.669 lower ≥ 0.598	0.784 (0.774)	−0.131 (12/12)	0.955
lung	495 / 4,297	0.674 (0.698)	0.662–0.684 lower ≥ 0.636	0.786 (0.786)	−0.112 (12/12)	0.966
lung (ext.)	539 / 4,292	0.672 (0.693)	0.656–0.680 lower ≥ 0.631	0.798 (0.796)	−0.126 (12/12)	0.959
kidney A	45 / 1,414	0.871 (0.848)	0.793–0.902 lower ≥ 0.728	0.876 (0.844)	−0.005 (0/12)	0.964

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Table S9 (continued)

C. Lineage-specific TF sets: detection-stratified AUROC of set members against non-TF genes (layer mean)

Extraction	Immune set: AUROC [95% CI]	Immune set: observed (random)	Kidney set: AUROC [95% CI]	Kidney set: observed (random)	Lung set: AUROC [95% CI]	Lung set: observed (random)
lung	0.911 [0.859, 0.984]	28 (0.595)	0.795 [0.653, 1.000]	13 (0.560)	0.903[h] [0.836, 1.000]	16 (0.551)
lung (ext.)	0.908 [0.860, 0.973]	27 (0.589)	0.711 [0.545, 0.962]	10 (0.539)	0.916[h] [0.858, 1.000]	16 (0.548)
kidney B	0.873 [0.816, 0.932]	29 (0.585)	0.962[h] [0.941, 0.991]	21 (0.570)	0.937 [0.860, 1.000][d]	8 (0.507)
immune	0.709[h] [0.543, 0.911]	20 (0.604)	–	3	–	0
kidney A	0.763 [0.667, 0.877]	21 (0.697)	–	2	–	3
kidney (whole-human ckpt)	0.890 [0.844, 0.950]	30 (0.570)	0.961[h] [0.940, 0.998]	21 (0.565)	0.885 [0.705, 1.000][d]	6 (0.512)
kidney (kidney ckpt)	0.776 [0.693, 0.872]	30 (0.538)	0.716[h] [0.582, 0.872]	21 (0.537)	0.883 [0.715, 1.000][d]	6 (0.448)
kidney (brain ckpt)	0.796 [0.709, 0.906]	30 (0.515)	0.829[h] [0.751, 0.947]	21 (0.495)	0.614 [0.322, 1.000][d]	6 (0.422)
immune, 8 robustness extractions	0.665–0.755	19–20	1 of 8 evaluable	2–5	–	0

D. Detection alone: AUROC of $\log n_{\text{obs}}$ for separating TFs, and the extraction regime

Extraction	$\lvert V\rvert$; cells (input > 512 genes)	Median non-zero genes per cell	Median n_{obs}, Lambert TFs / non-TFs	Lambert TFs vs non-TFs: AUROC [95% CI]	TRRUST TFs vs target-only: AUROC [95% CI]	Lambert TFs with $n_{\text{obs}} \geq 10$
lung	5,648; 256 (249)	1,668	6 / 8	0.458 [0.434, 0.482]	0.519 [0.492, 0.548]	38%
lung (ext.)	5,616; 256 (220)	935	9 / 12	0.454 [0.432, 0.476]	0.534 [0.507, 0.560]	48%
kidney B	4,780; 192 (191)	1,054.5	6 / 7	0.461 [0.436, 0.486]	0.535 [0.506, 0.564]	39%
immune	2,039; 256 (2)	220	40 / 6	0.650 [0.590, 0.706]	0.705 [0.623, 0.779]	68%
kidney A	1,972; 256 (65)	436.5	20.5 / 11	0.555 [0.509, 0.599]	0.545 [0.458, 0.632]	71%
immune (seed 43)	2,080; 256 (2)	221	47 / 6	0.698 [0.643, 0.748]	0.730 [0.651, 0.803]	71%
immune (seed 44)	1,972; 256 (2)	238	49 / 7	0.677 [0.621, 0.733]	0.683 [0.603, 0.758]	70%
immune (donor TSP14)	1,964; 256 (2)	254.5	45 / 9	0.676 [0.619, 0.735]	0.724 [0.639, 0.798]	75%
immune (donor TSP21)	1,765; 256 (1)	216	64 / 9	0.706 [0.651, 0.758]	0.719 [0.638, 0.796]	77%
immune (donor TSP25)	1,791; 256 (5)	199	59.5 / 9	0.709 [0.651, 0.762]	0.707 [0.621, 0.784]	76%
immune (B cells)	1,398; 256 (0)	197	54 / 4	0.708 [0.648, 0.763]	0.703 [0.621, 0.789]	72%
immune (CD4 T cells)	1,465; 256 (0)	185	66 / 4	0.720 [0.659, 0.780]	0.739 [0.657, 0.812]	73%
immune (CD8 T cells)	1,595; 256 (0)	201	43.5 / 4	0.650 [0.587, 0.708]	0.682 [0.598, 0.763]	59%
kidney (whole-human ckpt)	4,923; 192 (188)	998	5 / 7	0.450 [0.427, 0.475]	0.525 [0.495, 0.553]	36%
kidney (kidney ckpt)	4,923; 192 (188)	998	5 / 7	0.450 [0.427, 0.475]	0.525 [0.497, 0.552]	36%
kidney (brain ckpt)	4,923; 192 (188)	998	5 / 7	0.450 [0.427, 0.474]	0.525 [0.497, 0.552]	36%

E. The two kidney extractions (one donor, different gene axes): mechanism, matched re-evaluation and the curated-gene comparison

Quantity	kidney A	kidney B	Shared / difference A − B [95% CI]
Gene axis G	4,803	8,143	2,766 shared
Observed genes $\lvert V\rvert$	1,972	4,780	1,126 observed in both (M)
Lambert TFs in V	90	511	76 in M; 49 TRRUST TFs / 103 target-only
Cells (input > 512 genes)	256 (65)	192 (191)	12 cells drawn in both
Median non-zero genes per cell	436.5	1,054.5	–
Mean fraction of a cell's detected genes not on the other axis	0.229	0.680	–

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Table S9 (continued)

Median n_{obs} on M: Lambert TFs / TRRUST TFs / non-TFs	24 / 24 / 32	14 / 15 / 17	Spearman ρ of n_{obs} on M 0.970
Matched genes M: S2b, band SV_2–SV_7, each extraction's own SVD	0.628	0.573	+0.055 [−0.033, 0.134]
Matched genes M: S2b, band SV_2–SV_7, SVD fitted on M	0.664	0.679	−0.016 [−0.057, 0.024]
Matched genes M: S2b, full dimension	0.676	0.646	+0.030 [−0.023, 0.088]
Matched genes M: S2, own detection strata	0.647	0.637	+0.011 [−0.035, 0.056]
Matched genes M: S2, shared detection strata	0.637	0.640	−0.003 [−0.039, 0.035]
Basis instability: $\lvert\Delta_{\text{band}}\rvert - \lvert\Delta_{\text{full}}\rvert$ (own SVD)	–	–	+0.025 [−0.053, 0.092]
TRRUST comparison on the curated genes	kidney A, 187 genes (65 / 122)	kidney A on kidney B's 167 genes	kidney B, 167 genes (59 / 108)
Band SV_2–SV_7, SVD including never-observed zero rows (the screen's convention), layer mean	0.612	0.636	0.590 L_0–L_3: 0.493, 0.500, 0.529, 0.551; L_4–L_{11}: 0.589–0.664
Same band, observed-gene SVD, layer mean	0.609	0.637	0.583 L_0–L_3: 0.677, 0.665, 0.648, 0.665
Full dimension, layer mean	0.676	0.681	0.644
Hanley–McNeil SE of one AUROC	0.044 (at 0.61)	–	0.047 (at 0.55)

Table S10: **Cross-model comparison: gene sets, eleven static representations on 17,693 genes, scGPT's residual stream against static representations, and the bias of a 186-gene curated set.** Unless stated otherwise, values in Panels B–D are AUROCs of full-dimensional vectors centred over the analysed gene set and compared by cosine. *Lambert*: leave-one-out (LOO) mean cosine to the census transcription factors (TFs) of Lambert et al., TFs against all other genes; *TRRUST*: the same score, TRRUST TFs against genes that are only targets; *PPI*: STRING physical pairs (score ≥ 700) against non-pairs from one degree-preserving rewiring; *edge*: within-TF AUROC of the TF–target cosine, TRRUST edges against candidate non-edges (targets of other TFs, absent from TRRUST and DoRothEA). 95% intervals: node tasks, gene bootstrap ($B = 1{,}000$); PPI, vertex subsampling ($m = n/2$, $B = 200$) with delete-d scaling; edge, two-way TF × target bootstrap ($B = 2{,}000$); paired differences reuse the same genes and draws for both representations. d, dimension; LM, language model; HPA, Human Protein Atlas; G, gene axis; V, observed genes. **A**: Tier A, genes in all 11 vocabularies; Tier B, $V \cap$ Tier A; the curated set starts from the automated screen's kidney A TRRUST edge list. **B**: token table, the model's input gene-embedding table ([a]with its LayerNorm applied); [b]gene outputs of one forward pass on the HPA mean-expression profile (no gene-token table, so not like-for-like with the token tables); ESM2-15B, the 5,120-dimensional per-protein embedding distributed with UCE; ESM2-650M, mean-pooled over the canonical sequence; HPA, single-cell expression over 154 cell types, 128 principal components. Lambert is pooled within lung detection deciles (unstratified in parentheses); the 12,467 genes not observed in lung all sit in the median decile, so this stratification is largely nominal (Panel C controls detection); within-decile permutation $p = 1/2{,}001$ throughout. TRRUST depends on the centring set: all Tier A genes, or only the 2,753 labelled genes (775 TFs, 1,978 target-only; point estimates; the convention of Panel D). Labelled-gene centring raises the single-cell models other than scFoundation by 0.048–0.119 and moves the others by at most 0.010, roughly halving the gap between protein and expression-trained representations (last two rows: paired differences from scGPT [95% CI], for Lambert of the stratified values). **C**: scGPT whole-human residual stream, one extraction per context (192–256 cells), centred over the Tier B set; Lambert is stratified within the extraction's deciles of $\log n_{\text{obs}}$ (the H2 statistic), the other tasks are unstratified. Positives / comparison set: TFs / other genes (Lambert), TFs / target-only genes (TRRUST), pairs / rewired non-pairs (PPI), documented edges / candidate non-edges (edge). Best layer: highest AUROC, chosen post hoc. Static columns: the representation's AUROC on the same genes; range over the 12 layers of the paired difference, static minus layer; k_+/k_-, layers whose interval lies entirely above / below 0. **D**: TRRUST AUROC (centred over the subset) on the curated set and in 10,000 random subsets of the 2,753 labelled Tier A genes matched on size, TF:target-only ratio (64:122 for 186) and lung mean-expression decile (degree-matched: TRRUST-degree quartiles and lung mean-expression quintiles; degree, TRRUST out-degree of a TF or in-degree of a target-only gene; pool, the labelled genes). Percentile: share of null subsets below the curated value (ties half); two-sided p with +1 correction (floor 2.0×10^{-4}); excess, curated value minus null mean; [c]scored on the 178 genes in tGPT's vocabulary. Kendall τ: agreement of the representation orders, with a paired gene-bootstrap interval ($B = 1{,}000$). Projection: Gaussian random projection (5 seeds) at a 5:1 ratio, scored as in Panel B (Lambert unstratified). All comparisons are sensitivity analyses outside the pre-registered hypotheses. For PPI, the vertex-subsampling intervals with the pre-registered scaling factor $\sqrt{m/n}$, which omits the finite-population term and so gives half-widths smaller by a factor $\sqrt{2}$, are in the CSV (columns `ci_lo_preregistered`/`ci_hi_preregistered`). Full values: `Table_S10_cross_model.csv`.

A. Gene sets: vocabularies, intersections, the two tiers and the curated set

Gene set	Genes	Composition, or genes lost at this step
Vocabulary of each static representation (genes with a non-zero vector)	19,264–60,694	scGPT (whole-human) 60,694; Geneformer V1-10M 21,243; Geneformer V2-104M 20,271; Geneformer V2-316M 20,271; tGPT 21,046; AIDO.Cell-100M 19,264; scFoundation 19,264; ESM2-15B 19,790; ESM2-650M 20,394; 32 sequence descriptors 20,213 (181 all-zero rows dropped); HPA co-expression 20,151

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Table S10 (continued)

Geneformer V1-10M ∩ V2-104M ∩ ESM2-15B	18,907	the vocabularies used to define the 186-gene curated set (below)
Tier A: all 11 vocabularies	17,693	Lambert TFs 1,587 vs 16,106 other genes; TRRUST TFs 775 vs 1,978 target-only genes; STRING physical pairs 73,430 vs 72,416 rewired non-pairs; TRRUST edges 7,593 (430 TFs) vs 70,882 candidate non-edges; 12,467 genes not observed in lung (assigned the median lung detection decile)
Tier B, lung	5,226	gene axis G 8,181; observed V 5,648; Lambert TFs in Tier B 579; genes in a TRRUST edge within Tier B 1,912; 256 cells
Tier B, lung (ext.)	5,154	gene axis G 8,229; observed V 5,616; Lambert TFs in Tier B 618; genes in a TRRUST edge within Tier B 1,948; 256 cells
Tier B, kidney B	4,441	gene axis G 8,143; observed V 4,780; Lambert TFs in Tier B 509; genes in a TRRUST edge within Tier B 1,542; 192 cells
Tier B, immune	1,338	gene axis G 4,941; observed V 2,039; Lambert TFs in Tier B 111; genes in a TRRUST edge within Tier B 281; 256 cells
Tier B, kidney A	1,298	gene axis G 4,803; observed V 1,972; Lambert TFs in Tier B 90; genes in a TRRUST edge within Tier B 172; 256 cells
Genes named in the curated kidney A edge list	209	1,152 TRRUST edge and sampled non-edge rows, so the list includes genes that occur only in non-edges; all 209 are on the kidney A gene axis
observed in kidney A	187	22 never-observed genes dropped
∩ Geneformer V1-10M, V2-104M, ESM2-15B: the 186-gene set	186	lost: NEAT1 (lncRNA); 64 TRRUST TFs, 122 target-only genes; 13 genes not observed in lung
∩ Tier A: the 178-gene set	178	lost (absent from tGPT): HLA-A, HLA-B, HLA-C, HLA-DMB, HLA-DOB, HLA-DPB1, HLA-DRA, HLA-DRB1; 64 TRRUST TFs, 114 target-only genes
Kidney A V ∩ Tier A, in a TRRUST edge	172	the most a kidney A-only analysis of the TRRUST tasks could use (kidney A observes 1,972 of its 4,803 genes)

B. Tier A (17,693 genes): every static representation, full dimension, centred over the analysed genes

Representation	Type; d	Lambert: stratified (unstratified) [95% CI]	TRRUST, centred over Tier A [95% CI]	TRRUST, centred over labelled genes	PPI [95% CI, delete-d]	Edge, within TF [95% CI]
scGPT (whole-human)	token table; 512	0.747 (0.704) [0.734, 0.760]	0.684 [0.660, 0.708]	0.789	0.669 [0.658, 0.679]	0.575 [0.534, 0.610]
Geneformer V1-10M	token table[a]; 256	0.776 (0.742) [0.761, 0.789]	0.688 [0.659, 0.715]	0.755	0.588 [0.579, 0.597]	0.529 [0.502, 0.562]
Geneformer V2-104M	token table[a]; 768	0.864 (0.836) [0.851, 0.876]	0.774 [0.749, 0.796]	0.821	0.662 [0.650, 0.671]	0.580 [0.549, 0.610]
Geneformer V2-316M	token table[a]; 1,152	0.855 (0.817) [0.842, 0.867]	0.743 [0.717, 0.765]	0.802	0.659 [0.650, 0.670]	0.578 [0.542, 0.611]
tGPT	token table; 1,024	0.690 (0.639) [0.678, 0.703]	0.520 [0.497, 0.543]	0.639	0.571 [0.561, 0.582]	0.544 [0.520, 0.570]
AIDO.Cell-100M	forward pass[b]; 640	0.623 (0.589) [0.611, 0.636]	0.579 [0.560, 0.600]	0.690	0.603 [0.590, 0.614]	0.552 [0.499, 0.591]
scFoundation	forward pass[b]; 512	0.554 (0.532) [0.537, 0.570]	0.487 [0.458, 0.511]	0.496	0.559 [0.544, 0.571]	0.500 [0.473, 0.530]
ESM2-15B	protein LM (UCE file); 5,120	0.902 (0.889) [0.893, 0.911]	0.870 [0.856, 0.884]	0.877	0.647 [0.638, 0.657]	0.491 [0.466, 0.523]
ESM2-650M	protein LM, mean-pooled; 1,280	0.861 (0.859) [0.851, 0.870]	0.826 [0.811, 0.843]	0.826	0.610 [0.600, 0.618]	0.492 [0.466, 0.530]
32 sequence descriptors	hand-computed; 32	0.916 (0.902) [0.908, 0.924]	0.869 [0.855, 0.883]	0.876	0.671 [0.660, 0.680]	0.497 [0.465, 0.538]
HPA co-expression	154 cell types, PCA; 128	0.617 (0.576) [0.604, 0.630]	0.615 [0.590, 0.640]	0.618	0.598 [0.588, 0.608]	0.518 [0.485, 0.560]
ESM2-15B minus scGPT (paired)		+0.155 [0.141, 0.168]	+0.186 [0.159, 0.212]	+0.088 [0.066, 0.111]	−0.022 [−0.032, −0.011]	−0.084 [−0.124, −0.042]
Geneformer V2-104M minus scGPT (paired)		+0.117 [0.104, 0.130]	+0.089 [0.066, 0.111]	+0.032 [0.014, 0.051]	−0.007 [−0.012, −0.002]	+0.005 [−0.021, 0.029]

C. Tier B: scGPT's residual stream (L_0–L_{11}) against three static representations on each extraction's observed genes

Extraction: task	Positives / comparison set	Residual stream: range over 12 layers	Best layer: AUROC [95% CI]	ESM2-15B: AUROC; minus layer; k_+/k_-	scGPT token table: AUROC; minus layer; k_+/k_-	Geneformer V2-104M: AUROC; minus layer; k_+/k_-
lung: Lambert (stratified)	579 / 4,647	0.638–0.668	L_1: 0.668 [0.652, 0.687]	0.880 0.212–0.241 12/0	0.751 0.083–0.113 12/0	0.825 0.157–0.187 12/0
lung: TRRUST	597 / 1,454	0.658–0.718	L_2: 0.718 [0.699, 0.737]	0.857 0.139–0.199 12/0	0.779 0.060–0.121 12/0	0.797 0.079–0.139 12/0

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Table S10 (continued)

lung: PPI	13,965 / 13,822	0.616–0.674	L_1: 0.674 [0.658, 0.690]	0.639 −0.036 to +0.022 1/5	0.661 −0.013 to +0.044 5/4	0.651 −0.023 to +0.035 4/4
lung: edge	5,131 / 45,609 (304 TFs)	0.515–0.542	L_1: 0.542 [0.494, 0.589]	0.498 −0.043 to −0.017 0/3	0.566 0.024–0.051 4/0	0.562 0.020–0.047 3/0
lung (ext.): Lambert (stratified)	618 / 4,536	0.631–0.675	L_1: 0.675 [0.654, 0.698]	0.881 0.206–0.250 12/0	0.732 0.056–0.100 12/0	0.833 0.158–0.201 12/0
lung (ext.): TRRUST	621 / 1,450	0.672–0.725	L_1: 0.725 [0.707, 0.745]	0.857 0.132–0.185 12/0	0.775 0.050–0.103 12/0	0.807 0.082–0.135 12/0
lung (ext.): PPI	13,714 / 13,561	0.633–0.678	L_2: 0.678 [0.660, 0.694]	0.643 −0.035 to +0.010 0/6	0.658 −0.020 to +0.026 5/4	0.652 −0.026 to +0.020 4/5
lung (ext.): edge	5,035 / 44,994 (311 TFs)	0.520–0.539	L_1: 0.539 [0.492, 0.592]	0.494 −0.045 to −0.025 0/3	0.562 0.023–0.042 4/0	0.569 0.029–0.049 3/0
kidney B: Lambert (stratified)	509 / 3,932	0.616–0.657	L_1: 0.657 [0.635, 0.680]	0.861 0.204–0.245 12/0	0.737 0.080–0.120 12/0	0.819 0.162–0.202 12/0
kidney B: TRRUST	548 / 1,188	0.637–0.711	L_1: 0.711 [0.689, 0.732]	0.843 0.132–0.206 12/0	0.754 0.043–0.117 12/0	0.791 0.080–0.154 12/0
kidney B: PPI	11,990 / 11,830	0.617–0.671	L_1: 0.671 [0.649, 0.688]	0.630 −0.041 to +0.012 0/6	0.659 −0.012 to +0.042 6/2	0.646 −0.025 to +0.028 4/4
kidney B: edge	3,569 / 31,590 (248 TFs)	0.522–0.553	L_0: 0.553 [0.500, 0.602]	0.500 −0.052 to −0.022 0/3	0.568 0.015–0.046 4/0	0.569 0.017–0.048 4/0
immune: Lambert (stratified)	111 / 1,227	0.616–0.746	L_2: 0.746 [0.677, 0.819]	0.933 0.187–0.317 12/0	0.811 0.065–0.195 12/0	0.815 0.069–0.199 11/0
immune: TRRUST	95 / 350	0.643–0.774	L_2: 0.774 [0.720, 0.824]	0.943 0.169–0.300 12/0	0.800 0.026–0.156 10/0	0.832 0.059–0.189 12/0
immune: PPI	1,185 / 1,155	0.673–0.742	L_2: 0.742 [0.701, 0.789]	0.693 −0.050 to +0.020 0/0	0.727 −0.015 to +0.054 4/0	0.720 −0.022 to +0.047 2/0
immune: edge	544 / 2,976 (36 TFs)	0.576–0.632	L_0: 0.632 [0.570, 0.706]	0.483 −0.149 to −0.093 0/8	0.650 0.018–0.075 0/0	0.634 0.002–0.058 0/0
kidney A: Lambert (stratified)	90 / 1,208	0.637–0.763	L_3: 0.763 [0.691, 0.830]	0.914 0.151–0.277 12/0	0.779 0.016–0.142 9/0	0.804 0.042–0.167 7/0
kidney A: TRRUST	91 / 288	0.707–0.828	L_2: 0.828 [0.758, 0.885]	0.940 0.112–0.233 12/0	0.831 0.003–0.124 7/0	0.839 0.011–0.131 6/0
kidney A: PPI	1,235 / 1,176	0.746–0.786	L_1: 0.786 [0.741, 0.831]	0.730 −0.055 to −0.016 0/2	0.785 −0.001 to +0.039 4/0	0.772 −0.014 to +0.025 0/0
kidney A: edge	201 / 1,329 (28 TFs)	0.618–0.664	L_0: 0.664 [0.570, 0.761]	0.529 −0.135 to −0.089 0/2	0.650 −0.014 to +0.032 0/0	0.636 −0.028 to +0.018 0/0

D. The 186- and 178-gene curated sets against matched random subsets of Tier A (TRRUST task), and a projection control

Representation	186 genes: AUROC	186: null mean [2.5%, 97.5%]	186: percentile; two-sided p	186: excess over null mean	178 genes: AUROC; percentile; excess	186, degree-matched: null mean; percentile; excess
scGPT (whole-human)	0.833	0.739 [0.661, 0.810]	0.9950 $p = 0.010$	+0.095	0.821; 0.9878 +0.085	0.770; 0.9787 +0.064
Geneformer V1-10M	0.802	0.646 [0.536, 0.743]	0.9996 $p = 1.0 \times 10^{-3}$	+0.156	0.797; 0.9992 +0.149	0.667; 0.9987 +0.135
Geneformer V2-104M	0.874	0.745 [0.655, 0.824]	0.9992 $p = 1.8 \times 10^{-3}$	+0.129	0.867; 0.9994 +0.122	0.771; 0.9996 +0.104
Geneformer V2-316M	0.848	0.715 [0.624, 0.795]	0.9996 $p = 1.0 \times 10^{-3}$	+0.133	0.834; 0.9989 +0.118	0.747; 0.9980 +0.101
tGPT	0.723[c]	0.587 [0.518, 0.669]	0.9996 $p = 1.0 \times 10^{-3}$	+0.136	0.723; 0.9997 +0.139	0.622; 0.9947 +0.101
AIDO.Cell-100M	0.769	0.648 [0.582, 0.716]	0.9997 $p = 8.0 \times 10^{-4}$	+0.121	0.759; 0.9990 +0.109	0.688; 0.9946 +0.082
scFoundation	0.549	0.300 [0.179, 0.434]	0.9994 $p = 1.4 \times 10^{-3}$	+0.250	0.533; 0.9972 +0.248	0.301; 0.9994 +0.249
ESM2-15B	0.954	0.862 [0.809, 0.911]	1.0000 $p = 2.0 \times 10^{-4}$	+0.092	0.949; 1.0000 +0.087	0.892; 0.9989 +0.062

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Table S10 (continued)

ESM2-650M	0.907	0.819 [0.763, 0.872]	0.9999 $p = 4.0 \times 10^{-4}$	+0.088	0.902; 0.9994 +0.083	0.847; 0.9942 +0.060
32 sequence descriptors	0.966	0.865 [0.811, 0.913]	1.0000 $p = 2.0 \times 10^{-4}$	+0.101	0.962; 1.0000 +0.097	0.894; 1.0000 +0.071
HPA co-expression	0.688	0.594 [0.535, 0.649]	0.9997 $p = 8.0 \times 10^{-4}$	+0.094	0.685; 0.9970 +0.075	0.624; 0.9892 +0.063
Kendall τ, order of the representations, curated set vs Tier A		186 genes (10 representations, without tGPT): 0.867 [0.689, 0.956]; 178 genes (11): 0.855 [0.673, 0.964]; 178 genes vs Tier A centred over its TRRUST-labelled genes: 0.964 (point estimate)				
Mean TRRUST degree, 186-gene set vs pool		TFs 16.6 vs 10.8; target-only genes 7.0 vs 3.2				

Projection control (Tier A): representation, task	d: full → projected	Full: AUROC [95% CI]	Projected: mean (SD) over 5 seeds	Projected: min–max	Projected minus full	scGPT token table, full 512-d
ESM2-15B: Lambert (unstratified)	5,120 → 1,024	0.889 [0.882, 0.897]	0.887 (0.007)	0.877–0.895	−0.002	0.704
ESM2-15B: TRRUST	5,120 → 1,024	0.870 [0.856, 0.884]	0.870 (0.007)	0.861–0.877	−0.0005	0.684
ESM2-15B: PPI	5,120 → 1,024	0.647 [0.638, 0.657]	0.648 (0.004)	0.641–0.650	+0.001	0.669
ESM2-15B: edge	5,120 → 1,024	0.491 [0.466, 0.523]	0.490 (0.002)	0.487–0.493	−0.001	0.575
scGPT (whole-human): Lambert (unstratified)	512 → 102	0.704 [0.694, 0.713]	0.678 (0.005)	0.672–0.685	−0.027	–
scGPT (whole-human): TRRUST	512 → 102	0.684 [0.660, 0.708]	0.635 (0.025)	0.601–0.667	−0.049	–
scGPT (whole-human): PPI	512 → 102	0.669 [0.658, 0.679]	0.634 (0.002)	0.631–0.636	−0.036	–
scGPT (whole-human): edge	512 → 102	0.575 [0.534, 0.610]	0.556 (0.012)	0.540–0.572	−0.019	–

Table S11: **A co-occurrence-only instrument: Word2Vec trained on the token sets scGPT receives, compared with scGPT's residual stream on H1–H3.** Word2Vec: skip-gram, 256 dimensions, 10 negative samples, trained on every cell of each extraction's population (lung 15,000, lung (ext.) 9,000, kidney B 8,000, immune 20,000, kidney A 11,376 cells), each tokenised as scGPT tokenises it (non-zero genes, the 512 most expressed), with token and cell order reshuffled on every pass. Every representation is mean-centred over the observed genes V and scored on identical genes, pairs, labels and resampling draws; scGPT is the whole-human checkpoint. H1b, AUROC of the cosine of STRING physical pairs (combined score ≥ 700) against one degree-preserving rewired non-pair set, with vertex-subsampling intervals ($m = n/2$, $B = 2{,}000$) under delete-d scaling; H2, detection-stratified AUROC of the LOO-cosine TF score (Lambert TFs against all other observed genes), within-stratum permutation null ($N = 2{,}000$; floor 5.0×10^{-4}) and gene-bootstrap intervals ($B = 2{,}000$, scores held fixed); H3, within-TF stratified AUROC of the TF–target cosine with two-way TF × target bootstrap intervals ($B = 2{,}000$); H1a, assortativity of SV_1–SV_{10} of the observed-gene SVD on the STRING network with a Maslov–Sneppen null ($N = 1{,}000$; max-T over 10 directions for Word2Vec and over the 120 direction–layer tests for scGPT). These p-values are not those of the primary H1/H2 tests. Differences are paired (Word2Vec minus scGPT on the same draws); the best layer is the scGPT layer with the highest value. Under the within-stratum null the LOO score is not centred at 0.5 (null means 0.27–0.50 for Word2Vec), so H2 is judged against its own null, and the null-centred difference subtracts each instrument's null mean (a descriptive adjustment). Seed variation is reported separately and is not propagated into the intervals. **B**: one seed; [a]the 10-epoch paired differences are in Panel A (mean of 5 seeds). **C**: the five-token window used for cost against a window spanning the whole cell (512), and the five-token window with its number of pair updates matched to the 512-token window (571 epochs for lung, 365 for immune). Pair updates, the number of (centre, context) updates in training. Per-seed values, every layer, the extraction-cells-only models for all extractions and all CKA values are in the CSV. For H1b, the vertex-subsampling intervals with the pre-registered scaling factor $\sqrt{m/n}$, which omits the finite-population term and so gives half-widths smaller by a factor $\sqrt{2}$, are also in the CSV (columns without the suffix `_deleted`, beside the reported `_deleted` columns). Full values: `Table_S11_cooccurrence.csv`.

A. Word2Vec trained on the full population corpus (five-token window, 10 epochs, mean of 5 seeds) against scGPT's residual stream, per extraction

Task and quantity	lung	lung (ext.)	kidney B	immune	kidney A
H1b: STRING pair-cosine AUROC against rewired non-pairs (delete-d vertex-subsampling intervals)					
Word2Vec [95% CI]	0.605 [0.589, 0.621]	0.597 [0.581, 0.613]	0.605 [0.586, 0.622]	0.704 [0.656, 0.751]	0.739 [0.690, 0.794]
scGPT, 12 layers (best layer)	0.619–0.680 (L_1)	0.630–0.677 (L_1)	0.621–0.675 (L_1)	0.668–0.743 (L_2)	0.749–0.814 (L_1)
Word2Vec minus scGPT at its best layer	−0.075 [−0.090, −0.059]	−0.080 [−0.097, −0.062]	−0.071 [−0.089, −0.052]	−0.040 [−0.082, 0.005]	−0.074 [−0.107, −0.042]
Word2Vec minus scGPT at L_{11}	−0.014 [−0.027, −0.000]	−0.032 [−0.049, −0.015]	−0.016 [−0.029, −0.002]	+0.036 [−0.005, 0.078]	−0.015 [−0.037, 0.008]
Layers with the difference's CI < 0 (scGPT higher) / > 0	12 / 0	12 / 0	12 / 0	0 / 0	8 / 0
H2: detection-stratified TF AUROC (LOO cosine; gene-bootstrap intervals)					
Word2Vec [95% CI]	0.632 [0.607, 0.656]	0.602 [0.577, 0.628]	0.601 [0.573, 0.629]	0.511 [0.454, 0.569]	0.469 [0.411, 0.529]
Word2Vec permutation p (null mean)	5.0×10^{-4} (0.494)	5.0×10^{-4} (0.497)	5.0×10^{-4} (0.494)	0.22 (0.485)	0.28 (0.444)

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Table S11 (continued)

scGPT, 12 layers (best layer)	0.625–0.665 (L_5)	0.627–0.662 (L_1)	0.615–0.654 (L_1)	0.566–0.720 (L_2)	0.678–0.783 (L_3)
Word2Vec minus scGPT at its best layer, raw	−0.033 [−0.060, −0.007]	−0.060 [−0.081, −0.038]	−0.053 [−0.078, −0.028]	−0.208 [−0.285, −0.124]	−0.314 [−0.384, −0.238]
Same, null-centred (each instrument's permutation-null mean subtracted)	−0.033 [−0.060, −0.007]	−0.061 [−0.082, −0.039]	−0.051 [−0.075, −0.025]	−0.204 [−0.281, −0.120]	−0.284 [−0.354, −0.208]
Layers with the null-centred CI < 0 (scGPT higher) / > 0	7 / 0	12 / 0	8 / 0	10 / 0	12 / 0
H3: within-TF stratified edge AUROC (two-way TF × target bootstrap intervals)					
Word2Vec [95% CI]	0.512 [0.481, 0.549]	0.534 [0.498, 0.570]	0.528 [0.476, 0.582]	0.603 [0.533, 0.696]	0.577 [0.470, 0.695]
scGPT, 12 layers (best layer)	0.516–0.542 (L_1)	0.518–0.538 (L_1)	0.524–0.554 (L_0)	0.577–0.638 (L_0)	0.601–0.652 (L_1)
Word2Vec minus scGPT at its best layer	−0.030 [−0.074, 0.011]	−0.005 [−0.051, 0.036]	−0.026 [−0.101, 0.029]	−0.035 [−0.100, 0.033]	−0.076 [−0.174, 0.021]
Layers with the difference's CI < 0 / > 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
H1a: assortativity of SV_1–SV_{10} *on the STRING network (descriptive; directions not index-matched)*					
Word2Vec: directions with max-T $p < 0.05$ in all 5 seeds (largest mean r)	10/10 (0.274)	10/10 (0.295)	10/10 (0.265)	10/10 (0.620)	9/10 (0.693)
scGPT: direction–layer tests with max-T $p < 0.05$ (largest r)	120/120 (0.435)	120/120 (0.407)	120/120 (0.404)	118/120 (0.566)	120/120 (0.702)
Largest seed SD of Word2Vec (H1b, H2, H3)	0.003	0.003	0.004	0.008	0.006

B. Saturation: full-corpus Word2Vec, seed 0, after 10 and 40 epochs

Quantity	lung, 10 epochs	lung, 40 epochs	immune, 10 epochs	immune, 40 epochs
Pair updates	4.5×10^8	1.8×10^9	2.9×10^8	1.2×10^9
H1b [95% CI, delete-d]	0.606 [0.589, 0.622]	0.612 [0.596, 0.628]	0.705 [0.657, 0.752]	0.702 [0.656, 0.748]
H2 [95% CI]	0.630 [0.605, 0.654]	0.664 [0.639, 0.687]	0.520 [0.461, 0.580]	0.515 [0.454, 0.577]
H2 permutation p (null mean)	5.0×10^{-4} (0.494)	5.0×10^{-4} (0.493)	0.16 (0.485)	0.19 (0.486)
H3 [95% CI]	0.512 [0.481, 0.547]	0.511 [0.476, 0.551]	0.608 [0.537, 0.699]	0.614 [0.546, 0.707]
H1a: largest r (directions with max-T $p < 0.05$)	0.277 (10/10)	0.282 (10/10)	0.620 (10/10)	0.616 (10/10)
Linear CKA between the 10- and 40-epoch models		0.944		0.945
H2, Word2Vec minus scGPT at its best layer, null-centred	[a]	+0.000 (L_5) [−0.024, 0.022]	[a]	−0.202 (L_2) [−0.283, −0.116]
H2, Word2Vec minus scGPT at L_{11}, null-centred	[a]	+0.039 [0.014, 0.063]	[a]	−0.049 [−0.119, 0.024]
H2, layers whose null-centred CI includes 0	[a]	11/12	[a]	3/12

C. Window calibration: Word2Vec trained on the extraction cells only (256 cells), seed 0

Quantity	lung: window 5, 10 epochs	lung: window 5, matched updates	lung: window 512, 10 epochs	immune: window 5, 10 epochs	immune: window 5, matched updates	immune: window 512, 10 epochs
Epochs; pair updates	10; 7.8×10^6	571; 4.5×10^8	10; 4.5×10^8	10; 3.6×10^6	365; 1.3×10^8	10; 1.3×10^8
H1b [95% CI, delete-d]	0.551 [0.535, 0.567]	0.571 [0.555, 0.586]	0.571 [0.555, 0.588]	0.616 [0.569, 0.663]	0.689 [0.640, 0.735]	0.656 [0.605, 0.707]
H2 (permutation p)	0.487 (0.64)	0.611 (5.0×10^{-4})	0.594 (5.0×10^{-4})	0.443 (0.29)	0.526 (0.12)	0.492 (0.43)
H3 [95% CI]	0.502 [0.476, 0.528]	0.516 [0.490, 0.545]	0.520 [0.495, 0.549]	0.512 [0.445, 0.598]	0.592 [0.523, 0.673]	0.569 [0.498, 0.661]
H1a: largest r (directions with max-T $p < 0.05$)	0.217 (7/10)	0.193 (9/10)	0.159 (10/10)	0.439 (7/10)	0.555 (9/10)	0.464 (10/10)
Linear CKA with the window-512 model	0.262	0.795	–	0.401	0.830	–

Table S12: **The 183 hypothesis records of the automated screen (iterations 1–63).** Each record is one hypothesis of the two-agent screen as archived in the iteration's JSON file, parsed together with the iteration's code. The first 63 iterations produced these 183 records, whose headline findings Kendiukhov (2026b) (arXiv:2602.22247) reported. *Panel A*, per JSON family (the 8 families the records use; Kendiukhov (2026b) regroups them into 13): n, records; spec., records involving a spectral decomposition; extr., records whose result string yields a primary p-value; = 0, of these, p stored as exactly 0 (permutation tests with $N = 50$–1,000 draws); Bonf. rec., records passing Bonferroni at 0.05/183 with the recorded p ($p \leq 0.05/183$, zeros kept); Bonf. fl., passing with the floored p ($p < 0.05/183$), where a zero is replaced by its resolution floor $1/(N+1)$, with N from the record's method text where stated and 1,000 otherwise; BH, passing Benjamini–Hochberg at $q = 0.05$ over the 112 extractable floored p-values; kid. A, records using the kidney A matrix (a Tabula Sapiens kidney extraction, 4,803 genes, described as immune in Kendiukhov (2026b)), with the number whose gene labels were misaligned in parentheses (mis.; iterations 27 and 29–39, where blank lines of the gene list were dropped so that all 209 labels sat on rows 0–208); imm., records using the immune extraction (one record uses both); other, neither (one Gaussian null matrix, one Geneformer token-embedding matrix); directions and bands, the five most frequent spectral directions or bands named in the family's spectral records (SV24 and SV57 are the records' own labels for SV_2–SV_4 and SV_5–SV_7; PC, principal component). The last two rows repeat the audit over all 409 records of the loop (137 JSON files; 3 records belong to an aborted duplicate run), with the Bonferroni denominator equal to the number of extractable p-values; this is a sensitivity analysis. *Panel B*, every record in iteration order: It., iteration; ID, hypothesis identifier within the iteration (⋆, record keyed to one of the 11 headline findings in the multiplicity audit); Fam., JSON family code (Panel A); Matrix, embedding matrix loaded (3 draws, the main extraction and two further 256-cell draws; lung attention, scGPT attention scores from a lung extraction; ^{m}, misaligned gene labels); Directions recorded, spectral directions or bands named in the record (—, none); p recorded, primary p as extracted from the result string (—, none extractable); p floored, after replacing zeros by $1/(N+1)$; Passes: R, Bonferroni at 0.05/183 with the recorded p; F, Bonferroni at 0.05/183 with the floored p; H, Benjamini–Hochberg at $q = 0.05$ with the floored p; Decision, the screen's own decision for the record. Of the 11 starred records, 8 carry R and 3 carry F; the headline-level count after flooring (5 of the 11) scores 5 of the headline findings by a p-value recomputed for the claim itself instead of the record's primary p shown here (for the transcription-factor classifier, a Stouffer combination of its 12 per-layer p-values, which is invalid because the layers share genes), and 2 of these pass; replacing the t-approximation $p = 1.4 \times 10^{-24}$ of the STRING confidence gradient by its exact value, 0.0167, and setting aside the Stouffer combination leaves 3 (Table S13, C23 and C28). Titles are truncated to about 55 characters. No intervals are reported: p-values are as computed by the screen and are displayed to three significant digits. Because each iteration's hypotheses were chosen after seeing earlier results, no correction over the recorded family bounds the error rate of the screen as a whole. The file also gives, for every record, its gene set and gene-row mapping, decomposition convention, reference sets, script provenance and the claims of Table S13 it supports, and the same fields for the 226 later records. Full values: `Table_S12_screen_records.csv`.

A. Records per JSON family and multiple-testing passes

			p-values		passes			matrix used			
JSON family (code)	n	spec.	extr.	= 0	Bonf. rec.	Bonf. fl.	BH	kid. A (mis.)	imm.	other	directions and bands involved (records)
`module_structure` (MOD)	52	37	33	5	12	7	21	35 (7)	17	0	SV_2–SV_4 (16), SV_5–SV_7 (16), SV_2 (7), SV_1 (6), SV_2–SV_7 (4), …
`manifold_distance` (MAN)	42	11	29	4	10	6	19	22 (10)	20	0	SV_5–SV_7 (9), SV_2–SV_4 (3), SV_5 (2), SV_7 (2), SV_2 (1), …
`intrinsic_dimensionality` (IDM)	41	32	24	4	10	6	16	20 (6)	21	0	SV_1 (13), SV_2 (6), SV_1/SV_2 (5), PC1 (4), SV_2/SV_3 (2), …
`graph_topology` (GRT)	16	11	9	1	3	2	4	14 (3)	2	0	SV_2 (3), SV_3 (2), SV_2+SV_3+SV_4 (2), SV_1 (1), SV_4 (1)
`null_sensitivity` (NUL)	15	6	7	2	3	1	5	13 (7)	1	1	SV_2 (3), SV_3/SV_4 (1), SV_3 (1), SV_4 (1), PC1 (1)
`topology_stability` (TST)	8	6	6	2	2	0	2	6 (3)	2	0	SV_1 (2), SV_5–SV_7 (2), SV57 (2), SV_2–SV_4 (2), SV_2/SV_3 (1), …
`cross_model_alignment` (CMA)	5	3	0	0	0	0	0	3 (0)	2	1	SV_5–SV_7 (1)
`persistent_homology` (PHO)	4	2	4	0	3	3	3	2 (0)	2	0	SV_2–SV_4 (2)
All 183 records (iterations 1–63)	183	108	112	18	43	25	70	115 (36)	67	2	Bonferroni threshold $0.05/183 = 2.73\times10^{-4}$
All 409 records (iterations 1–138)	409	—	130	18	—	28	79	—	—	—	Bonferroni at 0.05/130
excl. aborted duplicate run	406	—	127	18	—	27	76	—	—	—	Bonferroni at 0.05/127

B. The 183 records

It.	ID	Hypothesis (short title)	Fam.	Matrix	Directions recorded	p recorded	p floored	Passes	Decision
2	H01_graph_topology_cc	kNN Clustering Coefficient vs Feature-Shuffle Null	GRT	kidney A (3 draws)	—	—	—		promising
2	H02_graph_topology_tr	kNN Transitivity vs Feature-Shuffle Null	GRT	kidney A (3 draws)	—	—	—		negative
4	H01	TwoNN intrinsic dimensionality per layer	IDM	kidney A	—	—	—		promising
4	H02	TRRUST co-target distance clustering (random-group…	MOD	kidney A	—	—	—		inconclusive
4	H03	Cross-layer linear CKA alignment	CMA	kidney A	—	—	—		promising
5	H01	GO term embedding clustering across layers	MOD	kidney A	—	—	—		inconclusive

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Table S12 (continued)

It.	ID	Hypothesis (short title)	Fam.	Matrix	Directions recorded	p recorded	p floored	Passes	Decision
5	H02	Per-gene residual drift + GO enrichment	MOD	kidney A	—	0.0044	0.0044	H	promising
5	H03	Effective rank per layer (spectral cross-check vs. . .	IDM	kidney A	—	0.002	0.002	H	promising
6	H01	Full-vocabulary residual drift + GO enrichment	MOD	kidney A	—	0.0078	0.0078	H	promising
6	H02⋆	SVD biology of near-rank-1 layer-11 subspace	IDM	kidney A	SV_1, SV_1/SV_2	3×10^{-4}	3×10^{-4}	H	promising
6	H03	Effective rank curvature and compression breakpoint	IDM	kidney A	—	0.0021	0.0021	H	promising
7	H01	SV_1 feature-shuffle and gene-label-shuffle null	IDM	kidney A	SV_1	0.004	0.004	H	promising
7	H02	Layer-wise SVD trajectory (12 layers)	IDM	kidney A	SV_1/SV_2, SV_1	—	—		promising
7	H03	Drift gene-label-shuffle null + SV_1 negative-pole GO. . .	MOD	kidney A	SV_1	0.124	0.124		neutral
8	H01	SV_2/SV_3 axes GO enrichment + gene-label null at. . .	IDM	kidney A	SV_2/SV_3, SV_2, SV_3	0	9.99×10^{-4}	R H	promising
8	H02⋆	Layer-specific compartment transients: mitochondrion. . .	IDM	kidney A	SV_1	0	9.99×10^{-4}	R H	promising
8	H03	Signal peptide proxy vs SV_1 top-pole (layer-11)	MOD	kidney A	SV_1	0.002	0.002	H	promising
9	H01	SV_2 bottom-pole extracellular vesicle null (N=1000)	IDM	kidney A	SV_2	0	9.99×10^{-4}	R H	promising
9	H02	12-layer x 8-compartment systematic SV_1 scan	MOD	kidney A	SV_1	—	—		promising
9	H03	Cross-layer SV_1 rank stability (Spearman)	TST	kidney A	SV_1	0	9.99×10^{-4}	R H	promising
10	H01	SV_2/SV_3 12-layer × 9-compartment scan	IDM	kidney A	SV_2/SV_3, SV_2, SV_3	0.05	0.05		promising
10	H02⋆	TRRUST TF-target co-pole test in SVD geometry	MOD	kidney A	SV_1, SV_2	0	9.99×10^{-4}	R H	promising
10	H03	$SV_1/SV_2/SV_3$ spectral ratio profile + annotation. . .	IDM	kidney A	$SV_1/SV_2/SV_3$, SV_1/SV_2, SV_1, SV_2	—	—		promising
11	H01⋆	TRRUST co-pole x 12 layers + Activation/Repression. . .	MOD	kidney A	SV_2	0	9.99×10^{-4}	R H	promising
11	H02	SV_2 mito pole-flip gene tracking at L_3-to-L_4 transition	IDM	kidney A	SV_2	—	—		neutral
11	H03	GO Biological Process enrichment in SV_2 poles (broad. . .	MOD	kidney A	SV_2	1	1		negative
12	H01	Bootstrap CI on activation vs repression co-pole. . .	MOD	kidney A	—	0.05	0.05		negative
12	H02	Mean pairwise SV_2 distance — spatial concentration. . .	MAN	kidney A	SV_2	0.05	0.05		promising
12	H03	STRING PPI co-pole enrichment across 12 layers	MOD	kidney A	SV_2	0.05	0.05		promising
13	H01	SV axis specificity: STRING PPI co-pole on SV_1 vs SV_2. . .	GRT	kidney A	SV_1, SV_2, SV_3	—	—		promising
13	H02	Repression anti-pole: TRRUST repression pairs occupy. . .	MOD	kidney A	SV_2	—	—		negative
13	H03	Hub-degree control: does PPI co-pole signal survive in. . .	NUL	kidney A	SV_2	—	—		inconclusive
14	H01	GO enrichment of SV_3 poles across 12 layers	MOD	kidney A	SV_3	—	—		promising
14	H02	Layer-depth Spearman trend on SV_2/SV_3 PPI z-scores	TST	kidney A	SV_2/SV_3, $SV_1/SV_2/SV_3$, SV_2, SV_3, SV_1	0.059	0.059		neutral
14	H03	Expanded hub-degree control: non-hub SV_2 co-pole. . .	NUL	kidney A	SV_2	—	—		promising
15	H01	SV_4/SV_5 PPI co-pole test	MOD	kidney A	SV_4/SV_5, SV_4, SV_5	—	—		promising
15	H02⋆	STRING confidence gradient: co-pole z vs score quintile	GRT	kidney A	SV_2	1.4×10^{-24}	1.4×10^{-24}	R F H	promising
15	H03	GO co-annotation vs PPI as driver of SV_2 geometry	MOD	kidney A	SV_2	—	—		promising
16	H01	SV_3 STRING confidence gradient	GRT	kidney A	SV_3	0.037	0.037		promising
16	H02	$SV_2/SV_3/SV_4$ axis independence test	MOD	kidney A	$SV_2/SV_3/SV_4$, SV_2, SV_3, SV_4, SV_2–SV_3, SV_2–SV_4, SV_3–SV_4	—	—		promising
16	H03	SV_4 GO biology profile	MOD	kidney A	SV_4	8.2×10^{-4}	8.2×10^{-4}	H	promising
17	H01	SV_4 confidence quintile gradient	GRT	kidney A	SV_4	0.0374	0.0374		promising

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Table S12 (continued)

It.	ID	Hypothesis (short title)	Fam.	Matrix	Directions recorded	p recorded	p floored	Passes	Decision
17	H02	Axis-dominant pair GO composition (biological...	MOD	kidney A	$SV_2/SV_3/SV_4$, SV_2–SV_3, SV_2–SV_4, SV_3–SV_4	—	—		promising
17	H03	TRRUST signed regulation in SV_3/SV_4	NUL	kidney A	SV_3/SV_4, SV_3, SV_4	—	—		negative
18	H01	Random Gaussian embedding null control for SV_2 co-pole	NUL	Gaussian null	SV_2	—	—		negative
18	H02	Out-of-sample PPI precision@k prediction benchmark	GRT	kidney A	SV_2	—	—		neutral
18	H03	scGPT attention co-occurrence geometry for...	GRT	kidney A + lung attention	—	0.87	0.87		promising
19	H01	Geneformer static token embedding cross-model alignment	CMA	kidney A + Geneformer	—	—	—		promising
19	H02	Multi-axis composite co-pole P@k enrichment	GRT	kidney A	SV_2+SV_3+SV_4	—	—		promising
19	H03	TRRUST repression vs activation in scGPT attention	NUL	kidney A + lung attention	—	9.4×10^{-9}	9.4×10^{-9}	R F H	promising
20	H01	Shuffle null for multi-axis composite co-pole...	NUL	kidney A	—	0	9.99×10^{-4}	R H	promising
20	H02	Attention-SVD joint ROC predictor for STRING/TRRUST	GRT	kidney A + lung attention	SV_2+SV_3+SV_4	—	—		neutral
20	H03	Persistent homology H0 and pairwise distance geometry	PHO	kidney A	—	1×10^{-72}	1×10^{-72}	R F H	promising
21	H01	STRING confidence quintile x pairwise distance gradient	MAN	kidney A	—	0.05	0.05		inconclusive
21	H02	TRRUST pairs distance — specificity control	MAN	kidney A	—	—	—		promising
21	H03	H1 persistence trajectory + bootstrap CIs for...	PHO	kidney A	—	1×10^{-4}	1×10^{-4}	R F H	promising
22	H01	STRING continuous AUROC + TRRUST-exclusive proximity...	MAN	kidney A	—	1×10^{-5}	1×10^{-5}	R F H	promising
22	H02*	Cell-type marker gene cluster separation	MOD	kidney A	—	1×10^{-6}	1×10^{-6}	R F H	promising
22	H03	Per-layer bootstrap CIs for co-polarity enrichment +...	TST	kidney A	SV_2+SV_3+SV_4	0.16	0.16		promising
23	H01*	Cell-type marker separation — expanded panel +...	MOD	kidney A	—	0.004	0.004	H	promising
23	H02	GO Biological Process proximity — 3rd independent...	MOD	kidney A	—	1×10^{-19}	1×10^{-19}	R F H	promising
23	H03	TRRUST Activation vs Repression proximity asymmetry	MAN	kidney A	—	0.002	0.002	H	promising
24	H01	Dorothea confidence stratification	MAN	kidney A	—	0	9.99×10^{-4}	R H	promising
24	H02	TF activation hub centrality	MOD	kidney A	—	0.41	0.41		negative
24	H03	GO ontology comparison BP vs MF vs CC	MOD	kidney A	—	9×10^{-56}	9×10^{-56}	R F H	promising
25	H01	Multi-predictor joint model (STRING + Dorothea + GO CC...	MAN	kidney A	—	—	—		promising
25	H02	Layer-resolved encoding timeline across 5 biological...	MAN	kidney A	—	—	—		promising
25	H03	Chromosomal proximity negative control	NUL	kidney A	—	0.096	0.096		negative
26	H01	Immune gene family clustering AUROC	MOD	kidney A	—	—	—		promising
26	H02	Dorothea confidence-tier proximity gradient	MAN	kidney A	—	—	—		promising
26	H03	Intrinsic dimensionality progression (participation...	IDM	kidney A	—	1×10^{-8}	1×10^{-8}	R F H	promising
27	H01	PC1 Biological Identity at L_{11}	IDM	kidney A^m	PC1	0.42	0.42		negative
27	H02	TRRUST Activator/Repressor Geometric Polarity	MOD	kidney A^m	PC1	0.178	0.178		negative
27	H03	Dorothea Threshold Sensitivity + PR Elbow Alignment	NUL	kidney A^m	—	—	—		negative
28	H01	Hub Gene Embedding Centrality	MAN	kidney A	—	0.00135	0.00135	H	promising
28	H02	PC1 T-cell vs APC Cell-type Axis	IDM	kidney A	PC1	0.12	0.12		negative
28	H03	kNN Graph Spectral Gap Across Layers	GRT	kidney A	—	1.3×10^{-10}	1.3×10^{-10}	R F H	promising
29	H01	kNN Connected Components Biological Enrichment at L_{11}	GRT	kidney A^m	—	0.023	0.023	H	promising

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Table S12 (continued)

It.	ID	Hypothesis (short title)	Fam.	Matrix	Directions recorded	p recorded	p floored	Passes	Decision
29	H02	Gene Trajectory Clustering by Distance-to-Centroid...	MAN	kidney A^m	—	0.352	0.352		promising
29	H03	Spectral Gap k-Robustness Sweep (k=5–30)	TST	kidney A^m	—	—	—		promising
30	H01	GO and TRRUST enrichment of 14 diverging genes	MOD	kidney A^m	—	2.34×10^{-11}	2.34×10^{-11}	R F H	positive
30	H02	Layer-of-bifurcation anatomy (inter/intra ratio)	TST	kidney A^m	—	0	9.99×10^{-4}	R H	negative
30	H03	Trajectory slope vs STRING degree (continuous)	GRT	kidney A^m	—	0.683	0.683		negative
30	H04	OOV gene discovery in scGPT named gene set	NUL	kidney A^m	—	—	—		promising
31	H01	kNN Spectral Gap Monotonic Decrease (195 in-vocab...	GRT	kidney A^m	—	0	9.99×10^{-4}	R H	promising
31	H02	STRING Score → Embedding Distance (195 in-vocab genes...	MAN	kidney A^m	—	—	—		negative
31	H03	Participation Ratio (Intrinsic Dim) Collapse Across...	IDM	kidney A^m	—	0	9.99×10^{-4}	R H	promising
32	H01	Community Detection at L_{11} → Biological Annotation	MOD	kidney A^m	—	0.09	0.09		promising
32	H02	Dorothea Regulatory Pairs → Embedding Distance Decay...	MAN	kidney A^m	—	4×10^{-4}	4×10^{-4}	H	promising
32	H03	PC1 at L_{11} Separates TFs from B-cell Markers...	IDM	kidney A^m	PC1	0.041	0.041		inconclusive
33	H01	L_{11} Community Differential Analysis (B-cell vs...	MOD	kidney A^m	PC1	1×10^{-9}	1×10^{-9}	R F H	promising
33	H02	Community Partition Stability: k-sweep and Layer Sweep	TST	kidney A^m	—	0.167	0.167		promising
33	H03	PC1 Polarity Expanded Gene Sets — B-cell Stable Across...	IDM	kidney A^m	PC1	0.001	0.001	H	promising
34	H01	Layer-wise B-cell community purity emergence	MOD	kidney A^m	PC1	0.804	0.804		inconclusive
34	H02	PC2/PC3 cell-type axes at L_{11}	IDM	kidney A^m	PC2/PC3, PC2, PC3	0.168	0.168		negative
34	H03	Expression-level confound test for B-cell PC1 signal	NUL	kidney A^m	PC1	0	9.99×10^{-4}	R H	promising
35	H01	TRRUST co-regulation + GO co-annotation vs embedding...	MOD	kidney A^m	—	1×10^{-4}	1×10^{-4}	R F H	negative
35	H02	B-cell marker kNN precision@10 profile across layers	MAN	kidney A^m	—	0	9.99×10^{-4}	R H	promising
35	H03	B-cell vs non-B-cell centroid separation trajectory...	IDM	kidney A^m	—	0.088	0.088		promising
36	H01	Multi-cell-type kNN precision@10 specificity panel	MAN	kidney A^m	—	—	—		promising
36	H02	Gene-name permutation null for B-cell kNN precision	NUL	kidney A^m	—	0.005	0.005	H	promising
36	H03	B-cell neighborhood functional characterization	MOD	kidney A^m	—	—	—		promising
37	H01	Extended cell-type panel kNN precision@10 + centroid...	MAN	kidney A^m	—	0	9.99×10^{-4}	R H	promising
37	H02	Held-out B-cell marker generalization (germinal center...	MAN	kidney A^m	—	—	—		promising
37	H03	T-cell permutation null + B-cell neighborhood TF...	NUL	kidney A^m	—	0.96	0.96		promising
38	H01	GC-TF vs Plasma-TF centroid proximity divergence...	MAN	kidney A^m	—	0.024	0.024	H	promising
38	H02	B-cell centroid directional drift alignment with...	MAN	kidney A^m	—	1×10^{-4}	1×10^{-4}	R F H	inconclusive
38	H03	Master B-cell TF proximity + NK/myeloid specificity...	NUL	kidney A^m	—	—	—		inconclusive
39	H01	Drift target identification — B-cell centroid at L_{11}	MAN	kidney A^m	—	—	—		promising
39	H02	Extended plasma panel + BCL6/PAX5 GC screen...	MAN	immune	—	—	—		mixed
39	H03	LOO ablation of B-cell anchor genes on GC-TF proximity...	NUL	kidney A^m	—	—	—		promising
40	H01	Full 12-layer GC attractor onset scan + IRF4...	MAN	immune	—	2×10^{-6}	2×10^{-6}	R F H	promising
40	H02	Geneformer cross-model B-cell precision@10 (input...	CMA	Geneformer	—	—	—		negative
40	H03	GC-plasma subspace principal angle + CD19+BLK minimal...	MAN	immune	—	1×10^{-4}	1×10^{-4}	R F H	promising
41	H01	Multi-lineage attractor screen (T-cell + myeloid)	MAN	immune	—	1×10^{-6}	1×10^{-6}	R F H	promising

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Table S12 (continued)

It.	ID	Hypothesis (short title)	Fam.	Matrix	Directions recorded	p recorded	p floored	Passes	Decision
41	H02	BCL6 and PAX5 neighborhood characterization	MAN	immune	—	—	—		promising
41	H03	TwoNN intrinsic dimensionality by layer and lineage	IDM	immune	—	1×10^{-4}	1×10^{-4}	R F H	promising
42	H01	TwoNN change-point analysis at L_3 (B-cell ID…	IDM	immune	—	0.386	0.386		promising
42	H02	Metabolic isolation generalization — BCL6 specificity…	MAN	immune	—	—	—		promising
42	H03⋆	BATF/BACH2 convergence trajectory toward B-cell context	MAN	immune	—	1×10^{-4}	1×10^{-4}	R F H	promising
43	H01	GC repression circuit PRDM1/IRF4 anti-convergence test	MAN	immune	—	—	—		negative
43	H02	BCL6 divergence from B-cell centroid	MAN	immune	—	0	9.99×10^{-4}	R H	negative
43	H03	T-cell subtype ID stratification at L_7 breakpoint	IDM	immune	—	—	—		promising
44	H01	Cross-lineage ID compression law	IDM	immune	—	—	—		negative
44	H02	TCR activation circuit attractor test	MAN	immune	—	0.275	0.275		inconclusive
44	H03	BCL6 rank B-cell vs metabolic co-movement	MAN	immune	—	0.095	0.095		negative
45	H01	kNN lineage community purity at L_2 vs L_{11}	GRT	immune	—	0.958	0.958		negative
45	H02	Full-manifold TwoNN intrinsic dimensionality across 12…	IDM	immune	—	—	—		promising
45	H03	TCR signaling circuit convergence to T-cell identity…	MAN	immune	—	0.32	0.32		inconclusive
46	H01	Cross-seed ID compression replication + bootstrap CI	IDM	immune (3 draws)	—	—	—		promising
46	H02	Singular value spectrum concentration across layers	IDM	immune	—	—	—		promising
46	H03	Lineage centroid orthogonality vs null	GRT	immune	—	—	—		negative
47	H01	Zero-norm validity check: TwoNN and SVD on…	IDM	immune	—	—	—		promising
47	H02	SV_1 identity: what does the dominant direction encode?	IDM	immune	SV_1, SV_1/SV_2, SV_2	—	—		negative
47	H03	ID compression breakpoint detection across layers	IDM	immune	—	—	—		negative
48	H01	Gene annotation via edge dataset + SV_1/SV_2 identity	IDM	immune	SV_1/SV_2	—	—		neutral
48	H02	Full spectral decay curve across 12 layers	IDM	immune	SV_1	—	—		promising
48	H03	Wasserstein/transport distance between consecutive…	MAN	immune	—	0.15	0.15		inconclusive
49	H01	SV_1 gene loading direction stability across layers	IDM	immune	SV_1	—	—		promising
49	H02	SWD vs feature-shuffle null comparison	NUL	immune	—	0.006	0.006	H	promising
49	H03	SV_2–SV_4 subspace encodes TRRUST TF-target regulatory…	MOD	immune	SV_2–SV_4, SV_1, SV_1–SV_4	1×10^{-4}	1×10^{-4}	R F H	promising
50	H01	SV_2–SV_4 co-expression specificity control	MOD	immune	SV_2–SV_4	—	—		negative
50	H02	SV_1 identity: expression correlation across layers	IDM	immune	SV_1	—	—		promising
50	H03	SV_5–SV_{10} spectral axis biological screen (TF/target…	MOD	immune	SV_5–SV_{10}, SV_1, SV_2–SV_4, SV_5–SV_7, SV_8–SV_{10}	0.05	0.05		promising
51	H01	Residual TF signal after co-expression regression in…	MOD	immune	SV_2–SV_4	0.0016	0.0016	H	negative
51	H02	SV_1-high genes are depleted of TFs at both L_0 and L_{11}	IDM	immune	SV_1	1×10^{-4}	1×10^{-4}	R F H	promising
51	H03	0-dim PH cluster topology for circuit genes in SV_2–SV_4…	PHO	immune	SV_2–SV_4	1×10^{-6}	1×10^{-6}	R F H	inconclusive
52	H01	Housekeeping gene enrichment in SV_1-high	IDM	immune	SV_1	—	—		inconclusive
52	H02	H1 Betti loops on circuit genes at L_8	PHO	immune	SV_2–SV_4	0.953	0.953		negative
52	H03	SV_5–SV_7 regulatory signal after co-expression…	MOD	immune	SV_5–SV_7, SV_2–SV_4	0.002	0.002	H	promising

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Table S12 (continued)

It.	ID	Hypothesis (short title)	Fam.	Matrix	Directions recorded	p recorded	p floored	Passes	Decision
53	H01	Bootstrap + spectral decay scan of SV regulatory…	MOD	immune	SV_2–SV_4, SV_5–SV_7, SV_8–SV_{10}, SV_{11}–SV_{14}, SV_{15}–SV_{20}, SV_{21}–SV_{30}, SV_8–SV_{14}	—	—		promising
53	H02	TF→Target directional asymmetry in SV_5–SV_7	MAN	immune	SV_5–SV_7, SV_5, SV_6, SV_7	8×10^{-4}	8×10^{-4}	H	promising
53	H03	SV_1 loading vs TRRUST network degree	IDM	immune	SV_1	0.001	0.001	H	neutral
54	H01	LR classifier: TF vs target-only from SV_5–SV_7 L_0 coords	MOD	immune	SV_5–SV_7, SV_2–SV_4	0	0.0099	R H	promising
54	H02	BFS hierarchy depth vs SV_5–SV_7 dominant axis (L_0)	MAN	immune	SV_5–SV_7, SV_5, SV_7	0.0044	0.0044	H	neutral
54	H03	Layer-resolved TF→target directionality trajectory…	MAN	immune	SV_5–SV_7	0.05	0.05		promising
55	H01	Cross-seed replication: TF vs target-only AUROC…	MOD	immune (3 draws)	SV_5–SV_7	0.0099	0.0099	H	promising
55	H02	Layer-resolved AUROC: SV_5–SV_7 vs SV_2–SV_4 across all 12…	MOD	immune	SV_5–SV_7, SV_2–SV_4	—	—		promising
55	H03	Directionality trajectory SV_2–SV_4 vs SV_5–SV_7 (opposite…	MAN	immune	SV_2–SV_4, SV_5–SV_7	0.001	0.001	H	promising
56	H01⋆	Joint SV_2–SV_7 (6D) Layer-Stable TF/Target Classifier	MOD	immune	SV_2–SV_7, SV_2–SV_4, SV_5–SV_7, SV24, SV57	0	0.0099	R H	promising
56	H02	L_9 Directionality Crossover Cross-Seed Replication +…	MAN	immune (3 draws)	SV_2–SV_4, SV_5–SV_7, SV57, SV24	—	—		inconclusive
56	H03	SV_5–SV_7 Subspace Principal Angle vs AUROC Trajectory	TST	immune	SV_5–SV_7, SV57, SV_2–SV_4	0.417	0.417		negative
57	H01⋆	Joint 6D (SV_2–SV_7) TF/target classifier cross-seed…	MOD	immune (3 draws)	SV_2–SV_7, SV_2–SV_4, SV_5–SV_7	0	0.0196	R H	promising
57	H02	Seed43 SV basis alignment via principal angles	TST	immune (3 draws)	SV_2–SV_4, SV_5–SV_7, SV57	—	—		negative
57	H03	SV energy fraction as AUROC proxy	IDM	immune	SV_1, SV_2–SV_4, SV_5–SV_7, SV_2–SV_7	1×10^{-5}	1×10^{-5}	R F H	neutral
58	H01	TF boundary gene identity and family enrichment at…	MOD	immune (3 draws)	SV_2–SV_7	—	—		promising
58	H02	TRRUST graph Laplacian spectral alignment with SV_5–SV_7	MOD	immune	SV_5–SV_7, SV_2–SV_4	—	—		neutral
58	H03	Effective rank as cross-seed AUROC predictor	IDM	immune (3 draws)	SV_2–SV_7	1×10^{-4}	1×10^{-4}	R F H	promising
59	H01	Cross-seed TF boundary anchor stability + out-degree…	MOD	kidney A (3 draws)	SV_5–SV_7	0.9	0.9		neutral
59	H02	Pairwise TF-target 6D distance as edge probability…	MAN	kidney A	SV_5–SV_7	0.002	0.002	H	negative
59	H03	Partial Spearman: effective rank vs AUROC controlling…	IDM	immune (3 draws)	SV_5–SV_7	0.795	0.795		negative
60	H01	Signed displacement projection AUROC (directional…	MAN	kidney A	SV_5–SV_7	0.001	0.001	H	negative
60	H02	TF family-stratified margin trajectory across 12 layers	MOD	kidney A	SV_5–SV_7	—	—		neutral
60	H03	Feed-forward loop triangle geometry in SV_5–SV_7	MOD	kidney A	SV_5–SV_7	0.002	0.002	H	inconclusive
61	H01	Layer-to-layer CKA trajectory (L_8 boundary test)	CMA	immune (3 draws)	—	—	—		negative
61	H02	Dual-role gene (TF intersect target) margin trajectory	MOD	immune (3 draws)	SV_5–SV_7	0.084	0.084		neutral
61	H03	FFL geometry expanded (N=264) with multi-subspace and…	MOD	immune (3 draws)	SV_2–SV_4, SV_5–SV_7, SV_2–SV_7	0.297	0.297		negative
62	H01⋆	cycle4_immune edge-level AUROC (735 pairs…	MAN	immune (3 draws)	SV_5–SV_7/SV_2–SV_4, SV_5–SV_7, SV_2–SV_4	0.045	0.045		promising
62	H02	AUROC mechanism diagnostic: sparsity vs depth as…	IDM	immune	—	9.5×10^{-7}	9.5×10^{-7}	R F H	promising
62	H03	Cross-cycle Procrustes alignment transfer (cycle1 to…	CMA	kidney A + immune	SV_5–SV_7	—	—		negative
63	H63-A	SV_2–SV_4 repulsion mechanistic probe: TF class separation	MOD	immune	SV_2–SV_4, SV_5–SV_7	—	—		promising

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Table S12 (continued)

It.	ID	Hypothesis (short title)	Fam.	Matrix	Directions recorded	p recorded	p floored	Passes	Decision
63	H63-G	Signed regulation AUROC split: Activation vs. . .	MOD	immune	SV_5–SV_7, SV_2–SV_4	—	—		promising
63	H63-B	Systematic subspace scan SV(k,k+2) for k=0..9 at L_0. . .	MAN	immune	SV_k–SV_{k+2}, SV_8–SV_{10}, SV_5–SV_7	0.28	0.28		inconclusive

Table S13: **Claims ledger: the 61 numerical claims reported from the screen, traced and recomputed.** One row per numerical claim of Kendiukhov (2026b) (abstract, results, methods and tables), numbered C01–C61. ID, claim identifier (the file of Table S12 lists, for each record, the claims it supports); Claim, what was claimed; Source records, iteration/hypothesis of the record(s) behind the claim, with the matrix the record used (kidney A, the Tabula Sapiens kidney extraction with 4,803 genes, described as immune in Kendiukhov (2026b); immune, the immune extraction; 3 draws, the main extraction and two further 256-cell draws; earlier loop, iterations of an earlier run of the loop whose folders were overwritten) and its decomposition: 209-gene subset (22 zero rows), decomposition fitted to the 209 named genes at their true rows, 22 of which are never-observed all-zero rows; misaligned rows 0–208, the same with every label on the wrong row (blank lines of the gene list dropped before numbering); observed rows, only genes observed in at least one sampled cell, mean-centred over them (the convention of this paper); full matrix (zero rows), all rows including the never-observed ones, whose first direction is then the observation indicator. As reported, the value in Kendiukhov (2026b). Screen's procedure, recomputed, the value recomputed here with the screen's own code conventions on the matrix and gene rows the record used (brackets in C02: minimum–maximum over 50 random subsamples). Status: *reproduces*, exactly or within resampling error (25 claims); *partly* (2); *does not reproduce* (5); *provenance only*, traced to its record and code but not recomputed (27); *not recomputable*, records not preserved (2). Observed-gene value; this paper: the same quantity recomputed on the observed genes or on the labels' true rows, where computed, and the section and table of the main text that examine the corresponding structure under the convention of this paper (—, not examined here). Abbreviations: OR, odds ratio; ER, effective rank; TwoNN, two-nearest-neighbour intrinsic dimension; emp p, empirical permutation p; r_{rb} and rbc, rank-biserial correlation; OOV, out of vocabulary; GC, germinal centre; FFL, feed-forward loop; CKA, centred kernel alignment; U, left singular vectors; S, singular values; TF, transcription factor; PPI, protein–protein interaction; GO BP, Gene Ontology biological process; EV, extracellular vesicle; mito, mitochondrion; DC, dendritic cell; CV, cross-validation; OOF, out-of-fold. No new intervals are computed for this table. Full values: `Table_S13_claims_ledger.csv`.

ID	Claim	Source records; data	As reported	Screen's procedure, recomputed	Status	Observed-gene value; this paper
C01	Size of the screen	all records, iterations 1–63	63 iterations; 183 hypotheses; 13 families	183 records in 61 JSON files (63 iteration dirs; iter_0001 and iter_0003 have no JSON); 8 JSON families	*reproduces*: 183/63; no (families: 8 in the JSON, 13 is Kendiukhov (2026b)'s own regrouping)	Sec. 2.9; Sec. 4.16 (Table S12)
C02	Effective-rank collapse	46/H02 immune; 2,000-row subsample (zero rows)	23.6 (L_0) → 1.6 (L_{11}), 14.4-fold, 'computed on the full 4,803-gene vocabulary'	50 random subsamples: L_0 23.92 [22.44–25.24], L_{11} 1.67; fold 14.3; full 4,941 rows: 25.46→1.67 (15.2-fold); kidney A full 4,803 rows (what the text says): 19.54→1.63 (12.0-fold)	*reproduces*: on immune subsample (approx.; seed unknown); no on the matrix the text names	*Observed genes / true rows:* immune observed 2,039 rows: 236.9→48.7 (4.86-fold); Sec. 2.2; Table 2; Table 3 (Table S1)
C03	SV_1 variance fraction	46/H02, 7/H02 immune; zero rows included	93.4% at L_{11} (53.7% at L_0), 'full vocabulary'	immune subsample mean L_0 53.5%, L_{11} 93.1%; kidney A full 4,803: L_0 57.2%, L_{11} 93.2%	*reproduces*: approx.	*Observed genes / true rows:* immune observed: 5.6% → 18.6%; kidney A observed: 8.1% → 23.1%; Sec. 2.2; Table 2; Table 3
C04	SV_1 variance on the named-gene submatrix	9/H03, 10/H03 kidney A; 209-gene subset (22 zero rows)	19% (L_0) → 77% (L_{11}), '195-gene submatrix'	19.05% → 76.74%	*reproduces*: on 209 rows incl. 22 zero rows	*Observed genes / true rows:* 187 observed named genes: 7.1% → 27.5%; misaligned '195': 8.0% → 25.9%; Sec. 2.2 (all observed genes)
C05	TwoNN intrinsic dimension	45/H02, 46/H01 immune; not spectral	32.6 → 18.1 (44.6% decrease), 2,000 random genes, three seeds	iter_0042 estimator, seeds 42/0/1: L_0 28.4/27.8/34.4, L_{11} 18.7/16.9/20.3	*does not reproduce*: not exactly; seed and estimator of iter_0045 unknown, value depends on the sample: L_0 range 27.8–34.4 across 3 seeds	—
C06	Participation ratio	31/H03 kidney A; misaligned rows 0–208	58 → 9.5 (6.1-fold), '195-gene submatrix'	58.09 → 9.53 (misaligned rows, as executed)	*reproduces*: exactly, on the misaligned rows	*Observed genes / true rows:* true 187 observed named genes: 59.47 → 10.28; Sec. 2.2 (participation ratio, all observed genes)

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Table S13 (continued)

ID	Claim	Source records; data	As reported	Screen's procedure, recomputed	Status	Observed-gene value; this paper
C07	Feature-shuffle control for effective rank	46/H02 immune; 2,000-row subsample (zero rows)	ER 28.9 at L_{11} (17.6× above observed)	mean of 10 shuffles: 29.2	*reproduces*: approx.	Sec. 2.2 (feature-permutation control)
C08	SV_1 extracellular pole	6/H02 kidney A; full matrix (zero rows)	GO:0005615 OR = 6.37, $p = 2.6\times10^{-4}$ (SV_1, L_{11})	OR = 6.3677, $p = 2.57\times10^{-4}$ (21 vs 5 genes)	*reproduces*: exactly	*Observed genes / true rows:* observed-gene SVD, 187 genes, quartiles: OR = 1.00, $p = 0.60$; Sec. 2.3; Table 3 (Table S2)
C09	SV_1 cytosolic pole	7/H03 kidney A; full matrix (zero rows)	GO:0005829 OR = 2.96, $p = 0.010$	OR = 2.956, $p = 0.0104$	*reproduces*: exactly	Sec. 2.3 (Table S2)
C10	SV_1 gene-label-shuffle null	7/H01 kidney A; full matrix (zero rows)	$N = 500$, empirical $p = 0.004$	—	*provenance only*: null = permuting named genes over rows of a matrix whose SV_1 is the zero-row axis; the statistic is the min Fisher p over 296 terms	Sec. 2.3 (max-T over 288 tests)
C11	Compartment transients on SV_1	8/H02, 9/H02 kidney A; 209-gene subset (22 zero rows)	mito OR 23.3 at L_3; endoplasmic-reticulum lumen OR 4.7 → 18.5 (L_1–L_{11})	—	*provenance only*	Sec. 2.3 (compartment gradient)
C12	SV_2 / compartment enrichments	8/H01, 9/H01, 10/H01 kidney A; 209-gene subset (22 zero rows)	SV_2 IL-4 emp $p<0.001$; EV OR 7.99; cytoskeleton OR 20.4	—	*provenance only*	Sec. 2.3 (first twelve directions tested)
C13	TRRUST co-pole on SV_2	10/H02, 11/H01 kidney A; 209-gene subset (22 zero rows)	observed 0.206 vs null 0.122; $p < 0.001$ at all 12 layers	L_{11} SV_2 co-pole = 0.2055 over 326 rows (SV_1: 0.1626)	*reproduces*: 0.206	Sec. 2.6 (within-factor edge benchmark)
C14	Sign specificity of TRRUST co-pole	11/H01 kidney A; 209-gene subset (22 zero rows)	activation 12/12 layers; repression 1/12	—	*provenance only*	—
C15	SV_3 signalling axis	14/H01 kidney A; 209-gene subset (22 zero rows)	all layers p ≤ 0.017	—	*provenance only*	—
C16	SV_1 stability and annotation-density check	9/H03, 10/H03 kidney A; 209-gene subset (22 zero rows)	mean adjacent-layer Spearman $r = 0.929$; zero annotation-density confounders	mean adjacent-layer Spearman (raw signs) = 0.9283	*reproduces*	—
C17	SV_1 depleted of TFs	51/H02 immune; observed rows	top-quartile SV_1: ∼9× fewer TFs, OR = 0.108, $p < 1\times10^{-4}$	—	*provenance only*	Sec. 2.5 (single directions, Table S6)
C18	Cell-type marker clustering	23/H01 kidney A; not spectral	AUROC = 0.851 (12/12 layers $p < 1\times10^{-8}$); random-partition control 0.488	mean AUROC = 0.8508 (float32, as executed; float64 gives 0.8504)	*reproduces*: exactly in float32	*Observed genes / true rows:* dropping the 3 never-observed markers (DCN, COL1A1, MUC1): mean AUROC 0.843 (T and B only)
C19	Marker AUROC trend with depth	23/H01 kidney A; not spectral	slope = +0.036 per layer, $r = 0.765$, $p = 0.004$ ('increases with depth')	slope of logit(AUROC) on layer = −0.0357, $r = -0.765$, $p = 0.0038$	*does not reproduce*: sign reversed in Kendiukhov (2026b): AUROC decreases with depth	Table 3
C20	Gene-family clustering	26/H01 kidney A; not spectral	HLA-I 1.000; AP1 0.969; RUNX 0.925; IL-2 0.869	—	*provenance only*	—
C21	STRING PPI co-pole by axis (Tables 1 and S4 of Kendiukhov 2026b)	12/H03, 13/H01 kidney A; 209-gene subset (22 zero rows)	1,022 pairs (≥ 0.7); SV_2 mean 0.226 vs null 0.124 (12/12); SV_3 0.198 (12/12); SV_1 0.119 (2/12)	pairs = 1022; mean co-pole SV_2 0.226, SV_3 0.198, SV_1 0.128; SV_2 per-layer = Table S4 of Kendiukhov (2026b) exactly	*reproduces*: for SV_2/SV_3 and N pairs; no for SV_1 (0.128 vs 0.119)	Sec. 2.4; Table 3 (Tables S3, S4)

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Table S13 (continued)

ID	Claim	Source records; data	As reported	Screen's procedure, recomputed	Status	Observed-gene value; this paper
C22	SV_4 / SV_5 PPI co-pole	15/H01 kidney A; 209-gene subset (22 zero rows)	SV_4 0.158 (7/12); SV_5 0.149 (6/12) on 3,092 pairs (≥ 0.4)	pair count = 3092; rates not recomputed	*partly*: pair count	Sec. 2.9 (Table S14)
C23	STRING confidence gradient	15/H02 kidney A; 209-gene subset (22 zero rows)	quintile z 1.00/1.48/ 2.09/3.18/5.02; Spearman $\rho = 1.000$, $n = 5$, $p = 0.017$	exact two-sided p for $\rho = 1$ with $n = 5$: 0.0167; the record's $p = 1.4\times10^{-24}$ is scipy's t-approximation (1.40×10^{-24}) at $\rho = 1$	*reproduces*: $p =$ 0.017 is right; the screen's 1.4×10^{-24} is an artefact	Sec. 2.4; Table 3 (STRING confidence bands)
C24	PPI-only vs GO-only pairs	15/H03 kidney A; 209-gene subset (22 zero rows)	PPI-only $N = 670$, $z = 2.66$ (12/12); GO-only $z = 1.89$ (7/ 12)	—	*provenance only*	—
C25	Hub control	14/H03 kidney A; 209-gene subset (22 zero rows)	low-degree $z = 3.1$ vs hub $z = 2.6$	—	*provenance only*	Sec. 2.4 (hub removal, degree-stratified null; Table S3)
C26	Orthogonality of SV_2–SV_4 PPI axes	16/H02 kidney A; 209-gene subset (22 zero rows)	pairwise Pearson $r <$ 0.25	—	*provenance only*	—
C27	TRRUST pairs absent from STRING	22/H01 kidney A; not spectral	$N = 141$, AUROC = 0.573, 12/12 layers $p <$ 0.007	—	*provenance only*	Sec. 2.6 (within-factor edge benchmark)
C28	TF vs target classifier, joint 6-D SV_2–SV_7	56/H01 immune; observed rows	AUROC = 0.744 (mean over layers), max 0.789 at L_3, all 12 layers $p < 0.01$ (100 perms); $n = 51$ TFs vs 144 targets, cosine-to-TF score, 195-gene set	mean 0.747, max 0.792 (L_2); pooled-OOF mean 0.745; L_3 100-perm $p =$ 0.0099 (= floor 1/101)	*reproduces*: approx.; ±0.01, CV split seed unknown	Sec. 2.5; Table 3 (Table S6)
C29	Joint beats single subspaces; cross-seed	56/H01, 57/H01 immune (3 draws); observed rows	joint > both in 11/12 layers; cross-seed means 0.744/0.753/ 0.757 (SD ∼0.016)	joint > both in 10/12; seed43 mean 0.749, seed44 0.756	*reproduces*: approx.	Sec. 2.5 (all 78 contiguous bands)
C30	Depth division of labour	55/H02, 56/H01 immune; observed rows	SV_5–SV_7 dominates L_0–L_3 (∼0.71); SV_2–SV_4 L_4–L_8 (∼0.72)	SV_5–SV_7 L_0–L_3 mean 0.716; SV_2–SV_4 L_4–L_8 mean 0.737	*reproduces*: approx.	Sec. 2.5 (all 78 contiguous bands)
C31	SV_2–SV_4 signal explained by co-expression	51/H01 immune; observed rows	residual non-significant at 0/ 12 layers	layers with $p < 0.05$: 1/12 (L_8 rbc = 0.083, $p =$ 0.0016)	*does not reproduce*: 1/12, not 0/12 (the JSON record says 1/12)	Sec. 2.6 (co-expression comparator)
C32	SV_5–SV_7 regulatory proximity after co-expression residualisation	52/H03, 53/H01 immune; observed rows	$r_{rb} = 0.148$, $p <$ 0.001 at L_0; 100/100 bootstrap positive	L_0 rbc = 0.148, L_1 0.119, L_2 0.083; significant layers 3/12; L_5–L_{11} rbc < 0	*reproduces*: exactly	Sec. 2.6; Sec. 2.7
C33	Edge-level TF-target AUROC	62/H01 immune; observed rows	0.602 at L_0 (SV_5–SV_7); 589 positives vs 2,351 negatives	L_0 0.6017, L_1 0.6006 (U coords, 1,523 candidates); with U*S: 0.6053	*reproduces*: exactly, with 1,523 candidate pairs	Sec. 2.6; Table 3 (Table S7)
C34	Edge AUROC decays with depth	62/H01, 62/H02 immune; observed rows	Spearman $\rho =$ -0.958, $p =$ 9.5×10^{-7}; L_0–L_8 significant, L_9–L_{11} not	$\rho = -0.958$, $p = 9.54\times10^{-7}$	*reproduces*: exactly	Sec. 2.6 (every layer, Table S7)
C35	SV_2–SV_4 edge AUROC	62/H01 immune; observed rows	mean 0.492 (near/ below chance)	mean 0.492	*reproduces*	Sec. 2.6; Sec. 2.7
C36	Non-zero embeddings	62/H02 immune	constant at 2,039 of the 4,803 vocabulary genes	immune: 2039 of 4941; kidney A: 1972 of 4803	*does not reproduce*: 2,039 is the immune count; the 4,803-gene matrix has 1,972	Sec. 2.1; Table 1
C37	Repression more separable than activation	63/H63-G immune; observed rows	$\Delta = +0.021$ (SV_5–SV_7), +0.084 (SV_2–SV_4)	—	*provenance only*	—

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Table S13 (continued)

ID	Claim	Source records; data	As reported	Screen's procedure, recomputed	Status	Observed-gene value; this paper
C38	Mechanistic probe	63/H63-A immune; observed rows	TF-TF 0.539 / TF-target 0.485 (SV_2–SV_4); 0.528 / 0.599 (SV_5–SV_7)	—	*provenance only*	—
C39	B-cell kNN clustering	37/H01 kidney A; not spectral; misaligned rows	precision@10 $z = 7.55$ ($p < 0.001$); T-cell $z = -1.37$; DC $z = -0.87$	misaligned (as executed): L_2 $z = 7.55$	*reproduces*: exactly, on misaligned rows	*Observed genes / true rows:* true rows (187 observed): $z = L_0$ 7.16, L_2 8.22, L_5 8.30, L_8 7.65, L_{11} 6.31; Table 3
C40	B-cell signal not driven by norm	34/H03 kidney A; misaligned rows 0–208	survives L_2-norm regression (residual $p = 0.002$)	—	*provenance only*	—
C41	BATF/BACH2 convergence to the B-cell centroid	42/H03 immune; not spectral	BATF rank 1,510 → 189, dist to PAX5 18.3 → 5.1, $\rho = -0.972$; BACH2 611 → 146, 17.4 → 5.2, $\rho = -0.844$	BATF ranks 1510 → 189, dist 18.32 → 5.13, ρ −0.972; BACH2 611 → 146, 17.41 → 5.15, ρ −0.844	*reproduces*: exactly	—
C42	PAX5 as stable anchor	40/H01, 41/H02 immune; not spectral	rank ∼66 from L_0; convergence onset L_3	PAX5 rank (genes closer) $L_0 = 65$, range 28–72	*reproduces*: $65 = 66 - 1$	—
C43	Whole GC circuit converges	43/H01 immune; not spectral	PRDM1 $\rho = -1.000$; BACH2-PRDM1 tightest pair (3.94) at L_{11}	—	*provenance only*	—
C44	GC vs plasma programmes diverge	38/H02 kidney A; not spectral; misaligned rows	angle 77° (L_0) → 94° (L_{11})	misaligned (as executed): 77.1 → 94.4°	*reproduces*: exactly, on misaligned rows	*Observed genes / true rows:* true rows: 62.7 → 55.0° (the trend reverses); Table 3
C45	BCL6 metabolic neighbourhood	41/H02, 42/H02 immune; not spectral	10 of 20 nearest neighbours metabolic at every layer	—	*provenance only*	—
C46	B-cell manifold compression	41/H03 immune; not spectral	TwoNN 8.2 → 5.1, $\rho = -0.951$; T-cell $\rho = +0.287$ ($p = 0.37$); myeloid +0.699 ($p = 0.011$)	—	*provenance only*	—
C47	Lineage-specific delay	41/H01 immune; not spectral	T-cell / myeloid TFs start at ranks 170 / 86; GC $\rho = -0.993$ ($p < 1\times10^{-6}$)	—	*provenance only*	—
C48	Persistent homology collapses under rewiring	historical loop iter_0003–0008 (folders overwritten) earlier loop	11/12 layers $p < 0.05$ (feature shuffle); 0/24 under degree-preserving rewiring	—	*not recomputable*: iteration folders overwritten by an earlier loop	—
C49	Cross-model alignment partial	historical loop iter_0003 earlier loop	cosine 0.825; permutation $p > 0.3$	—	*not recomputable*: iteration folders overwritten by an earlier loop	—
C50	Geneformer static embeddings encode STRING	19/H01 Geneformer + kidney A	$p = 7.8\times10^{-127}$	—	*provenance only*	Sec. 2.7 (static embeddings as comparators)
C51	Geneformer B-cell structure absent	40/H02 Geneformer	precision@10 = 0.000	—	*provenance only*	—
C52	ER-AUROC correlation confounded by layer	58/H03, 59/H03 immune (3 draws); observed rows	$\rho = 0.855$; partial $\rho = -0.045$	—	*provenance only*	—
C53	GO BP not encoded in SV_2 poles	11/H03 kidney A; 209-gene subset (22 zero rows)	591 terms, zero significant	—	*provenance only*	—
C54	Feed-forward-loop geometry rejected	61/H03 immune; observed rows	264 FFLs, no significant result after permutation correction	—	*provenance only*	—

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Table S13 (continued)

ID	Claim	Source records; data	As reported	Screen's procedure, recomputed	Status	Observed-gene value; this paper
C55	Cross-seed CKA	61/H01 immune (3 draws); not spectral	0.979 (L_0) → 0.779 (L_{11})	—	*provenance only*	Sec. 2.8 (variance across cell draws)
C56	Attention vs residual geometry	18/H03 lung attention, kidney A labels	attention TRRUST 2× ($p = 9.9\times10^{-9}$); STRING $p = 0.091$	—	*provenance only*	—
C57	Graph topology	2/H01_graph_topology_cc kidney A (3 draws); not spectral	clustering coefficient $z > 9$ at all 36 tests	—	*provenance only*	—
C58	TRRUST signed subsets on the edge benchmark	63/H63-G immune	589 pairs: 270 activation-only, 141 repression-only, 178 mixed/ambiguous	any-Activation 270, any-Repression 141 (overlap 31); activation-only 239, repression-only 110, unknown-only 209	*partly*: 589 yes; 270/141 are 'any' counts that overlap by 31, so they are not 'only' sets and 178 = 589 − 270 − 141 is not a count of anything	—
C59	Named genes	30/H04 kidney A	209 named; 14 OOV (zero norm); 195 in-vocabulary genes used for all biological analyses	true rows: 22 zero, 187 observed; misaligned mapping: 14 zero (FOS, HLA-A, HLA-DPB1, JUNB, KLF6, LDHA, LGALS1, NCAM1, NCOA3, NR4A3, PAX5, PTGS2, TBXAS1, TNF) → 195	*does not reproduce*: 195 / 14 OOV are artefacts of the misaligned mapping; truth is 187 observed / 22 never-observed	Sec. 2.1; Sec. 2.9 (observed genes)
C60	TRRUST pairs in the named set	10/H02 kidney A	326 TF-target pairs (screen record; Kendiukhov (2026b) uses the resulting co-pole rate)	326 TRRUST rows = 288 distinct pairs; 245 with both genes observed	*reproduces*: as a row count; 288 distinct pairs	Sec. 2.6; Table 1 (edge benchmark counts)
C61	Front-loaded compression	6/H03 kidney A; 209-gene subset (22 zero rows)	effective rank below half its initial value by layer 4	—	*provenance only*	Sec. 2.2

Table S14: **Correspondence of other spectral bases with the observed-gene basis, and provenance of the screen's three bands.** *Panel A*, correspondence of single directions, from the right singular vectors (512-dimensional feature space; $|\cos|$ is sign-free) at each layer. kidney A, full matrix: the decomposition of all 4,803 rows, mean-centred over all of them, including the 2,831 never-observed zero rows, whose first direction is therefore the observation indicator; it is compared with the decomposition of the 1,972 observed genes (this paper's convention). For its SV_k, $k = 2, \ldots, 12$: best = SV_{k-1}, number of directions whose best-matching observed direction (largest $|\cos|$ among observed SV_1–SV_{40}) is SV_{k-1}; mean and range of that best $|\cos|$ over the 11 directions; SV_1: best $|\cos|$ of its first direction with any observed direction (and that direction), showing that it has no counterpart. Pooled over the 132 layer–direction cases, the mean $|\cos|$ between full-matrix SV_k and observed SV_{k-1} exactly is 0.956 (independent re-derivation); at layers where all 11 best matches are SV_{k-1} it equals the mean shown. kidney A, 209-gene: mean best $|\cos|$ over SV_1–SV_{12} for the decomposition fitted to the 209 named genes at their true rows (22 of them zero rows), on which the screen assembled SV_2–SV_4. immune, screen's basis: the screen's decomposition of the 2,039 observed rows ("centered nonzero embeddings") against this paper's; best = SV_k, number of SV_1–SV_{12} whose best match is the same index, with the mean $|\cos|$ at the same index. *Panel B*, mapping of bands: band, the band's indices in that decomposition; matrix and decomposition (rows), the matrix and the rows the decomposition was fitted to (mean, mean embeddings; summed, the summed-embedding matrix the screen loaded in iterations 59–60); used in, the iterations of the screen whose records use the band in that decomposition (—, a reference basis: the full kidney A decomposition, never-observed rows included); mapped, the contiguous observed-gene band of the same width (ending at or before SV_{12}) with the largest mean $\cos^2$ of the principal angles between the two subspaces, averaged over the 12 layers ($\overline{\cos^2}$; range, its range over the 12 layers); identity, the same quantity for the band with the same indices; runner-up, the second-best band; layers won, number of layers at which the mapped band is the best (other winning bands in parentheses). The mapping uses the geometry of the matrices only; no outcome data (interaction, regulatory or label sets) were read to fix it. *Panel C*, provenance of the bands from the iteration files (JSON records, iteration reports, the language-model brainstormer's briefs and next-step notes): #, step; iteration, iteration and hypothesis or file; band, the band the step bears on (all, every direction); text in quotation marks is verbatim from those files. "sig" is the screen's count of layers with empirical permutation $p < 0.05$; emp_p and perm_p, empirical permutation p; mean_z, mean z over layers; rbc, rank-biserial correlation; TRRUST, STRING, reference networks of documented regulation and protein interaction; PPI, protein–protein interaction; TF, transcription factor. None of the three bands was specified before the results that motivated it were seen. No sampling intervals apply: the matrices are fixed and all values are exact geometric quantities. Full values: `Table_S14_band_crosswalk.csv`.

A. SV_k of the full-matrix, 209-gene and screen's immune decompositions against the observed-gene SV_k and SV_{k-1}, per layer

	kidney A: full matrix, zero rows included				kidney A: 209-gene	immune: screen's basis	
Layer	best = SV_{k-1}	mean $\lvert\cos\rvert$	range $\lvert\cos\rvert$	SV_1: best $\lvert\cos\rvert$	mean best $\lvert\cos\rvert$	best = SV_k	mean $\lvert\cos\rvert$
L_0	10/11	0.971	0.824–0.999	0.219 (SV_3)	0.488	12/12	1.000
L_1	11/11	0.994	0.983–1.000	0.233 (SV_{12})	0.483	12/12	1.000
L_2	11/11	0.992	0.952–1.000	0.270 (SV_{12})	0.485	12/12	1.000
L_3	9/11	0.941	0.686–1.000	0.224 (SV_{10})	0.501	12/12	1.000
L_4	11/11	0.982	0.916–1.000	0.261 (SV_8)	0.492	12/12	1.000
L_5	9/11	0.946	0.702–1.000	0.309 (SV_7)	0.473	12/12	1.000
L_6	9/11	0.969	0.825–1.000	0.356 (SV_6)	0.474	12/12	1.000
L_7	11/11	0.958	0.785–1.000	0.309 (SV_5)	0.473	12/12	1.000
L_8	11/11	0.973	0.901–1.000	0.265 (SV_5)	0.524	12/12	1.000
L_9	11/11	0.985	0.957–0.998	0.232 (SV_5)	0.543	12/12	1.000
L_{10}	11/11	0.976	0.904–0.999	0.211 (SV_5)	0.584	12/12	1.000
L_{11}	11/11	0.994	0.987–1.000	0.155 (SV_3)	0.555	12/12	1.000
all layers	125/132 (94.7%)	0.973	0.686–1.000	0.254 (mean)	0.506	144/144	1.000

B. The three bands in each decomposition, mapped to the observed-gene basis

Band	Matrix	Decomposition (rows)	Used in	Mapped	$\overline{\cos^2}$	range	identity	runner-up	layers won
SV_2–SV_4	kidney A	full matrix (2,831 zero rows)	— (reference basis)	SV_1–SV_3	0.990	0.966–0.999	0.659	SV_2–SV_4 (0.659)	12/12
SV_5–SV_7	kidney A	full matrix (2,831 zero rows)	— (reference basis)	SV_4–SV_6	0.938	0.697–0.993	0.649	SV_3–SV_5 (0.650)	12/12
SV_2–SV_7	kidney A	full matrix (2,831 zero rows)	— (reference basis)	SV_1–SV_6	0.966	0.848–0.992	0.820	SV_2–SV_7 (0.820)	12/12
SV_2–SV_4	kidney A	209 named genes (22 zero rows)	iterations 13–19 (co-pole axes)	SV_1–SV_3	0.489	0.350–0.640	0.452	SV_2–SV_4 (0.452)	8/12 (SV_2–SV_4 3, SV_5–SV_7 1)
SV_5–SV_7	kidney A	observed (1,972), mean	later iterations on kidney A	SV_5–SV_7	1.000	1.000–1.000	1.000	SV_4–SV_6 (0.667)	12/12
SV_5–SV_7	kidney A	observed (1,972), summed	iterations 59–60	SV_3–SV_5	0.212	0.146–0.319	0.198	SV_4–SV_6 (0.210)	2/12 (7 other bands)
SV_2–SV_4	immune	observed (2,039)	iterations 49+ (TRRUST, TF/target)	SV_2–SV_4	1.000	1.000–1.000	1.000	SV_1–SV_3 (0.667)	12/12
SV_5–SV_7	immune	observed (2,039)	iterations 50+	SV_5–SV_7	1.000	1.000–1.000	1.000	SV_4–SV_6 (0.667)	12/12
SV_2–SV_7	immune	observed (2,039)	iterations 56+ (joint classifier)	SV_2–SV_7	1.000	1.000–1.000	1.000	SV_1–SV_6 (0.833)	12/12

C. How the bands were chosen

#	Iteration	Band	Matrix, gene rows, convention	What was done or decided	Result or criterion (quotations verbatim)
1	8 H01	SV_2–SV_4	kidney A, 209 true rows, subset-fitted	first use of SV_2 and SV_3: "SVD of layer-11 mean-centered 209-gene embedding. Project onto SV_2 and SV_3."	SV_2 IL-4 term emp_p=0.000; SV_3 emp_p=0.088

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Table S14 (continued)

#	Iteration	Band	Matrix, gene rows, convention	What was done or decided	Result or criterion (quotations verbatim)
2	10 H02	SV_2–SV_4	same as above	TRRUST co-pole on SV_1 and SV_2 at L_{11} (326 TRRUST rows, i.e. 288 distinct pairs)	"SV_1: obs=0.163 vs null=0.123, emp_p=0.016; SV_2: obs=0.206 vs null=0.122, emp_p=0.000"
3	13 H01	SV_2–SV_4	same as above	STRING (score $\geq$ 0.7, 1,022 API pairs) co-pole on SV_1, SV_2 and SV_3	"SV_1 z=0.39 (2/12 sig), SV_2 z=10.04 (12/12 sig), SV_3 z=7.18 (12/12 sig)"
4	14 brief	SV_2–SV_4	—	brainstormer proposes extending to the next axes: "H01: SV_4/SV_5 PPI co-pole test", with the success criterion "SV_4 or SV_5 $z > 3$ in $\geq$6/12 layers"	—
5	15 H01	SV_2–SV_4	same, but 3,092 pairs at STRING $\geq$ 0.4, not the brief's 1,022 at $\geq$ 0.7	SV_4 and SV_5 tested	"SV_4 mean_z=2.21, sig=7/12; SV_5 mean_z=1.65, sig=6/12" ("sig" = emp_p $<$ 0.05). Against the stated criterion "$z > 3$ in $\geq$6/12", SV_4 passes at 6/12 with L_1 z = 3.00003, and SV_5 fails at 2/12
6	15 brief	SV_2–SV_4	—	SV_5 dropped: "SV_5 (deprioritized; rescue only if SV_6+ shows new pattern)"	SV_4 kept by a z-margin of 3×10^{-5} at one layer; the group becomes SV_2, SV_3, SV_4
7	16 H02, 19 H02, 20	SV_2–SV_4	same as above	"SV_2/SV_3/SV_4 axis independence"; "multi-axis co-polarity count (SV_2+SV_3+SV_4)"	the SV_2–SV_4 triple becomes a unit, defined by PPI co-pole on the kidney 209-gene subset
8	47 H01/ H02	all	immune, 2,039 observed	the screen finds the zero rows: "Found 2902 genes have zero norm at ALL layers", and that the earlier "93.4%" SV_1 figure "was dominated by zero vectors"	switch to observed rows only
9	48 brief	SV_2–SV_4	immune, observed	the kidney band is carried over unchanged: "H02 (HIGH VALUE): SV_2–SV_4 PPI Geometry in 2039-Gene Active Set", comparing with "prior ρ values (SV_2: 1.0, SV_3: 0.9, SV_4: 0.7)" from the kidney matrix	—
10	49 H03	SV_2–SV_4	immune, observed; coordinates = left singular vectors	first band-level test of SV_2–SV_4 on TRRUST pairs	"Best: L_8 p<0.0001 effect=0.268; significant (p<0.05) at 8/12 layers"
11	49 next steps	SV_5–SV_7	—	**origin of** SV_5–SV_7: "Q: Is the effect specific to SV_2–SV_4, or does it extend further into the spectrum? Test SV_5–SV_7, SV_8–SV_{10} subspaces"	fixed-width windows of three, chosen after the SV_2–SV_4 result
12	50 H03	SV_5–SV_7	immune, observed	first recorded SV_5–SV_7 (with SV_1, SV_2–SV_4, SV_8–SV_{10})	"SV_2–SV_4: rbc=+0.229 (10/12 layers p<0.05); SV_5–SV_7: rbc=+0.152 (7/12 layers); SV_8–SV_{10}: rbc=+0.123 (5/12 layers)"
13	50 H01, 51 H01	SV_2–SV_4	immune, observed	SV_2–SV_4 TRRUST proximity tested against "co-expression" (in practice the cosine of the full 512-d embeddings, not expression correlation)	"0/12 layers specific"; residualised: "1/12 layers significant (L_8: rbc=0.083, p=0.0016)"
14	51 brain-stormer	SV_5–SV_7	—	SV_5–SV_7 promoted as the alternative: "SV_5–SV_7 regulatory signal (independent of co-expression) | UNTESTED | High potential"	—
15	52 H03	SV_5–SV_7	immune, observed	SV_5–SV_7 residualised	"rbc_resid=0.148 (L_0), 0.119 (L_1), 0.083 (L_2); all p<0.002. L_3+ lose significance". Recomputed exactly (0.148 / 0.119 / 0.083); layers 5–11 have *negative* rbc
16	53 H01	SV_5–SV_7	immune, observed	other bands scanned only after SV_5–SV_7 had been chosen: "SV_2–SV_4, SV_5–SV_7, SV_8–SV_{10}, SV_{11}–SV_{14}, SV_{15}–SV_{20}, SV_{21}–SV_{30}" at $L_0/L_1/L_2/L_8$	"SV_5–SV_7 dominant at L_0–L_2; SV_2–SV_4 dominant at L_8"
17	55 H02	SV_2–SV_7	immune, observed	layer profiles of SV_5–SV_7 and SV_2–SV_4 classifiers	"Complementary encoding: SV_5–SV_7 high early+late, SV_2–SV_4 high mid-depth"
18	55 brain-stormer	SV_2–SV_7	—	**origin of** SV_2–SV_7: "can we dissect the joint SV_2–SV_7 space to produce a layer-stable classifier?"; decision threshold "joint AUROC $\geq$ 0.72 at 10+ of 12 layers"	—
19	56 H01	SV_2–SV_7	immune, observed	joint 6-D classifier	"joint $\geq$ 0.72 at 9/12 layers". The threshold was then relaxed: "Lower threshold for criterion since 9/12 vs 10/12 is minor; note claim should state 'most layers"'
20	63 H63-B	all	immune, observed	the screen's own systematic scan of SV_k–SV_{k+2}, k = 0…9, at L_0 with Bonferroni over 10 windows	"Best raw AUROC: SV_8–SV_{10}=0.619 (perm_p=0.67); SV_5–SV_7=0.611 (perm_p=0.28); no window survives Bonferroni"